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A Smart Maximum Power Point Tracking for LLC Converter

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Abstract

Enhancing the performance of photovoltaic (PV) systems under challenging conditions, such as partial shading, requires advanced control strategies and efficient power converters. Under shading, the P–V curve presents multiple local peaks, complicating maximum power extraction. Global maximum power point tracking (GMPPT) algorithms are essential for locating the true peak. Recently, deep learning techniques, particularly neural networks, have shown strong potential in improving GMPPT by offering faster convergence, greater accuracy, and better adaptability to dynamic irradiance conditions.

In this study, a two stage GMPPT method is proposed for a three series one parallel (3S1P) PV configuration that uses an LLC resonant converter. In the first stage, artificial neural network (ANN) and one-dimensional convolutional neural network (1D-CNN) are developed and is used to predict the GMPP. After the prediction, two initialization techniques based are used to initiate the second stage conventional perturb and observe (P&O) algorithm to fine tune the tracking, correcting any small errors and maximizing power output.

The proposed method was evaluated through both simulation and hardware testing under a wide range of partial shading and temperature conditions, demonstrating excellent robustness and adaptability. The ANN and 1D-CNN models accurately identified the correct power peak with classification accuracies of 94.2% and 97.8%, respectively, across 1851 test samples. The system consistently achieved tracking efficiencies above 99% in simulation and up to 99.53% in hardware experiments, with a maximum tracking time of just 0.17s in both cases, confirming its suitability for real-time PV applications. Furthermore, the proposed method operates without relying on irradiance or temperature sensors, significantly reducing hardware complexity and overall system cost.

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List of Abbreviations

Abbreviations	Definition
1D-CNN	One-dimensional convolutional neural network
3S1P	3 series 1 parallel
ANN	Artificial neural network
FHA	First harmonic approximation
GMPP	Global maximum power point
GMPPT	Global maximum power point tracking
I-V	Current-voltage
LMPP	Local maximum power point
MPP	Maximum power point
MPPT	Maximum power point tracking
P&O	Perturb and observe
PFM	Pulse frequency modulation
PSC	Partial shading conditions
PV	Photovoltaic
P-V	Power-voltage
SS-PSM	Secondary side phase shift modulation
ZVS	Zero voltage switching

Chapter 1. Introduction

1.1 Background

The accelerating shift towards renewable energy is driven by the global need to reduce carbon emissions, address the depletion of fossil fuel reserves, and meet the rising demand for sustainable electricity. Fossil fuels have been the main source of energy for many years, but they are now considered unsustainable because their use has harmed the environment and their supply is limited. Studies show that these fossil fuel reserves could be significantly depleted by the end of the century, highlighting the urgent need for alternative energy solutions [1]. In response to these challenges, renewable energy sources (RES) such as wind, hydro, and particularly solar photovoltaic (PV) systems have emerged as key solutions due to their sustainability and environmental benefits [2]. Among these technologies, solar PV is especially prominent for its scalability, declining installation costs, and ability to harness abundant solar energy. Global initiatives, including the United Nations' Sustainable Development Goal 7 (SDG 7) [3], which advocates for affordable and clean energy access, have accelerated PV adoption worldwide. This momentum was reinforced at COP28, where over 130 nations pledged to triple renewable energy capacity by 2030, highlighting the critical role of solar PV and wind energy in meeting future energy demands [4].

Despite significant technological advancements, PV systems still face efficiency challenges, especially when it comes to accurately tracking the maximum power point (MPP). This is mainly due to the nonlinear nature of their current-voltage (I-V) characteristics. maximum power point algorithm (MPPT) algorithms are designed to tackle this issue by continuously adjusting the operating point of the solar panels to ensure they deliver the highest possible power under changing environmental conditions. Traditionally, MPPT algorithms such as perturb and observe (P&O) [5] and incremental conductance (INC) [6] have been paired with hard-switched DC-DC converters like buck and boost topologies. While these methods are simple and commonly used, they come with drawbacks such as high switching losses and increased electromagnetic interference (EMI). They also struggle to adapt in rapidly changing conditions. To address these challenges, a smart MPPT method is needed for LLC resonant converters, which are recognized as a more efficient alternative. These converters reduce switching losses by using soft-switching techniques such as zero voltage switching (ZVS) to improve overall system efficiency and reliability [7].

Furthermore, the complexity increases further under non-uniform conditions, particularly with fluctuations in irradiance and temperature, which directly impact output power from the PV system and make consistent harvesting challenging. This is especially evident under partial shading conditions (PSC), where multiple local maximum power points (LMPPs) appear on the power-voltage (P-V) curve, further complicating efficient energy extraction. In such scenarios, traditional MPPT methods often fail to locate the true global maximum power point (GMPP), leading to significant tracking losses. This again highlights the need for an advanced and smart MPPT algorithm capable of accurately identifying the GMPP to improve tracking efficiency when integrated with the LLC resonant converter.

1.2 Problem Statement

Unlike duty cycle modulated converters, LLC resonant converters utilize different modulation strategies, each with their own advantages and disadvantages. As a result, conventional MPPT algorithms, which are designed for duty cycle control, are incompatible with LLC converters.

Another challenge arises under PSC and varying temperatures. Changes in irradiance can create multiple local maximum power points on the P-V curve, while an increase in temperature causes the voltage and power output to decrease and the current to rise, making the tracking process more challenging [8], [9]. Failing to accurately identify the true GMPP under these conditions can lead to significant losses in tracking efficiency and reduce the overall performance of the PV system. Recent progress in artificial intelligence, especially deep learning, offers practical ways to solve the challenges faced in solar power systems. Deep neural networks can understand complex patterns and adjust to changes in the environment and system conditions. Although deep learning methods for MPPT have shown good results in regular converters, their use in LLC resonant converters, particularly under partial shading, is still not well explored. Furthermore, many existing AI approaches also depend on cloud computing, which can cause delays, rely on internet connection, and raise security concerns. This makes them less suitable for real time control. Edge computing solves these problems by allowing the AI model to run directly on small devices like microcontrollers. This ensures faster response, better safety, and more stable operation without needing an internet connection. This project aims to close these gaps by creating an efficient smart MPPT method by exploring using both frequency and phase shift control, supported by deep neural networks and edge computing, to improve power tracking in LLC based solar systems.

1.3 Aims and Objectives

The aim of this project is to design and develop a real-time, high-efficiency smart MPPT algorithm for LLC resonant converter-based PV systems. The algorithm will use frequency and phase shift modulation separately combined with DNN and edge computing to improve GMPPT under changing environmental conditions. Special focus will be given to handling the effects of partial shading and temperature changes, which are known to create multiple power peaks and complicate the tracking process. The solution aims to provide adaptive control that responds effectively to these challenges, while remaining scalable for different PV system setups. The main objectives of this project are listed below:

i. To analyze the operation and modulation strategies of LLC resonant converters used in PV-based microgrids.
ii. To develop a smart MPPT algorithm that maintains high efficiency under partial shading, temperature changes.
iii. To validate the algorithm through simulation using software simulation tool.
iv. To implement and test the algorithm on hardware using a programmable DC supply to simulate varying PV conditions.
v. To compare the performance of the developed algorithm with MPPT methods from the literature, particularly the method presented in [10].

1.4 Project Scope and Limitations

This project focuses on the design, development, and implementation of a smart MPPT algorithm integrated with DNN for LLC resonant converter-based PV systems. The algorithm is specifically tailored to the hardware constraints, particularly the maximum capacity of the programmable DC power supply and the existing ratings of the LLC converter prototype. As a result, the PV rating and configuration are defined within these limitations to ensure seamless compatibility during both simulation and hardware testing. Synthetic datasets replicating real-world PV conditions,

including PSC and temperature variations, will be generated to support effective model training and validation.

The passive component values used in the simulation are selected to closely match those of the hardware prototype. However, achieving exact equivalence is beyond the scope of this project due to the complexity and additional effort required for precise tuning. To ensure fairness in comparing simulation and hardware results, the evaluation will focus primarily on GMPP prediction accuracy and tracking efficiency, rather than the transient response of the converter.

All testing and validation activities will be performed at the Final Year Project Lab (DA18), University of Nottingham Malaysia. While standard practice connects the DC-DC converter output to a programmable DC load rated at 400V, this project operates within the laboratory constraint of a 120V electronic load. Consequently, both simulation and hardware testing will be conducted using a constant 120V load setting. Furthermore, the scope of this project is confined to algorithm development, simulation, hardware implementation, and performance evaluation. The design of new converter hardware, development of voltage and current sensors, or large-scale PV array configurations lies outside the scope of this work.

1.5 Deliverables

The following are the major deliverables of this project:

i. A highly accurate DNN-based model for GMPP prediction.
ii. A complete smart MPPT algorithm specifically designed for LLC resonant converters.
iii. A fully functional implementation of the smart MPPT algorithm validated through software simulation tool.
iv. A working hardware prototype demonstrating real-time operation of the smart MPPT algorithm on the LLC converter.

1.6 Project Timeline

Successful engineering projects rely on structured planning, systematic execution, and continuous monitoring to ensure completion within the designated timeframe. Project management methodologies facilitate this process by providing frameworks to efficiently organize, schedule, and oversee tasks. In this project, the Critical Path Method (CPM) has been employed to optimize task management and ensure timely delivery.

CPM is a project scheduling technique that identifies essential tasks whose timely completion is critical to the overall success of the project. According to *Atin* and *Lubis.* [11], CPM highlights activities with fixed deadlines, where delays could compromise project outcomes or lead to failure. By mapping task sequences and dependencies, CPM determines the critical path the longest chain of dependent tasks that defines the shortest possible project duration. Any delay along this path directly extends the overall timeline, as these tasks lack flexibility.

The straightforward nature of CPM, which involves summing task durations and identifying key dependencies, makes it an effective tool for managing complex workflows. Table 1.1 presents the critical tasks for this project along with their respective deadlines, providing a clear framework for tracking progress and maintaining schedule adherence.

Table 1.1. Deadlines of critical tasks.

Critical Tasks	Deadline
Subsmission of Project Outline Plaanning Report	31st October 2024
Proposed Smart MPPT Algorithm	9th March 2025
Software Completion	30th March 2025
Hardware Completion	30th March 2025
Submission of Final Thesis	30th April 2025

After identifying the key tasks, a revised Gantt chart was created, as shown in Appendix A, to more accurately reflect the project timeline. The work was divided into six main phases: project initiation and planning, analysis of the LLC resonant converter for microgrid use, design and development of the two-stage GMPPT algorithm, software simulation, edge computing deployment, hardware testing, and final thesis documentation and presentation. The Gantt chart details the tasks within each phase, with estimated durations that include a buffer of at least one additional weeks per task to accommodate potential uncertainties. This buffer serves as a contingency to address risks arising from factors beyond direct control, such as availability of human resources, procurement delays, and working environment constraints. Among these factors, human resource availability posed the greatest risk, particularly during hardware testing, where assistance from a PhD student was required for troubleshooting issues such as circuit board failures (as my role is to deploy the algorithm but not designing the hardware converter). Therefore, careful coordination of hardware testing activities was essential to prevent delays in subsequent tasks.

Despite thorough planning, delays occurred between weeks fourteen and twenty-four. These delays were recovered during the semester break and the early weeks of the spring semester. An additional delay was encountered after the Hari Raya break due to the need to access a programmable DC power supply located at another university. Transporting the equipment, along with the time required for unpacking, setup, minor circuit enhancements, and adjusting to an unfamiliar working environment, extended the task duration beyond initial estimates. This situation highlighted the importance of anticipating logistical challenges and allowing flexibility in scheduling. Furthermore, consistent and effective communication with all parties involved was crucial in managing these delays. Establishing mutual understanding and maintaining clear communication channels contributed significantly to recovering from disruptions and keeping the project on track.

1.7 Industrial Relevance

As global demand for renewable energy continues to grow, PV systems play a vital role in reducing dependence on fossil fuels and supporting sustainable development. The adoption of PV technology has surged in recent years, driven by rising environmental awareness and supportive government policies. According to [4], China is expected to contribute nearly 60 percent of the global growth in renewable energy capacity by 2028. It is also on pace to achieve its 2030 renewable energy target six years early, playing a key role in the global effort to triple renewable capacity [4]. In Malaysia, this trend is reflected through initiatives like the Malaysia Renewable Energy Roadmap (MyRER) and the National Energy Transition Roadmap (NETR), both aiming to significantly expand solar integration. Organizations such as the Malaysian Photovoltaic Industry Association (MPIA) also play a key role in driving national PV adoption and innovation.

In parallel with the growth of solar energy, the demand for efficient power conversion technologies has also increased. LLC resonant converters are gaining popularity due to their high efficiency, soft-switching behavior, and reduced electromagnetic interference (EMI) [7]. Their ability to reduce switching losses makes them well-suited for compact and reliable power conversion systems. However, integrating advanced control strategies like MPPT into LLC converters remains a challenge, especially under dynamic environmental conditions such as partial shading and temperature fluctuations.

This research also aligns with the growing role of AI in the energy sector. Deep learning techniques are increasingly used to improve energy management, grid operations, and automation [12]. Incorporating intelligent control strategies, machine learning, and edge computing supports the transition toward smart grids and echoes the goals of the Fourth Industrial Revolution (IR4.0). This project contributes not only to enhanced PV and power converter performance but also to broader sustainability efforts by promoting the adoption of AI-driven solutions in renewable energy and power electronics.

To address these challenges, this project proposes a smart, high-efficiency MPPT algorithm tailored for LLC resonant converters. By integrating deep learning and edge computing, the system enables adaptive, real-time control that improves energy harvesting, reduces system losses, and operates independently of cloud infrastructure.

1.8 Thesis Outline

Chapter 1. This chapter presents the background, problem statement, project aims and objectives, scope and limitations, industrial relevance, project deliverables, and project timeline.

Chapter 2. This chapter introduces the fundamental concepts and prior research relevant to this study. It covers PV systems, LLC resonant converters with various modulation strategies, MPPT algorithms and background on deep neural network.

Chapter 3. This chapter outlines the methodology used to achieve the project objectives. It describes the development of the two-stage GMPPT algorithm, including synthetic data generation, model training, and deployment on the edge computing platform. It also details the setup of software simulations and hardware implementation of the LLC resonant converter.

Chapter 4. This chapter presents the simulation and hardware testing results of the proposed smart MPPT system. It includes quantitative analysis of the deep learning models and a detailed comparison of the proposed algorithm's performance against existing methods.

Chapter 5. This chapter summarizes the key findings of the project and provides recommendations for future improvements to enhance system performance and expand research scope.

Chapter 2. Literature Review

2.1 Overview

This chapter provides the fundamental knowledge and background necessary for this project. It begins with an overview of PV systems and explores how the LLC resonance converter can be utilized as a microgrid converter. Next, various MPPT algorithms from existing literature are presented and compared. Since this project focuses on using neural networks as the MPPT algorithm, this chapter includes a discussion on the fundamentals of deep neural networks and their application in MPPT systems.

2.2 Photovoltaic System

2.2.1 Photovoltaic Models

Before exploring the deeper aspects of PV systems, it is important to acknowledge the origin of the concept. The journey began with Alexandre Edmond Becquerel's discovery of the photovoltaic effect in 1839 when he observed that sunlight could produce electric current in a solid material [13]. This process is explained by the photoelectric effect where certain materials convert light energy into electrical energy at the atomic level [13]. This happens inside a PV cell which is the basic building block of all photovoltaic systems.

A solar cell behaves similarly to a diode, but instead of being forward-biased by an external voltage, it generates current when exposed to light. Photons from sunlight energize the electrons, creating a photocurrent I_C that flows from the n-junction to the p-junction of the semiconductor material of PV cell [13]. In the equivalent circuit model shown in Figure 2.1, the solar cell is represented as a current source I_C in parallel with a diode (to represent internal structure of p-n junction). The total load current i_L delivered to the external circuit is the difference between the photocurrent and the diode's current i_D.

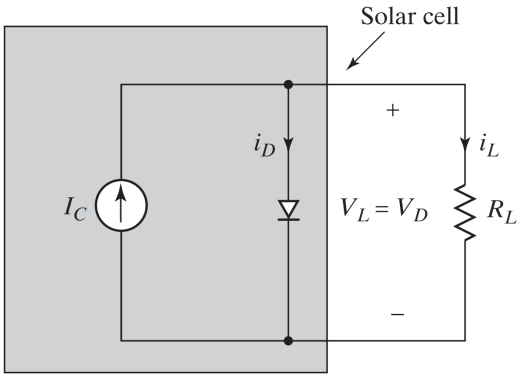


Figure 2.1. Ideal current source model of solar cell [13].

The load current can be expressed in terms of the diode current as follows.

$$i_L = I_C - I_S\left(e^{v_D/\eta V_T} - 1\right) \qquad \textit{Eq 2.1}$$

Where

- I_C is the current generated from solar cell.
- i_D is the current through the diode.
- v_D is the voltage across anode and cathode of the diode.
- I_S is the leakage or reverse saturation current of the diode.
- η is the emission coefficient or ideality factor.

Furthermore, the load lines shown in Figure 2.2 affect the operating point of the solar cell. This operating point depends on the value of the load resistance R_L and is determined by the intersection of the PV cell's I-V characteristic curve with the load line, defined by $i_L = v_L/R_L$ [13]. As the load resistance increases, the output voltage of the solar cell also increases. This concept is closely related to impedance matching, where the goal is to select a load resistance R_L that maximizes power transfer from the solar cell to the load [14].

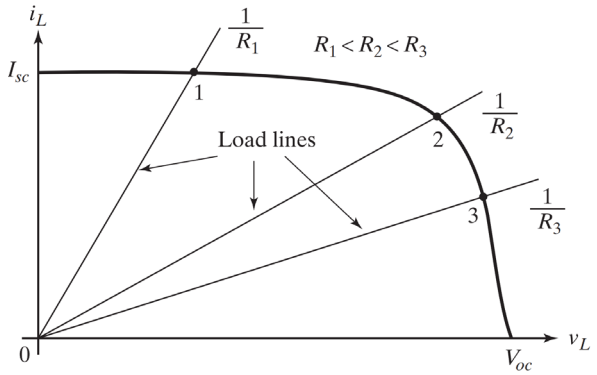


Figure 2.2. Load line representation and its effect on the solar cell operating point [13].

The I–V characteristics shown in Figures 2.3(a) and (b) are based on a specific irradiance ρ. An increase in irradiance directly raises the solar current I_C, the short-circuit current I_{SC} , and the open-circuit voltage V_{OC} , as illustrated in Figure 2.3(a). Additionally, the maximum power output of the solar cell also increases with higher irradiance, as shown in Figure 2.3(b).

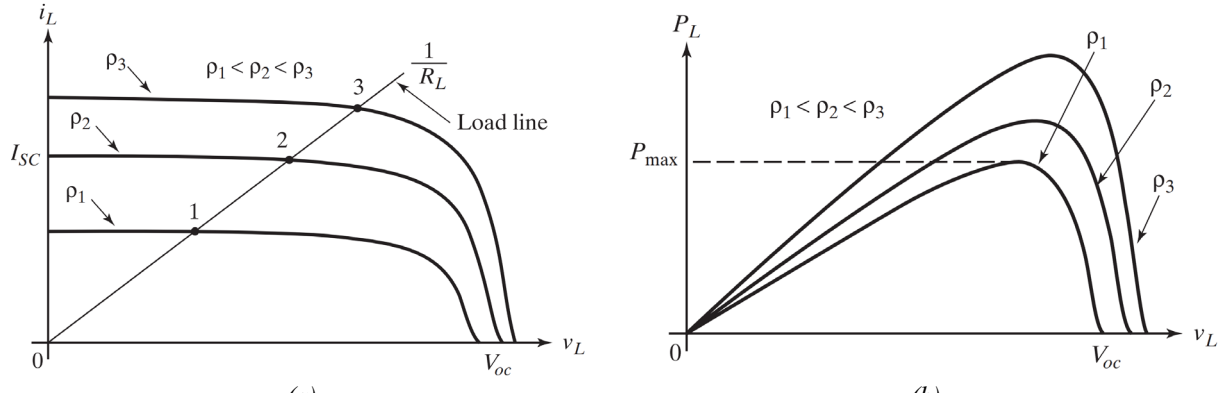


Figure 2.3. Effect of different irradiance to operating point of PV system [13]. *(a) I-V curve. (b) P-V curve.*

Moreover, the saturation current I_S is highly sensitive to temperature changes. The load current, as described in Equation 2.1, depends nonlinearly on temperature. As the temperature increases, the

open-circuit voltage V_{OC} decreases, as illustrated in Figure 2.4. Hence, it can be concluded that a PV system responds nonlinearly to changes in both temperature and irradiance.

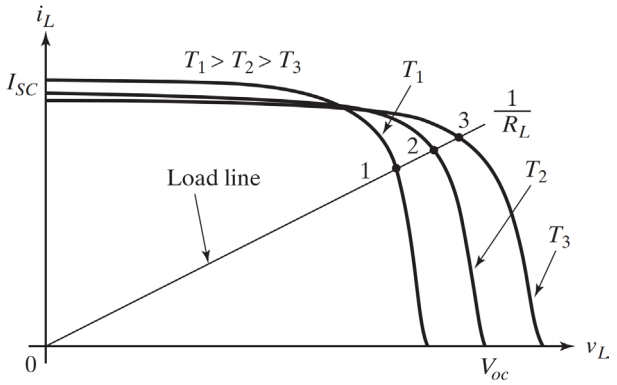


Figure 2.4. Effect of different temperature to operating point of PV system [13].

In a practical solar cell, electrical losses occur due to external wiring, which are represented by a series resistance R_s, as shown in Figure 2.5. The value of R_s typically falls within a few milliohms. Additionally, the internal resistance of the crystal is modeled by a parallel resistance R_p, usually in the range of a few kilohms. These resistances impact the performance of the cell, as described in Equation 2.2.

Hence, an equivalent one-diode circuit model, as shown in Figure 2.5 will be used in this study to represent the electrical behavior of a PV cell. This model includes current generated for solar cell I_C, a diode current i_D, a shunt resistance R_p that accounts for leakage current, and a series resistance R_s, which represents both the internal resistance of the cell and the resistance between cells.

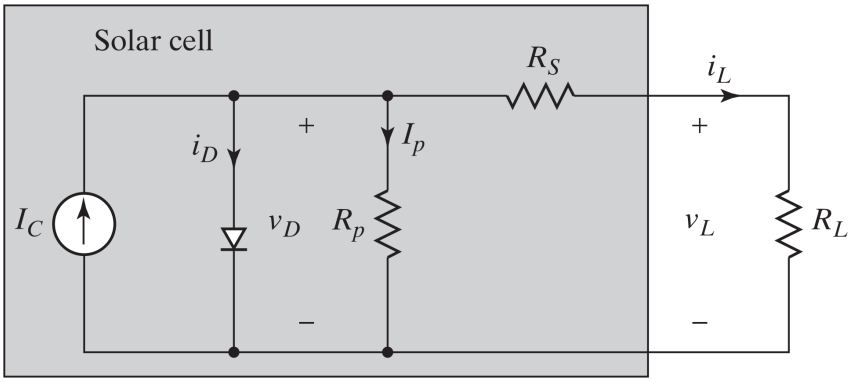


Figure 2.5. Equivalent circuit of a practical solar cell [13].

The load voltage V_L can be expressed in terms of the diode current as follows.

$$V_L = v_D - R_s i_{solar} = v_D - R_S \left[I_C - I_S \left(e^{\frac{v_D}{\eta V_T} - 1} \right) - \frac{v_D}{R_p} \right] \qquad \textit{Eq 2.2}$$

A single PV cell is typically very small, usually producing only 1 to 2 watts of power [13]. To increase the power output, multiple PV cells are connected in series or parallel to form a larger

unit known as a module. When i cells are connected in series, the voltage of the module becomes $V_{module} = i \cdot V_{cell}$. Modules can then be interconnected to form an even larger unit called a PV array, which is used to generate higher power. When modules are connected in series, the total voltage increases, and the array voltage is given by $V_{array} = j \cdot V$, where j is the number of modules connected in series. On the other hand, when modules are connected in parallel, the total current increases, and the array current is given by $I_{array} = k \cdot I$, where k is the number of modules connected in parallel. The configuration for the PV array are normally constructed based on the electric power demand.

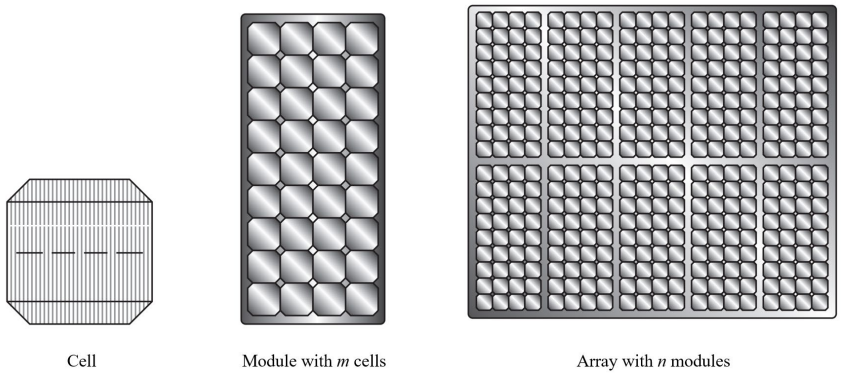


Figure 2.6. PV cell, module and array [13].

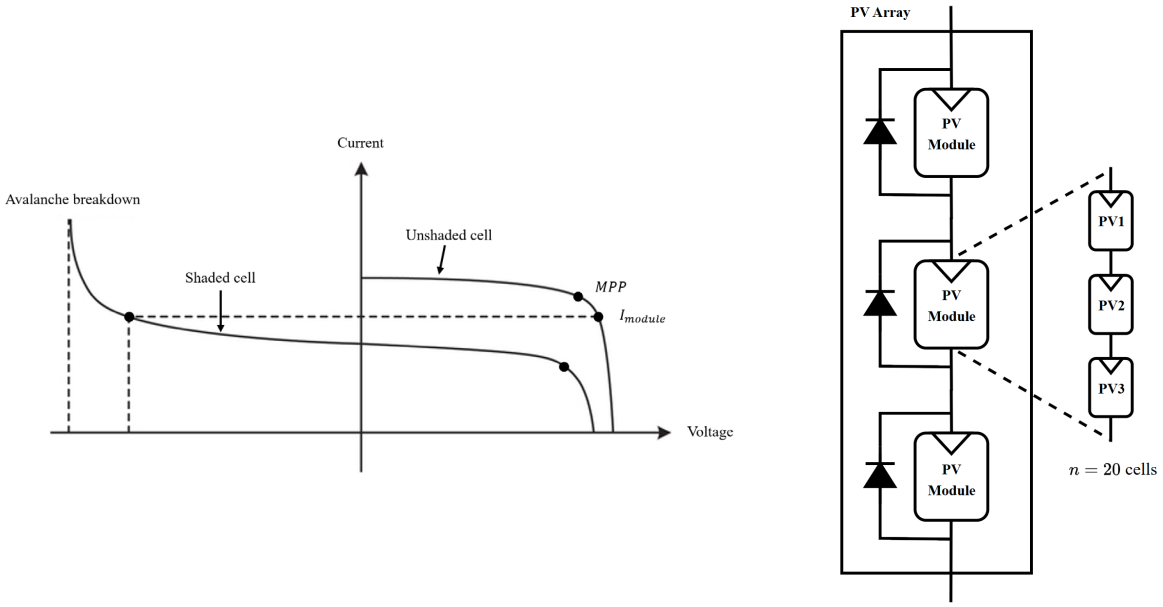
Furthermore, the P-V curve of the solar array will have similar shape compared to the one shown in Figure 2.3(b). Hence, to obtain the maximum output power, the load resistance R_L is dynamically adjusted to ensure the slope of the P–V curve is zero (indicating maximum point) [14]. This process is commonly referred to as the MPPT algorithm. In this project, the MPPT will be implemented using a DC–DC converter, specifically the LLC resonant converter, to perform the required adjustments and achieve optimal power extraction. A detailed explanation of various MPPT algorithms will be provided in Section 2.4.

2.2.2 Partial Shading Condition

Partial shading condition (PSC) in PV systems can be grouped into two types. One is static shading, which occurs when a shadow remains on the PV surface for a long time [8], [9]. Examples include dust, or dirt. The other is dynamic shading, which happens due to temporary obstructions such as moving clouds, tree branches, nearby buildings, or smoke. The current generated by a PV cell depends on the amount of light it receives and the active area of the cell [9]. When a portion of a PV module is shaded, it produces less current compared to cells that are fully exposed to sunlight, as illustrated in Figure 2.7(a). This current mismatch causes the shaded cells to become reverse biased, effectively turning them into electrical loads [9]. As a result, the overall power output of the module is reduced. If this condition persists without proper protection, it can lead to hot-spot heating, which may cause irreversible damage to the PV system [9].

To prevent damage caused by shading, bypass diodes are connected across specific groups of PV cells, as illustrated in Figure 2.7(b) [8], [9]. These diodes provide an alternate path for the current to flow around the shaded cells, allowing the system to continue operating. Although this results in a reduction in power output, the PV system can still function safely, but at a lower voltage. During PSC, the P–V curve of the system may exhibit multiple peaks rather than a single smooth

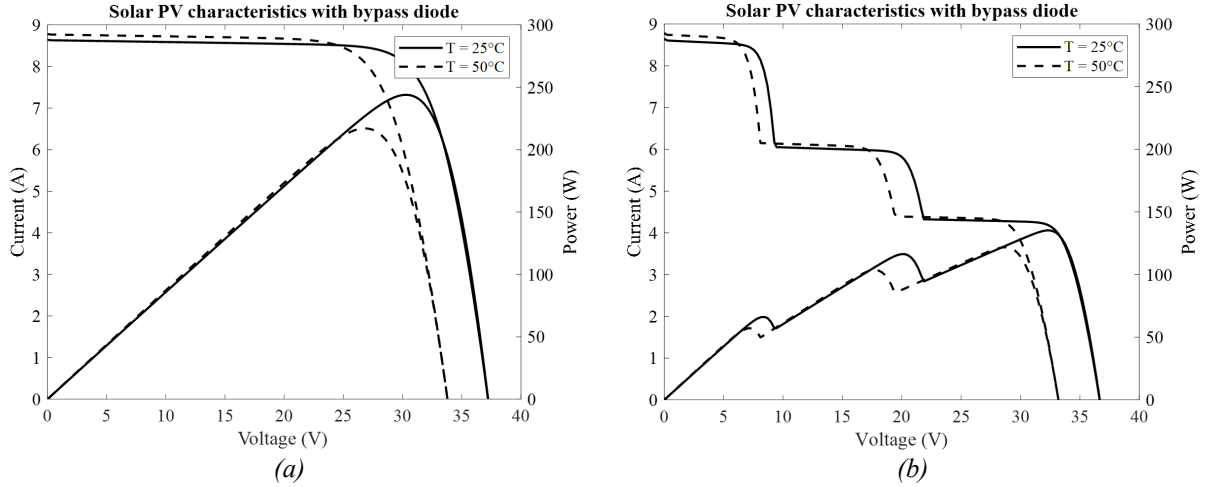
maximum. This phenomenon is a result of the bypass diodes switching on and off to protect shaded cells, which adds complexity to the system's electrical behavior.



(a) *(b)*

Figure 2.7. (a) Impact of partial shading on the I–V characteristics of a PV module,. (b) Schematic of a PV array with bypass diodes connected across individual PV modules.

When modules in a series-connected PV string are shaded, bypass diodes conduct to redirect current around the affected modules [9]. This causes varying power output among modules depending on shading levels. For example, if three modules receive different irradiance, the overall power becomes uneven, and multiple peaks appear on the P–V curve. These additional peaks make tracking the true maximum power point more complex. Although bypass diodes are essential for protection, they add complexity to the system's electrical behavior. Hence, advanced MPPT algorithms are required to accurately track the maximum point under PSC.



(a) *(b)*

Figure 2.8. I-V and P-V curve of PV module configured with 3S1P at 25°C (solid line) and 50°C (dotted line). (a) Single peak with irradiance condition of 1000 W/m². (b) Multiple peak with irradiance condition of 1000 W/m², 700 W/m², 500 W/m².

2.3 LLC Resonance Converter as Microgrid Converter

2.3.1 Microgrid Converter

The two main types of PV systems are standalone systems and grid-connected systems [14]. A standalone PV system typically consists of a PV array, a DC–DC converter (commonly referred to as a charge controller), DC loads, a battery bank, an inverter, and AC loads [14]. The DC–DC converter regulates power flow from the PV array to the battery and load, ensuring efficient energy transfer and battery protection. In a grid-connected PV system, the inverter converts the regulated DC output from the DC–DC converter into an AC waveform, that is fed into the grid [14]. For this project, a standalone PV system is used and will be modelled as illustrated in Figure 2.9.

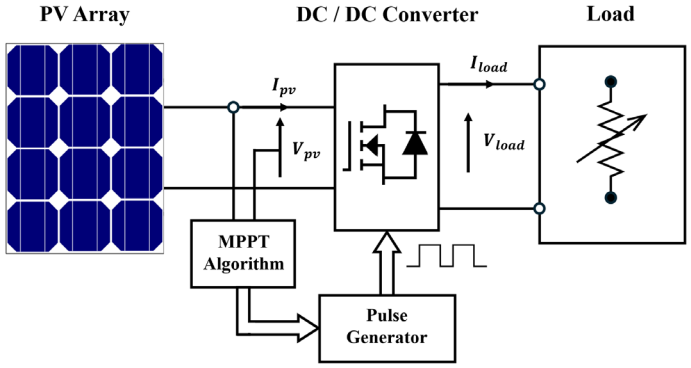


Figure 2.9. Block diagram of a PV system connected to a load through a DC/DC converter. The MPPT algorithm determines the optimal operating point and sends control signals to the pulse generator, which modulates the switching of the converter.

The following section discusses the background of a charge controller using the LLC resonant converter, which is designed to harvest the maximum available power from the PV array through a two-stage global maximum power point tracking method. The LLC converter is employed as the microgrid converter within the proposed system.

2.3.2 Topology of LLC Resonant Converter

The LLC resonant converter is one of the most widely used isolated DC–DC converters and has been extensively adopted in various applications, including PV systems [15], [16], [17], [18] and high-efficiency power supplies [19]. Figure 2.10 illustrates the topology of a LLC resonant converter with a full bridge switching network and a voltage doubler rectifier circuit.

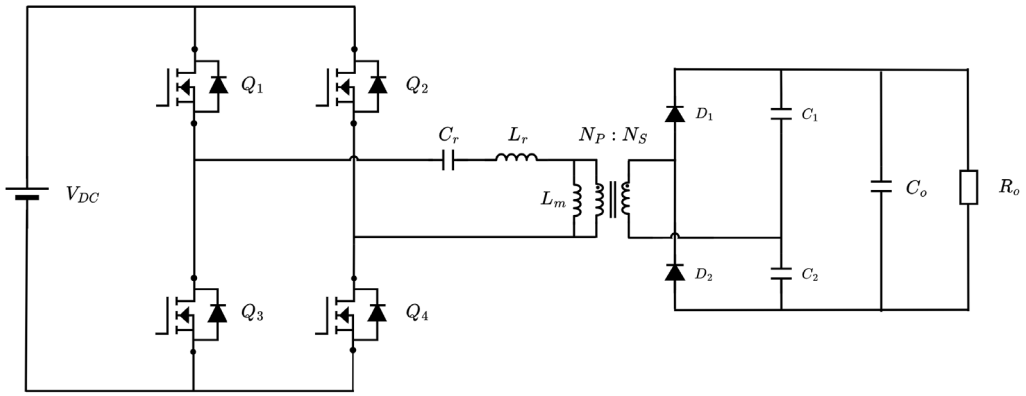


Figure 2.10. Topology of LLC converter with full-bridges switching network and full bridge rectifier circuit.

A typical LLC resonant converter is made up of three main stages which are the switching network, the resonant tank, and the output rectification stage [20]. The switching network uses semiconductor devices to generate the input signal required by the converter. In general, there are two types of most commonly used switching networks, namely half-bridge and full-bridge. When the half-bridge switching network is applied to an LLC converter, it offers both advantages and disadvantages. One advantage is that the RMS voltage applied to the resonant tank is lower compared to the full-bridge configuration, which allows for a lower voltage rating of the resonant capacitor [20]. This configuration is more cost-effective. On the downside, the transformer turn ratio needs to be doubled in order to achieve the same output voltage. Hence, the half-bridge structure is commonly used in low to medium power applications (below 1kW) [20].

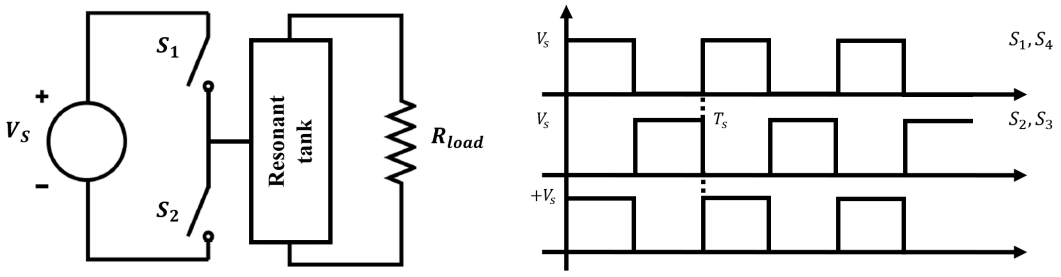


Figure 2.11. Circuit diagram of a half-bridge switching network (left) and corresponding switching waveforms (right). The top waveform represents the switching signal of switch 1, the middle waveform corresponds to switch 2, and the bottom waveform shows the voltage signal appearing across the resonant tank.

The full-bridge switching structure operates by combining two half-bridge stages that function in opposite phases [21]. This configuration allows power to be drawn from the input source during both switching cycles, which is especially beneficial in high-power or high-density converter applications [21]. Additionally, the effective input voltage seen by the transformer is doubled, reducing the required turn ratio by half [20]. This results in material savings such as less copper usage in the transformer windings, reduced physical size, and lower power losses. However, one drawback of the full-bridge topology is that it requires more switching components, which increases both cost and control complexity.

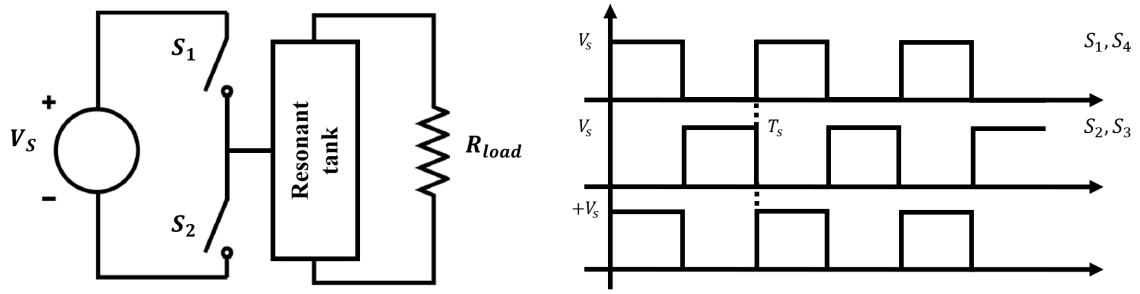
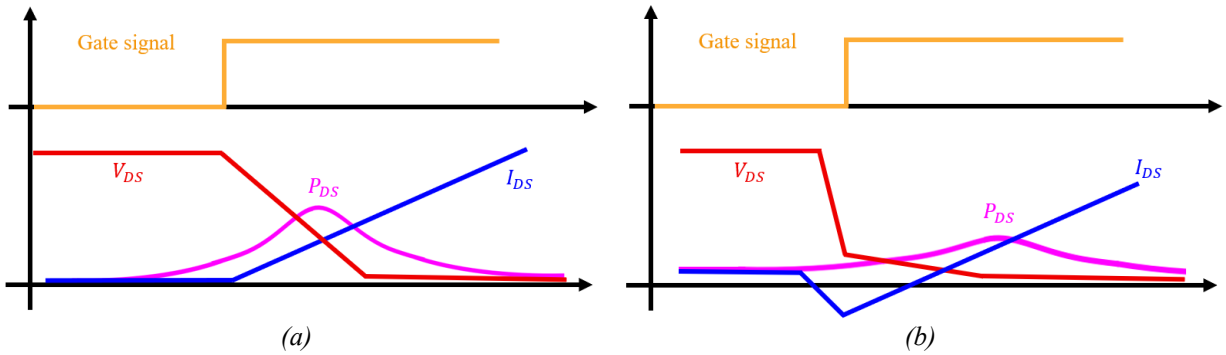


Figure 2.12. Circuit diagram of a full-bridge switching network (left) and corresponding switching waveforms (right). The top waveform represents the switching signal of switch 1 and 4, the middle waveform corresponds to switch 2 and 3, and the bottom waveform shows the voltage signal appearing across the resonant tank.

In a typical power switch, losses can be categorized into commutation losses and conduction losses [13], [21]. Commutation losses occur during switching transitions between the ON and OFF states. These are influenced by factors such as the reverse recovery of diodes and parasitic capacitances, including input capacitance C_{iss} and gate-source capacitance C_{gs} [13]. These losses are associated

with the dynamic behavior of the device. Conduction losses, on the other hand, occur when the switch is in the ON state and are determined by the series resistance $R_{DS(on)}$ of the power switch.

One of the main advantages of the LLC resonant converter is its ability to operate under ZVS, which is a form of soft switching [7] as shown in Figure 2.13(b). Unlike hard switching, where voltage and current overlap during transitions and cause significant switching losses, ZVS allows the switching events to occur when voltage is zero [7], [21]. This minimizes commutation losses and contributes to higher overall efficiency.



(a) *(b)*

Figure 2.13. Comparison between hard switching (a) and soft switching (b). In hard switching, voltage and current overlap during the transition, leading to high switching losses. In soft switching, transitions occur when voltage or current is near zero, minimizing power loss across the device.

In practical applications, the switches in a bridge circuit cannot change their states simultaneously. For example, if both switches in a half-bridge or all four switches in a full-bridge conduct at the same time, it creates a condition known as shoot-through, which is a destructive mode of operation [21]. To prevent this, a simple and effective solution is to introduce a delay between turning off one switch and turning on the other as shown in Figure 2.14. This delay allows enough time for the first switch to fully turn off before the next one is activated. This intentional delay is known as the dead time [21]. It is applied by delaying the rising edge of all gate signals by an appropriate amount of time to ensure safe operation and avoid short-circuit conditions within the bridge.

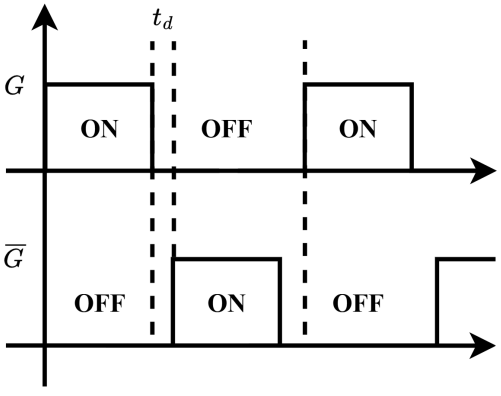


Figure 2.14. Gate signals with 50% duty cycle and dead time t_d inserted between complementary signals G and $\bar{G}$.

Moreover, the resonant tank contains three reactive components. These include a series capacitor C_r, a series inductor L_r, and a parallel magnetizing inductor L_m [20], [22]. Both the series and magnetizing inductors are typically integrated within the transformer.

Then, the AC output from the transformer is converted into a DC signal through the rectification stage. In the circuit shown in Figure 2.10, a voltage doubler rectifier is used. This type of rectifier effectively increases the output voltage, allowing the transformer to operate with a lower turns ratio [23]. A voltage doubler works by using a specific arrangement of diodes and capacitors to accumulate charge over successive half-cycles of the AC waveform [23]. Diodes D_1 and D_2 conduct alternately during each half-cycle of the transformer's output, charging capacitors C_1 and C_2, respectively. Since each capacitor charges to the peak of the transformer's half-cycle voltage, the total output across C_1 and C_2 adds up, effectively doubling the peak secondary voltage. This arrangement provides a higher output voltage without requiring a higher turns ratio or additional switching stages. Additionally, a DC link capacitor is connected at the output to further smooth the voltage. Voltage doublers are particularly useful in systems where increasing the transformer size is not practical, yet a higher DC output is necessary, such as in photovoltaic applications.

2.3.3 Pulse Frequency Modulation Technique

To determine the effective DC gain of the resonant tank of the LLC converter, methods based on the first harmonic approximation (FHA) are commonly used in literature. FHA approach assumes that power transfer from the source to the load occurs primarily through the fundamental harmonic component in the fourier expansion of voltages and currents in the resonant tank . Higher-order harmonics of the switching frequency are neglected, and the tank waveforms are approximated as purely sinusoidal at the fundamental switching frequency.

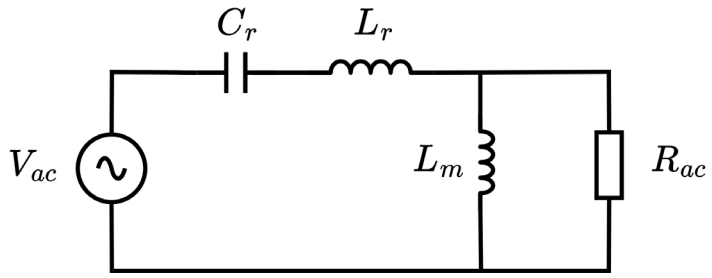


Figure 2.15. Equilvalent circuit of resonant tank using FHA method.

These elements create two distinct resonant frequencies. The first is series resonant frequency f_r which is mainly determined by the combination of L_r and C_r and the second is parallel resonant frequency f_{rp} [20], [22], [24].

$$f_r = \frac{1}{2\pi\sqrt{L_r C_r}} \qquad \textit{Eq 2.3}$$

$$f_{rp} = \frac{1}{2\pi\sqrt{(L_r + L_m)C_r}} \qquad \textit{Eq 2.4}$$

The behavior of the resonant inductor current I_{Lr} resonant converter depends on the operating mode. Figure 2.16(a) shows that when the resonant tank operating at resonant frequency, the resonant inductor current I_{Lr} completes a full half-sinusoidal cycle during each switching period, and it naturally aligns in phase with the magnetizing current I_{Lm} ensuring optimal energy transfer [20]. Figure 2.16(c) shows that when $f_{sw} > f_r$ (above resonance), switching occurs before I_{Lr}

completes its natural cycle, causing truncation, incomplete energy transfer, and increased switching losses due to hard commutation [20]. Figure 2.16(b) shows that when $f_{sw} < f_r$ (below resonance), I_{Lr} finishes early and enters a freewheeling mode where circulating current flows, leading to higher conduction losses [20]. A key reason I_{Lr} does not synchronize with the V_{GS} is because I_{Lr} follows the natural dynamics of the series resonant tank (determined by L_r and C_r), while V_{GS} is governed by the external switching control logic. Adjusting f_{sw} relative to f_r allows control of the voltage gain (step-up or step-down), but it inherently distorts I_{Lr}'s natural sinusoidal period.

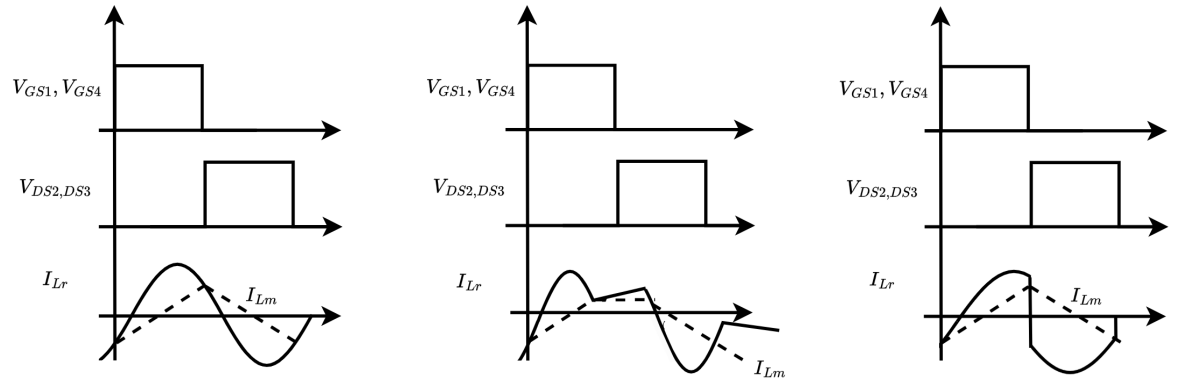


Figure 2.16. LLC converter mode of operation with gate-source voltage V_{GS} of switches 1 and 4 (Top), drain-source voltage of switches 2 and 3 (Middle), and resonant tank current I_{Lr} along with magnetizing current (Bottom). (a) f_{sw} at resonance frequency. (b) f_{sw} below resonance frequency. (c) f_{sw} above resonance frequency.

According to [18], the DC gain of the resonant tank can also be represented as shown with the assumption that the the tank waveforms only consist of the fundamental component.

$$G(f_{sw}, k, Q) = \frac{1}{\sqrt{\left(1 + \frac{1}{K} - \frac{f_r^2}{k f_{sw}^2}\right)^2 + Q^2 \left(\frac{f_{sw}}{f_r} - \frac{f_r}{f_{sw}}\right)^2}} \qquad \textit{Eq 2.5}$$

Where

- Inductance ratio $k = \frac{L_m}{L_r}$.
- Quality factor $Q = \frac{\sqrt{L_r/C_r}}{R_{ac}}$.

The DC gain characteristics of the resonant tank for the LLC resonant converter can be broadly classified into two main operating regions, as illustrated in Figure 2.17. When the switching frequency f_r is higher than the series resonant frequency, the voltage gain drops below unity. When the switching frequency f_{sw} equals the series resonant frequency f_r, the converter typically achieves a voltage gain close to unity. In contrast, operating below the resonant frequency places the converter in the boost region, where the voltage gain exceeds one. The peak of the gain curve in an LLC resonant converter represents the frequency at which voltage gain is at its highest for a specific load condition.

To maintain ZVS, the LLC resonant converter must operate at a switching frequency higher than the frequency where peak gain occurs [24]. If the switching frequency drops below this point, the converter enters the capacitive region, where the current leads the voltage. In this condition, it becomes difficult or impossible to maintain ZVS, leading to high switching losses. The peak gain frequency is not constant and shifts depending on the load. Under higher Q factor, it moves to a higher frequency, while under lower Q factor, it shifts to a lower frequency and the gain increases significantly [24]. At no load, the converter tends to operate near the parallel resonance point, where the impedance of the resonant tank is very high and the current is minimal, yet the voltage gain becomes sharply elevated [24]. On the other hand, when the switching frequency is above the resonant frequency, the converter operates in the inductive region, also known as the buck region, where the voltage gain is less than one [20]. In this mode, the current lags behind the voltage, allowing the switch voltage to naturally fall to zero before turning on [20]. This supports soft switching and helps to reduce switching losses. At the resonant frequency itself, the converter achieves unity gain across all load conditions, including short circuit. For stable and efficient operation, the converter should always operate above the peak gain frequency to remain in the inductive region. This ensures reliable ZVS, lower power losses, and consistent performance across varying load conditions.

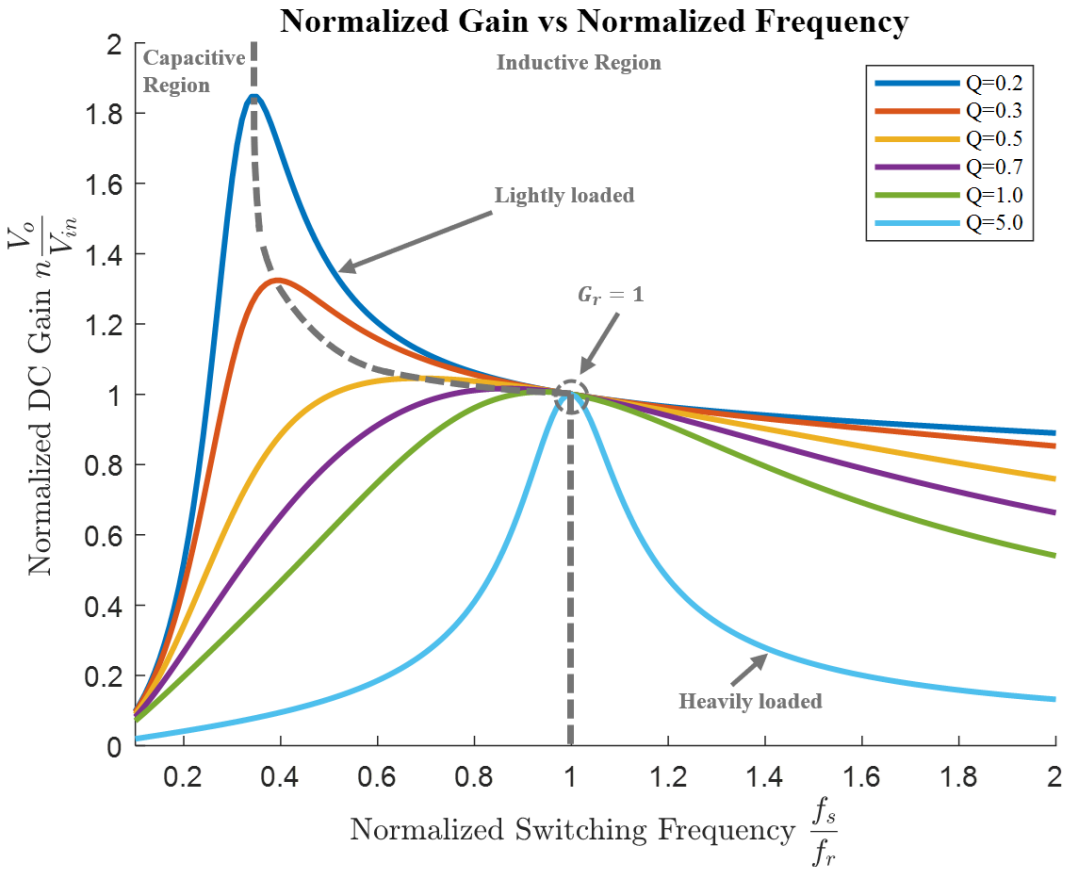


Figure 2.17. DC Gain characteristics LLC resonant converter using PFM method.

However, a limitation arises in the analysis of the resonant tank due to the three key assumptions made in the First Harmonic Approximation [24]. As a result, in applications such as photovoltaic

systems that require a wide voltage gain range, relying solely on FHA may lead to inaccurate estimations.

- The input square-wave voltage and current are approximated by their fundamental (first harmonic) components, while higher-order harmonics are ignored. This simplifies the analysis and aligns with the fact that the resonant tank naturally attenuates high-frequency harmonics near resonance, making the fundamental the dominant contributor to system behavior.
- The effect of the output capacitor is neglected by assuming that the output voltage remains constant over one switching cycle. This allows the system to be treated as a simplified AC model, making it easier to derive gain and impedance characteristics without involving the time-domain ripple on the output.
- By referring all voltages and currents from the secondary side of the transformer to the primary side, the system becomes easier to analyze in a unified domain. This eliminates the need to separately handle the transformer turns ratio during intermediate steps of the derivation.

Furthermore, when applied in a PV system, the output voltage of the LLC converter is typically clamped by the DC grid bus. As the PV curve is nonlinear and the load varies accordingly, the quality factor Q of the converter changes (when performing MPPT). This variation also affects the range of the capacitive region that the converter can reach before fully entering it. Once the converter enters the capacitive region, its behavior becomes less predictable, which complicates MPPT control. To avoid these issues, it is important to ensure that the converter operates within the inductive region. In this region, there is a consistent positive correlation between the switching frequency and the input voltage, which is essential for effective MPPT control. Although the LLC converter offers simple frequency-based control, it requires a more complex design for both the magnetic components and the gate driver circuit [25]. In addition, soft switching may not be sustained across a wide operating range [20]. This is because the switching frequency needs to vary widely to match the changing impedance of the PV module in order to extract maximum power. Despite these challenges, the method is still used in [16], [17], [18] due to its ease of control.

To address these limitations, *Wei et al.* [25] has suggested two strategies using the secondary-side equivalent impedance modulation techniques. These methods can increase the voltage gain while keeping the resonant tank operating at its resonant frequency, thus maintaining the benefits of the LLC topology. The first is known as the secondary-side PWM (SS-PWM) modulation strategy [26], and the second is the secondary-side phase shift modulation (SS-PSM) strategy [27]. Both methods allow the converter to operate at a fixed switching frequency while achieving ZVS for the primary switches and ZCS for the secondary rectifiers.

2.3.4 Secondary-Side Pulse Width Modulation Technique

The topology proposed by *Wang et al.* [26] using the SS-PWM LLC converter is shown in Figure 2.18, where an auxiliary switch is added on the secondary side. In SS-PWM operation, the primary-side switches operate at a fixed switching frequency (equal to the series resonant frequency) with a constant duty cycle of 50%. This method will ensure that ZVS at the primary side of the converter which follows the analysis of FHA. Then, the output voltage is regulated by adjusting the duty cycle of the secondary-side switching device Q_{mc}. When the duty cycle is set to zero, the converter

operates as a conventional LLC resonant converter. In contrast, when the duty cycle reaches one, the secondary rectifier behaves as a voltage doubler. By adjusting the secondary-side duty cycle, the voltage gain of the SS-PWM LLC converter can be continuously controlled within a range from one to two. According to [26], there is no closed-form expression available for the voltage gain in this configuration. It is also noteworthy that ZVS for the secondary switching device can still be maintained. Furthermore, the overall voltage gain remains independent of the Q factor [25].

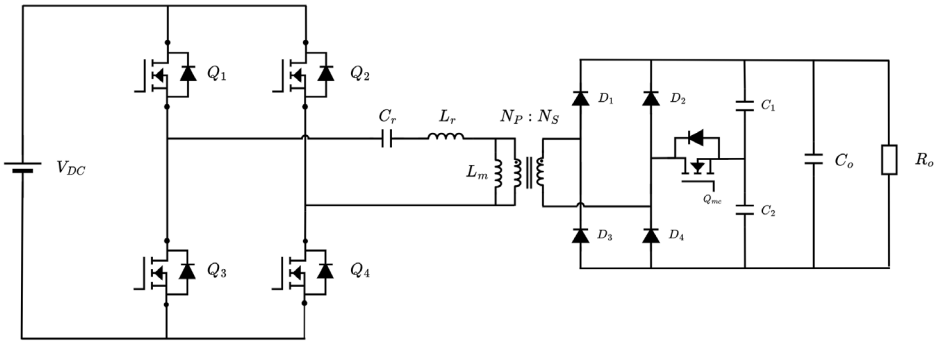


Figure 2.18. Topology of SS-PWM method for LLC resonant converter.

This method is good but the system voltage gain can only be in the range of 1 to 2 which is not suitable for PV system [25].

2.3.5 Secondary-Side Phase Shift Modulation Technique

The structure of the SS-PSM LLC proposed by *Wu et al.* [27] is shown in Figure 2.19, where the two bottom secondary rectifier diodes are replaced with switching devices with the complementary driving signals. The SS-PSC LLC resonant converter operates at a fixed series-resonant frequency f_r to maximize efficiency during normal operation [25], [27]. It employs a hybrid rectifier on the secondary side, consisting of two diodes and two synchronous switches (Q_{o1}, Q_{o2}) [27]. Compared to traditional frequency modulation techniques such as PFM, this modulation strategy reduces circulating currents and further decreases conduction losses during steady-state operation [27]. As a result, the topology enables efficient fixed-frequency operation while achieving dynamic voltage regulation through secondary-side phase-shift control.

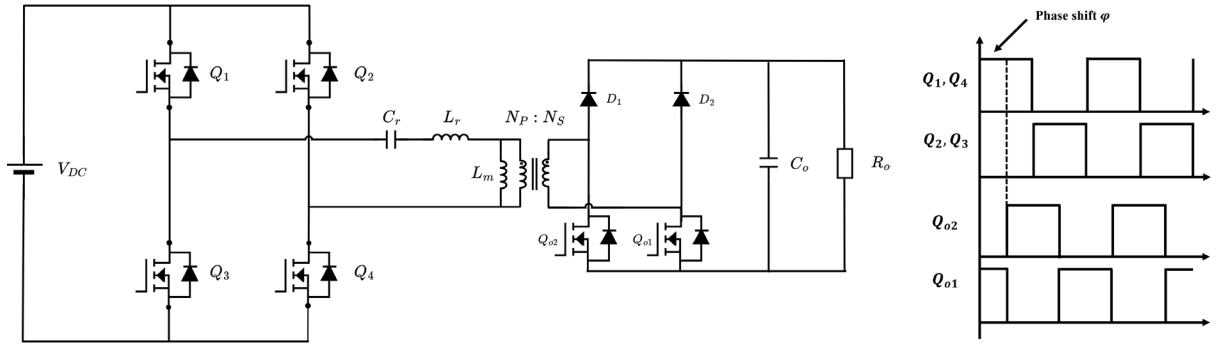


Figure 2.19. Topology of SS-PWM method for LLC resonant converter (Left) and the corresponding switching signal (Right).

The phase shift between the primary and secondary switches plays a critical role in adjusting the converter's effective voltage gain. This mechanism allows the converter to operate similarly to a

conventional LLC resonant converter while maintaining ZVS for all switches, thereby minimizing both switching and conduction losses [27]. For PV applications requiring variable voltage gain, a controlled phase shift is introduced between the secondary-side and primary-side switches. This phase shift alters the energy transfer timing by extending the charging period of the resonant inductor, effectively boosting the output voltage without deviating from the resonant frequency [25].

According to *Wu et al.* [27], the voltage gain of the SS-PSM LLC converter can be expressed as shown in Equation 2.6. From this equation, it is evident that when the phase-shift angle $\varphi = 0°$, the voltage gain equals unity, and the converter operates similarly to a conventional full-bridge LLC resonant converter. As the phase-shift angle φ increases within the range $[0, \pi]$, the term $\cos(\varphi)$ decreases, resulting in a voltage gain that increases proportionally with φ. Therefore, the voltage gain exceeds 1 when $\varphi > 0$. Similar to SS-PWM, the voltage gain of the SS-PSM LLC is independent of the quality factor, which simplifies controller design [25]. Therefore, this modulation technique is particularly beneficial in PV systems, where the input voltage from solar panels can vary widely. SS-PSM ensures stable power delivery to the grid or battery storage while preserving ZVS for minimal switching losses. However, SS-PSM requires two additional gate drivers to implement the secondary-side modulation.

$$G_{f_{sw}\approx f_r}(\phi, Q) = \frac{\sqrt{\pi Q} + \sqrt{-2(\cos\phi)^2 + \pi Q + 2}}{\sqrt{\pi Q}(1 + \cos\phi)} \qquad \textit{Eq 2.6}$$

In PV applications, PFM is wisely used for its wide voltage gain range but faces challenges with maintaining ZVS and increased conduction losses. SS-PWM offers simpler fixed-frequency control but is limited by its narrow gain range, making it less suitable for PV systems requiring flexibility. SS-PSM provides efficient fixed-frequency operation with a wider gain range, making it more adaptable for PV use, though it introduces additional complexity and hardware requirements. Therefore, this project will explore both PFM and SS PSM modulation techniques for deploying the MPPT algorithm.

2.4 Maximum Power Point Tracking Algorithm

2.4.1 Operating Principle of MPPT in PV Converters

As explained in Section 2.2.1, maximizing power output from a PV system requires adjusting the load resistance R_L so that the slope of the P–V curve becomes zero, indicating operation at the MPP. In typical PV applications, the output voltage of the DC-DC converter is held constant or clamped to a DC bus or grid [16]. Since the load voltage is fixed, direct adjustment of R_L is not possible. Instead, the converter modulates its operating parameters to vary the voltage gain, which indirectly changes the load resistance as seen by the PV panel [16].

The relationship between the PV side and the load side is governed by the converter's voltage gain [16]. By altering the modulation signal using duty cycle control in conventional converters or frequency adjustment in resonant converters, the converter changes the ratio of input to output voltage and current. This process effectively transforms a fixed physical load into a variable electrical load from the PV array's perspective. Increasing the voltage gain reduces the PV input voltage, shifting the operating point towards the lower voltage region of the P–V curve. Decreasing

the gain raises the PV voltage, moving towards the higher voltage region. This dynamic adjustment allows the converter to track the MPP by continuously presenting the optimal load condition to the PV array.

2.4.2 Conventional Methods MPPT Method

The perturb and observe (P&O) method elaborated by *Fermia et al.* in 2005 [5], also known as the hill-climbing method, is one of the most widely adopted MPPT techniques in PV systems due to its straightforward operation and ease of implementation. This algorithm operates by introducing small, periodic perturbations to the control parameter. In pulse width modulation (PWM) controlled DC-DC converters, such as buck, boost, buck-boost, Ćuk, and SEPIC converters, the perturbation is applied to the duty cycle. In contrast, for LLC resonant converters using pulse frequency modulation (PFM), the perturbation is applied by varying the switching frequency [18]. For any modulation technique, the objective of the P&O algorithm remains consistent. If the output power increases after a perturbation, the algorithm assumes that the operating point is approaching the MPP and continues to adjust the control parameter in the same direction [5]. Conversely, if the power decreases, it indicates that the MPP has been surpassed, prompting the algorithm to reverse the direction of the perturbation in the next cycle [5]. Through this iterative process, the P&O method progressively tracks the P-V curve, guiding the system towards the peak power point.

In frequency-based implementations of the P&O method, such as that proposed by [18], the control parameter is the switching frequency rather than the duty cycle. This approach is used for LLC resonant converter, where output regulation is achieved through frequency modulation. During each cycle, the PV voltage $V_{PV}(n)$ and current $I_{PV}(n)$ are measured to calculate the instantaneous power $P_{PV}(n)$. These values are then compared with the power and voltage from the previous cycle $P(n-1)$ and $V_{PV}(n-1)$ to determine whether the switching frequency perturbation Δf_{sw} should be increased or decreased. Figure 2.20(a) illustrates the flowchart of the frequency-based P&O method, along with a graphical representation of the hill-climbing process. A key characteristic of frequency-based P&O is that increasing the switching frequency reduces the effective voltage gain of the resonant tank, allowing the converter to operate at higher PV voltages. Conversely, decreasing the switching frequency increases the voltage gain, enabling the converter to access the lower voltage range of the PV panel.

Despite its simplicity and widespread use, the P&O method suffers from a trade-off between tracking speed and stability, as its performance depends on a fixed perturbation step size [28]. A smaller step improves accuracy but slows response, while a larger step speeds up tracking but increases oscillations and energy losses [28]. To address these limitations, variable step-size P&O methods have been proposed, offering faster tracking with reduced oscillations by dynamically adjusting the step size based on operating conditions [28], [29].

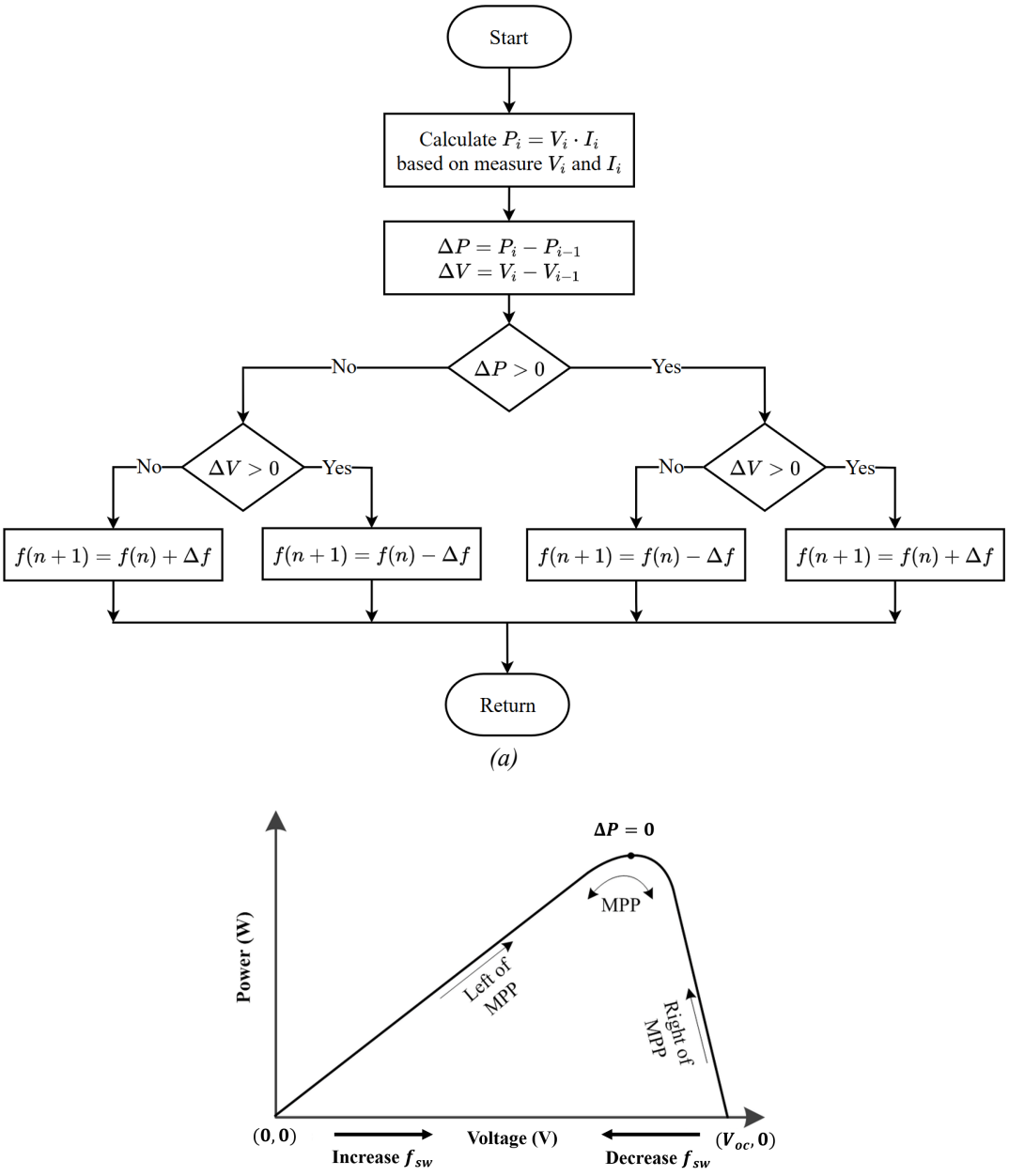


Figure 2.20. (a) Flowchart of the P&O algorithm used for MPPT in PV systems using PFM. (b) P-V curve illustrating how ΔP guides the algorithm to increase or decrease the converter switching frequency.

The incremental conductance (INC) [6] method is another widely adopted conventional MPPT technique with additional computational resources needed when comparing to the P&O algorithm. It operates on the principle that the slope of the power-voltage (P–V) curve is zero at the MPP, positive when operating to the left of the MPP, and negative when to the right [6]. Unlike the P&O method, which adjusts the operating point and observes power changes, the INC algorithm compares the instantaneous conductance I/V with the incremental conductance $\Delta I/\Delta V$ to accurately determine the location of the MPP [6]. The INC method identifies the MPP when $\Delta I/\Delta V = -\Delta I/\Delta V$ [6]. Once this condition is met, the algorithm halts further perturbation,

resulting in more stable operation compared to the continuous oscillations typical of the P&O method [6].

For LLC resonant converters, *Zhuang et at.* [16] proposed a modification of the Incremental Conductance (INC) algorithm by replacing the conventional voltage reference with a frequency reference, enabling MPPT control through frequency adjustment. The flowchart below illustrates this frequency-based INC algorithm, where the switching frequency is dynamically tuned to track the MPP. The algorithm calculates changes in voltage and current, then compares the incremental conductance $\Delta I/\Delta V$ with the negative instantaneous conductance $-\Delta I/\Delta V$ to determine whether the operating point lies on the positive or negative slope of the P–V curve. If the operating point is on the positive slope, the switching frequency is increased to shift towards the higher voltage region [16]. Conversely, if it is on the negative slope, the frequency is decreased to move towards the lower voltage region by increasing the converter's gain, thereby ensuring optimal power extraction [16].

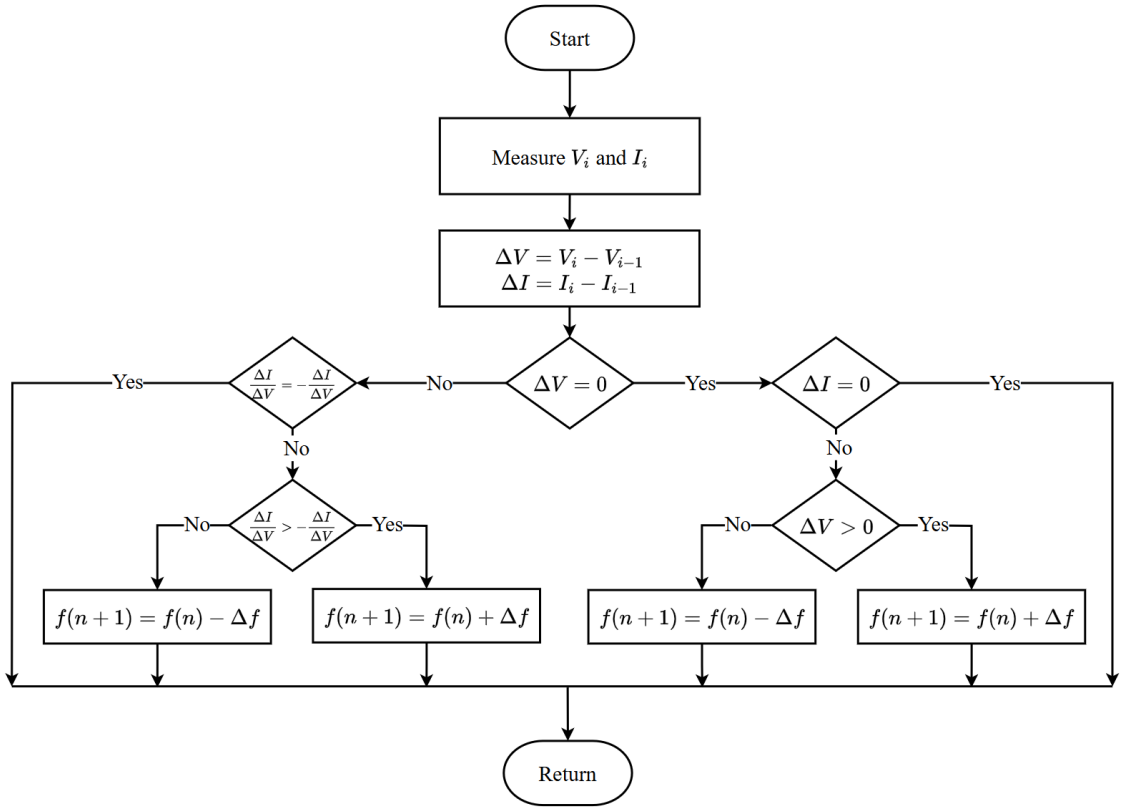


Figure 2.21. (a) Flowchart of the INC algorithm used for MPPT in PV systems using PFM.

The INC method will have slower tracking speed but with lower steady state oscillations compared to the commonly used P&O algorithm [30]. However, INC requires continuous monitoring and comparison of conductance values, which increases computational complexity and demands more from the hardware [30].

To overcome these challenges, researchers have proposed several improvements to traditional MPPT methods. For example *Kollimalla at al.* [31] proposed an adaptive step size P&O techniques to improve both tracking speed and accuracy, especially during sudden changes in irradiance. In another study, an adaptive sliding mode controller was proposed by *Ali et al.* [32] to eliminate the

chattering effect found in classical sliding mode control. This method combines sliding mode control with the P&O algorithm to offer better stability and performance. Another promising approach proposed by *Rezk et al.* [33] uses adaptive fuzzy logic control to improve maximum power point tracking in photovoltaic systems. This method typically uses two inputs, the error and the change in error which are calculated from the differences in power and voltage between two consecutive samples [33]. These values help the fuzzy controller decide the best direction and step size to adjust, improving performance under changing conditions [33]. Although fuzzy logic based MPPT methods are appreciated for their simplicity, speed, and accuracy, they still face limitations under rapidly changing light, especially during partial shading. In such cases, the algorithm may lock onto a local maximum instead of the true global maximum, which reduces overall energy harvesting efficiency.

The center point iteration (CPI) algorithm proposed by *Zhang et al.* [17] is a frequency based MPPT algorithm developed for LLC resonant converters. Unlike the Perturb and Observe method, CPI adjusts the switching frequency using variable steps, helping it reach the maximum power point more quickly and with fewer steady state oscillations. The method divides the power versus frequency curve into four regions using five key frequency points [17]. Power is measured at three locations within the range near the start, the center, and near the end [17]. If the center frequency produces the highest power, the algorithm narrows the search range around that point and updates the boundaries. This process continues by recalculating new midpoints until the difference in power between the tested points is small enough to indicate that the maximum power point has been found [17]. While CPI offers faster and more stable tracking than traditional methods, it has some limitations. Its accuracy drops under rapidly changing sunlight because early frequency decisions may no longer reflect the true maximum [18]. In addition, CPI has limited ability to recover if it initially moves away from the optimal path, and like many basic tracking methods, it often fails to detect the global maximum under partial shading conditions.

On the other hand, *Tang et al.* [18] proposed a hybrid MPPT algorithm for LLC resonant converters, combining frequency-modulated impedance matching (FMIM) with an enhanced P&O method to address limitations of conventional frequency-based MPPT techniques such as P&O and CPI. The FMIM stage estimates an optimal initial operating point using V_{oc} , I_{sc}, and converter parameters, while the refined P&O stage minimizes oscillations through smaller perturbations. This method shows improved tracking speed compared to CPI under changing irradiance, making it well suited for dynamic photovoltaic applications without requiring extra hardware. Although these methods are generally robust in locating the maximum power point, they fall under partial shading conditions because they are designed to find only a single peak and cannot detect multiple local maxima. This highlights the need for intelligence MPPT method in the initial stage of the algorithm which will be explain in next section.

Furthermore, one GMPP tracking method is the Full Scan (FS) method, where the entire P-V curve is scanned and the voltage at which the highest power is measured is given as a reference to the P&O. This simple brute force method works well, but there is a tradeoff between its tracking speed and accuracy, as both are dependent on the step size. Based on the P-V curves, *Ahmed et al.* [34] suggested an Enhanced Adaptive P&O (EA-P&O) with a skipping feature to skip through the measurements of voltage regions where the GMPP cannot exist, effectively improving upon the FS method.

2.4.3 Intelligent MPPT Methods

In addition to conventional techniques, various intelligent control methods have been developed. ant colony optimization (ACO) is a bio-inspired algorithm that has gained attention in MPPT research for its ability to perform global optimization in complex and dynamic environments. First introduced by *Dorigo* and *Gambardella* (1997), ACO mimics the natural foraging behavior of ants, where pheromone trails are used to guide others toward the most efficient path to a food source [35]. This adaptive search mechanism makes ACO suitable for handling the non-linear and multi-peak characteristics of PV systems, especially under partial shading conditions (PSC) [36]. Several studies have explored the application of ACO in MPPT. For instance, ACO has been successfully implemented in modular PV systems, demonstrating reliable tracking of the global MPP using only a single sensor, which reduces system complexity and cost [37]. Furthermore, ACO has been integrated with Fuzzy Logic Controllers (FLC) to optimize their parameters, resulting in improved steady-state accuracy and faster dynamic response compared to traditional FLC-based MPPT methods [38]. Despite these advantages, researchers have noted that ACO-based approaches can lead to higher computational complexity, particularly when combined with other intelligent control strategies, which may limit their suitability for real-time embedded applications [36].

Particle swarm optimization (PSO) is a popular optimization technique introduced by *Kennedy* and *Eberhart* in 1995 [39], inspired by the way birds flock together to find food. In MPPT applications, PSO is used to find the MPP by simulating a group of particles that "search" for the best solution, where each particle represents a possible operating point of the PV system. The particles move through the search space, adjusting their positions based on both their own experience and that of their neighbors, aiming to maximize the PV output power. PSO has been successfully applied to various PV configurations and has shown good performance under partial shading conditions due to its ability to avoid getting stuck at local maxima [40]. However, because PSO relies on random movements to explore solutions, it can sometimes result in slower or inconsistent tracking speeds [36].

Artificial neural networks (ANN) are among the most widely used soft computing techniques for MPPT due to their ability to model complex, non-linear behaviors similar to biological neural systems [36]. In ANN-based MPPT applications, input variables typically include PV system parameters such as short-circuit current, open-circuit voltage, terminal voltage, and output current, along with environmental factors like irradiance, temperature, and wind speed [36]. These inputs are processed through hidden layers, where the ANN learns the relationship between system conditions and the optimal operating point. The output layer then generates control signals, such as reference voltage or duty cycle, to guide the converter toward the MPP [36]. *Jyothy et al.* [41] an ANN that takes solar irradiance and temperature as inputs and provides the optimal duty ratio to control a DC–DC boost converter, ensuring the PV system operates at its maximum power point. The result were validated using MATLAB SIMULINK and according to [41] the ANN-based approach achieved faster tracking, higher output power, and reduced steady state oscillations, making it more efficient and reliable than traditional methods under changing environmental conditions. Additionally, *Mao et al.* [42] proposes a convolutional neural network (CNN) (one type of variation of neural network) method to quickly predict the MPP voltage in pavement solar panels affected by moving vehicle shadows. It uses irradiance and temperature data, converted into image-like formats, to make predictions without needing current or voltage sensors.

More recently, *Ye et al.* [10] introduced a two-stage ANN-based GMPPT method specifically designed for complex PSC environments. This approach is less computationally demanding when comparing to the one proposed by *Jyothy et al.* [41], as it performs a one-time prediction of the global maximum power point, followed by the use of the conventional P&O method for fine tracking. However, none of these intelligent MPPT methods have been widely applied to LLC resonant converters. To address this, adaptive neuro-fuzzy inference systems (ANFIS) have been proposed by *Ghosh et al.* [15] as an effective hybrid control strategy for MPPT in LLC converters. ANFIS combines the learning capabilities of neural networks with the rule-based decision-making of fuzzy logic, offering both adaptability and precision using pulse frequency modulation (PFM) and phase shift modulation (PSM) techniques.

By combining all the advantages the methods mentioned above, this study proposes a smart Global MPPT algorithm that combines the strengths of a neural network approach from *Ye et al.* [10] as the first stage of the algorithm. For second stage, *Tang et al.* [18] initialization method will be used to initialize the conventional P&O method.

2.5 Deep Learning Background

Deep neural networks (DNNs) have shown exceptional accuracy in complex tasks since the introduction of CNNs by *Lecun et al.* [43] in 1998 and *Krizhevsky et al.* [44] breakthrough with deep CNNs in the 2012 ImageNet competition. Leveraging the DNN's strength in handling non-linearities and dynamic conditions, DNNs are now emerging as a promising approach for intelligent MPPT in photovoltaic systems. This chapter presents the essential technical background on machine learning and neural networks required for the development of the smart GMPPT algorithm. The discussion is guided by concepts from the *Deep Learning* book by *Goodfellow et al.*, *Machine learning with neural networks* book by *Mehlig* [45], thesis by *Karpathy* [46] and with the proposed method by [10] used as a toy example to aid in the explanation.

2.5.1 Supervised Learning

Supervised learning is a type of machine learning where a computer learns to map inputs to corresponding outputs through example data. It involves finding a function $f(x) \rightarrow Y$, where X represents the input and Y is the desired output. For instance, in an MPPT application, X could be the sampled current values from a feedback sensor, while Y represents the voltage and power at the local maximum power points (LMPPs) under partial shading conditions (PSC). A training dataset consisting of known input-output pairs $\{(x_1, y_1), \dots (x_n, y_n)\}$ is provided, typically generated from the PV array, referred to as the data distribution D [46]. The objective is to train the computer to select a function that accurately maps inputs to outputs by minimizing errors on these examples [47]. This is achieved by evaluating different functions and using a loss function to measure the difference between predicted outputs ($\hat{y}$) and actual outputs (y) with the goal is to find the function f that minimizes this loss across the entire data distribution D, ensuring accurate predictions for new, unseen inputs [47].

$$f^* \approx \arg\min \frac{1}{N} \sum_{i=0}^{N} L(\hat{y}, y) \qquad \textit{Eq 2.7}$$

In MPPT application, once the learned f' function is identify, the original data can be discard and only the learned function , f' is keep to map the element, X to Y. A key challenge is that a model

can reduce training loss by simply memorizing the data without truly learning. For example, it might recall the exact y for every x in the training set but fail on any new input. This is called overfitting, where the model performs well on known data but poorly on unseen data [45], [47]. Another concern is that many functions can achieve low training loss, but only some will generalize well. The loss function alone cannot identify the best one. To solve this, regularization is used. It rewards simpler and more consistent solutions, helping the model avoid memorization and improve performance on new data [47]. The regularization equation is given as follows.

$$f' \approx \arg\min \frac{1}{N} \sum_{i=0}^{N} L(\hat{y}, y) + R(f) \qquad \textit{Eq 2.8}$$

Next, a commonly used loss function in regression to evaluate the performance of the learned learned f' function is the squared difference between the targeted value and the predicted value.

$$L(\hat{y}, y) = (\hat{y} - y)^2 \qquad \textit{Eq 2.9}$$

According to [47], the neural network for regression can be expressed as a linear function, where regularization plays a key role in ensuring good model behavior. Regularization is applied to prevent the neural network from overfitting by discouraging excessive weight values, which could cause certain input features to dominate the predictions [47]. This is controlled by the parameter λ, which balances fitting the training data with maintaining a simpler model for better generalization [47]. The objective function combines the error minimization term with a regularization term to penalize overly complex models. In addition to this, dropout proposed by *Srivastava et al.* [48] can be used within the hidden layers by randomly deactivating a fraction of neurons during training. This technique reduces dependency on specific neurons, encourages the network to learn more robust features, and further mitigates the risk of overfitting, enhancing the model's performance on unseen data.

$$f^* = \arg\min \left[\frac{1}{N} \sum_{i=0}^{N} L(\hat{y}, y) + R(f) \right] + [\lambda(w_1^2 + w_2^2)] \qquad \textit{Eq 2.10}$$

Therefore, the optimizer is guided towards a solution that not only fits the training data but is also more likely to perform well on unseen examples.

2.5.2 Artificial Neural Network

Neural networks are inspired by how biological neurons work in the brain [45], [47]. Each neuron in a neural network takes an input x, multiplies it by a weight w, and adds a bias b. The weights decide how strongly each input influences the neuron. Positive weights increase the signal, negative weights reduce it, and a weight of zero means the input has no effect [47]. The neuron adds up these weighted inputs to get a total signal. This signal is then passed through an activation function, which adds flexibility by allowing the neuron to handle more complex patterns instead of just simple calculations. The activation function also helps decide if the neuron should "fire" or

stay inactive, similar to how real neurons behave. Figure 2.22 below shows a simplified mathematical representation of a neuton.

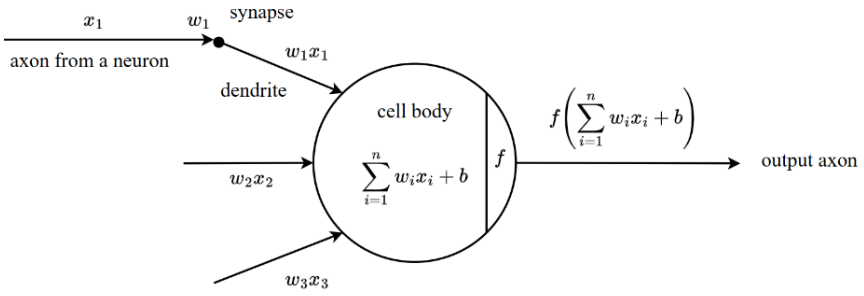


Figure 2.22. Simplified representation of a biological neuron modeled as an artificial neuron.

From Figure 2.23, a neuron receives multiple inputs, each multiplied by a weight that defines the connection strength. These weighted inputs are summed with a bias and passed through an activation function to produce an output signal [45]. This process allows the neuron to decide whether to activate based on the combined inputs. In neural networks, this operation is performed in parallel across layers using a weight matrix and bias, followed by an activation function. Activation functions such as *sigmoid*, *tanh*, and *ReLU* [49] introduce nonlinearity, enabling the network to learn complex patterns [45]. Among these, *sigmoid* and *ReLU* are the most widely used. A typical artificial neural network consists of an input layer, hidden layers, and an output layer. The figure below illustrates a compact representation of a neural network with 4 input neurons, 3 hidden layers where the first hidden layer has 5 neurons and the second and third hidden layers have 6 neurons each, followed by an output layer with 3 neurons. The hidden layers, positioned between the input and output layers, are crucial for enabling the network to learn and represent complex patterns in the data [45].

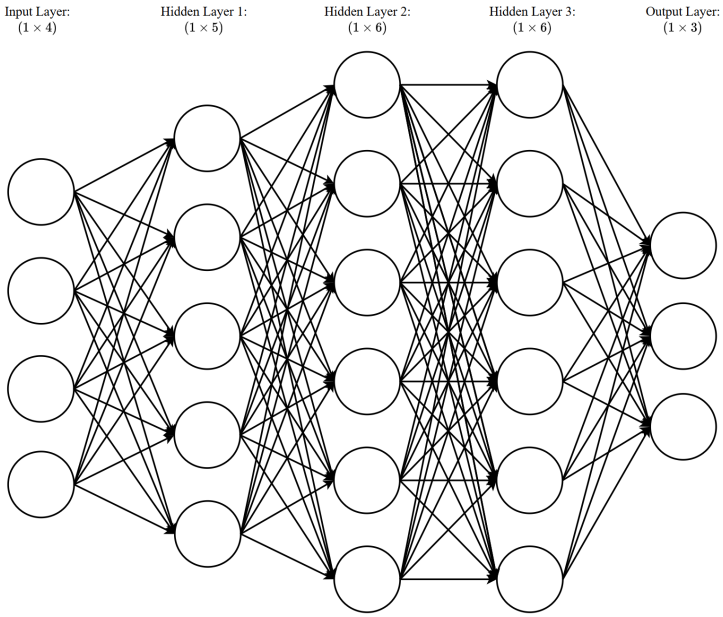


Figure 2.23. Simplified representation of a biological neuron modeled as an artificial neuron.

2.5.3 One-Dimensional Convolutional Neural Network

The commonly used 2D-Convolutional Neural Networks (CNNs) are designed to process data with spatial or sequential patterns, such as images [50], [51], [52], video [53]. On the other hand, the 1D-Convolutional Neural Network (1D-CNN) proposed by *Abdoli et al.* [54] has demonstrate its effectiveness in processing raw waveform data. For MPPT applications, where the data is one-dimensional, such as power-voltage or current-voltage curves, a 1D-CNN can be effectively used. As mentioned in [55], [56], 1D-CNNs are well suited for processing time series and signal-based data, making them ideal for capturing important features from PV signals. In this study, the 1D-CNN is used to identify key patterns in the PV curves, particularly the peaks that represent LMPPs. This is especially useful under PSC, where multiple peaks may appear, compared to a single peak under uniform irradiance. The ability of 1D-CNNs to detect such patterns makes them a powerful tool for improving MPPT accuracy in complex partial shading condition (PSC) environments.

When adapting a 2D-CNN to a 1D-CNN, the input tensor originally shaped as $(W \times H) \times D$ is restructured into a one-dimensional form $(1 \times L) \times D$, where L represents the length of the input sequence and D is the depth or number of channels. A 1D-CNN applies filters of size K across the sequence to extract local patterns. Applying a $(1 \times 3) \times 1$ filter to an input sequence with size of $(1 \times 10) \times 1$ with a stride of 1 and no padding will results in an output tensor of $(1 \times 8) \times 1$. At each position, the filter performs a dot product with the input segment, generating a (1×8) map. When using multiple filters D, such as four as shown in Figure 2.24 will produce four maps that are stacked to form an output tensor of size $(1 \times 8) \times 4$. According [56], the output length L can be calculated with using Equation 2.10 below.

$$L_{output} = (L_{input} - F + 2P) \,/\, S + 1. \qquad \textit{Eq 2.11}$$

Where:

- F: Filter size (spatial extent).
- S: Stride (step size for moving the filter).
- P: Padding (number of zeros added around the input border).

Each filter in the 1D-CNN is trained through backpropagation to recognize specific patterns, such as LMPPs in PV data. The same filter is applied across the full depth of the input, a process known as parameter sharing [45]. This significantly reduces the number of parameters and helps lower the risk of overfitting. To further simplify the model and improve efficiency, pooling layers are added after the convolution layers [47]. These layers reduce the size of the data by retaining only the most important features—typically through max pooling, where only the highest value in a group is kept [47]. Although this results in some loss of detail, it helps the model generalize better and improves overall performance.

Figure 2.24 below shows the 1D-CNN architecture with multiple convolutional layers followed by a flattening layer that converts the output of the convolutional layers into a one-dimensional vector. This is then passed through dense (fully connected) layers for final output processing. This layered design allows the network to progressively extract more meaningful and abstract features from sequential data, making it especially effective for tasks like MPPT, where identifying patterns in PV curves is crucial.

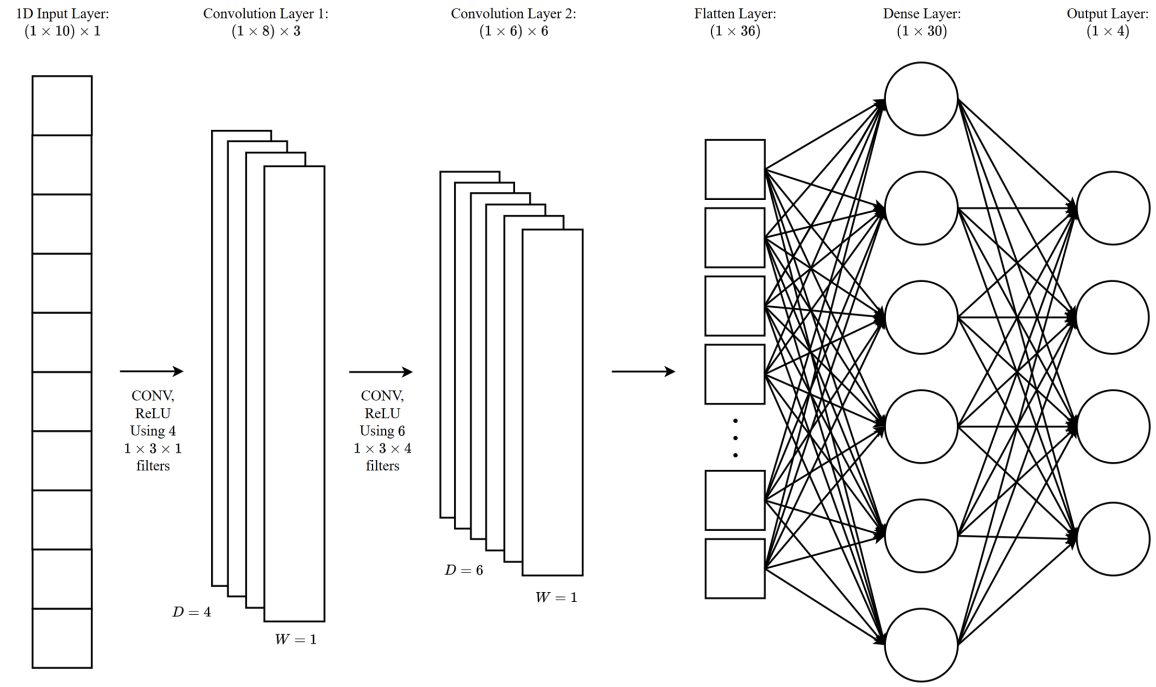


Figure 2.24. Simplified representation of a 1D-CNN model with convolutional layers, flattening layer and fully connected layer.

2.5.4 Optimization

Gradient Descent (GD) is a foundational optimization algorithm used to train deep neural networks by minimizing a cost function $f(\theta)$ [47]. It works by iteratively updating model parameters θ in the opposite direction of the gradient, moving the model towards reduced error [47]. A critical factor in this process is the learning rate α, which determines the size of each step. If set too high, the algorithm may overshoot the minimum or diverge entirely; if too low, convergence becomes slow and inefficient [45]. There are two main variations of gradient descent where Batch Gradient Descent computes the gradient using the entire training dataset before each update. While this approach provides stable and accurate updates, it becomes computationally expensive when dealing with large datasets.

$$\theta_{i+1} = \theta_i - \alpha \nabla f(\theta_i)$$

Stochastic Gradient Descent (SGD) [57] is normally used to improve efficiency by updating parameters using individual training samples. This allows for faster iterations and makes SGD suitable for large-scale problems. However, the frequent updates introduce variability, causing the optimization path to fluctuate, which may slow convergence or make it difficult to settle precisely at the minimum. For both GD and especially SGD, selecting an appropriate learning rate α is crucial. If set too high, the algorithm may fail to converge or diverge. If too low, training becomes unnecessarily slow. A common strategy suggested by [47] is to experiment with different learning rates to find the highest value that avoids divergence, then reduce it slightly to ensure stable training. It is also beneficial to gradually decrease the learning rate during training, a technique known as annealing [46]. This approach enables faster learning in the early stages and finer adjustments as the model approaches optimal performance. Annealing can be implemented by reducing the learning rate at fixed intervals or by applying a decay schedule where it decreases

progressively over time [46]. To further enhance optimization, several advanced techniques build upon SGD. Methods such as Momentum, AdaGrad, RMSProp, and Adam have been developed to improve speed, stability, and adaptability:

Momentum accelerates gradient descent by reducing fluctuations during optimization [46]. Rather than relying solely on the current gradient, it accumulates a fraction of past gradients, allowing the optimizer to build speed in consistent directions [46]. This approach is similar to how a rolling ball gains momentum as it moves through a valley, helping it push through small obstacles. For instance on a bumpy slope. Without momentum, SGD may get stuck bouncing around small bumps. Momentum helps push through these bumps by maintaining direction. especially in regions where gradients change direction frequently.

Adagrad and **RMSProp** are optimization algorithms that adjust the learning rate automatically during training to improve learning efficiency [46]. Adagrad reduces the learning rate for parameters that are updated frequently while keeping a higher learning rate for those that are updated less often, making it especially useful for handling sparse data where certain features appear rarely [46]. For example, if a feature is only occasionally used, Adagrad allows larger steps to ensure it still learns effectively. However, a major limitation of Adagrad is that the learning rate keeps decreasing throughout training, which can slow down progress or even stop learning before reaching the best result [46]. RMSProp addresses this issue by using a moving average of recent squared gradients instead of accumulating all past gradients. This approach allows RMSProp to focus on recent updates and "forget" older ones, keeping the learning rate stable and effective over time. As a result, RMSProp is more adaptive to changing data patterns and is particularly useful for tasks like training recurrent neural networks where the nature of the data can shift during training.

Adam [58] combines the advantages of both Momentum and RMSProp by calculating the first moment, which is the average of past gradients, and the second moment, which is the average of squared gradients. This approach allows Adam to adjust the learning rate for each parameter based on the optimization history while also smoothing the updates for greater stability and efficiency [46]. Similar to a smart driver who accelerates on smooth, consistent roads and slows down on rough, unpredictable terrain, Adam adapts its step size depending on whether the optimization path is steady or noisy. This ability to balance speed and control makes Adam highly effective in handling complex optimization problems. It is widely used in machine learning because it works well across various tasks with minimal need for manual tuning of hyperparameters.

2.5.5 Backpropagation

The backpropagation algorithm [59] is a method used to train neural networks by adjusting the weights and biases to reduce errors. It applies a basic concept from calculus called the chain rule [59], [60]. When training a neural network, the objective is to find the best weights and biases that minimize the loss function which is explain in earlier part of this section. During the forward pass, input data flows through the network layer by layer, with each layer performing calculations to produce an output. At the end of this process, the network makes a prediction and the loss function calculates the error. Once the error is known, backpropagation begins the backward pass.

Backpropagation uses the chain rule to pass gradients backward through the network step by step, helping the network understand how each input, weight, and bias affects the final error [45], [46], [47]. For example, if a layer produces an output z based on inputs x and y, backpropagation calculates how changes in x or y impact the overall loss L. As shown in Figure 2.25, the forward pass computes $z = f(x, y)$, and once the loss is known, the backward pass starts with the gradient of the loss with respect to z ($\delta L/\delta z$). To see how x and y contribute to the loss, backpropagation multiplies this gradient by the local gradients ($\delta z/\delta x$ and $\delta z/\delta y$) to compute $\delta L/\delta x$ and $\delta L/\delta y$. This process continues through each layer, calculating how much each weight and bias contributed to the error. The resulting gradients indicate both the direction and the amount each parameter should be adjusted to reduce the error in future predictions.

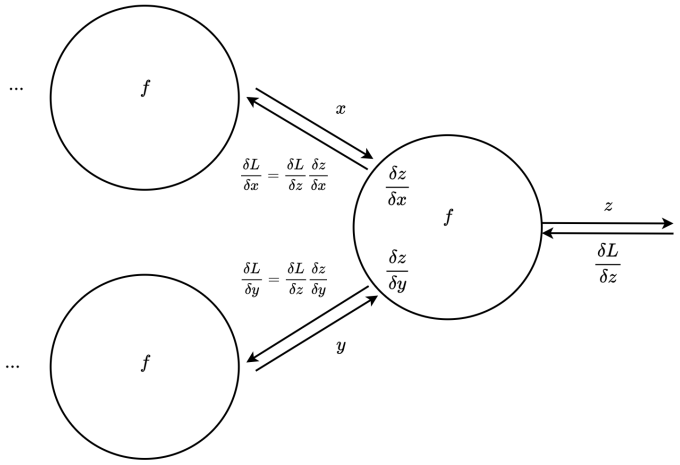


Figure 2.25. Illustration of backpropagation in a computational graph.

In simple terms, the forward pass calculates the network's output and checks how far it is from the correct answer by measuring the error. The backward pass then uses backpropagation to figure out how each weight should be adjusted to reduce this error. An optimizer, such as Adam [58], applies these adjustments by updating the weights in the right direction to improve future predictions. This cycle of forward pass, backpropagation, and weight updates continues until the network reaches a point where the error is minimized. Backpropagation can be thought of as working backwards from the final error, identifying which parts of the network contributed to it, and making small corrections to improve accuracy step by step.

2.5.6 Training Neural Network

When training a machine learning model, hyperparameters and configurations must be carefully tuned to ensure prevent overfitting or underfitting, and achieve optimal performance. Deep learning relies on iterative optimization, where models are refined over multiple cycles by adjusting parameters to minimize errors and improve generalization to unseen data.

Overfitting and underfitting are common challenges in machine learning that impact a model's ability to generalize. Overfitting occurs when a model learns both the underlying patterns and the noise in the training data, leading to poor performance on unseen data [61]. Underfitting happens when the model is too simple or not properly trained, failing to capture essential patterns, resulting in low accuracy on both training and test sets [61]. These issues can be detected by comparing training and validation errors, where a large gap indicates overfitting. Techniques such as early

stopping help prevent overfitting by halting training when validation performance stops improving [61]. For larger datasets, regularization methods mentioned earlier are often used to control model complexity and improve generalization [61].

Batch size is a crucial factor in optimization, determining how many data samples are processed before updating model parameters [45], [47]. Smaller batch sizes enable faster, more frequent updates but can introduce noise and instability, while larger batch sizes offer smoother, more stable updates but require greater computational resources and may miss subtle patterns in the data. To balance these trade-offs, mini-batch SGD is commonly used, where small subsets of data are processed in each iteration, followed by forward propagation, loss calculation, backpropagation, and parameter updates using optimizers like Adam. This method ensures a balance between training speed, stability, and memory efficiency, while also helping to reduce the risk of overfitting.

Data normalization and weight initialization are essential for stable and efficient training in deep learning models. When input features like x range from 1 to 10 and y from 1000 to 10000, the difference in scale causes the gradient in the xy-plane to be steep and biased towards y. This unbalanced gradient forces the model to take larger steps in the direction of y, making training unstable and highly sensitive to small changes in input [45]. Even minor fluctuations in x can lead to significant shifts in the output. By applying normalization, both x and y are scaled to have a mean of zero and unit variance, which flattens and balances the gradient across all directions. A balanced gradient allows the model to take smoother, controlled steps toward the optimal solution, reducing sensitivity to noise and improving generalization. Additionally, proper weight initialization, such as the Glorot Uniform initializer method [62] (or commonly called as Xavier initialization method), ensures consistent signal flow through the network by preventing vanishing or exploding gradients. Together, normalization and initialization enhance training stability, speed up convergence, and improve performance on unseen data.

Chapter 3. Methodology

3.1 Overview of Methodology

This chapter provides the steps used in developing the smart maximum power point tracking (MPPT) system using an LLC resonance converter. An LLC resonance converter prototype was built by my supervisor's PhD student, with the components soldered onto the board. This prototype will be used in the development process.

Figure 3.1 below shows the system level diagram of the proposed two stage GMPPT method. The system begins with a photovoltaic (PV) array connected to the input of the LLC converter. The proposed MPPT algorithm is deployed on an STM32-B-U585I-IOT02A microcontroller, which acquires real-time voltage and current data from the PV array via dedicated feedback circuits connected to its analogue digital converter (ADC) modules. Based on the sensed values, the microcontroller generates appropriate switching signals to dynamically adjust the circuit's DC voltage gain, reducing the mismatch between the PV source and the converter. This enhances tracking efficiency and maximizes power extraction.

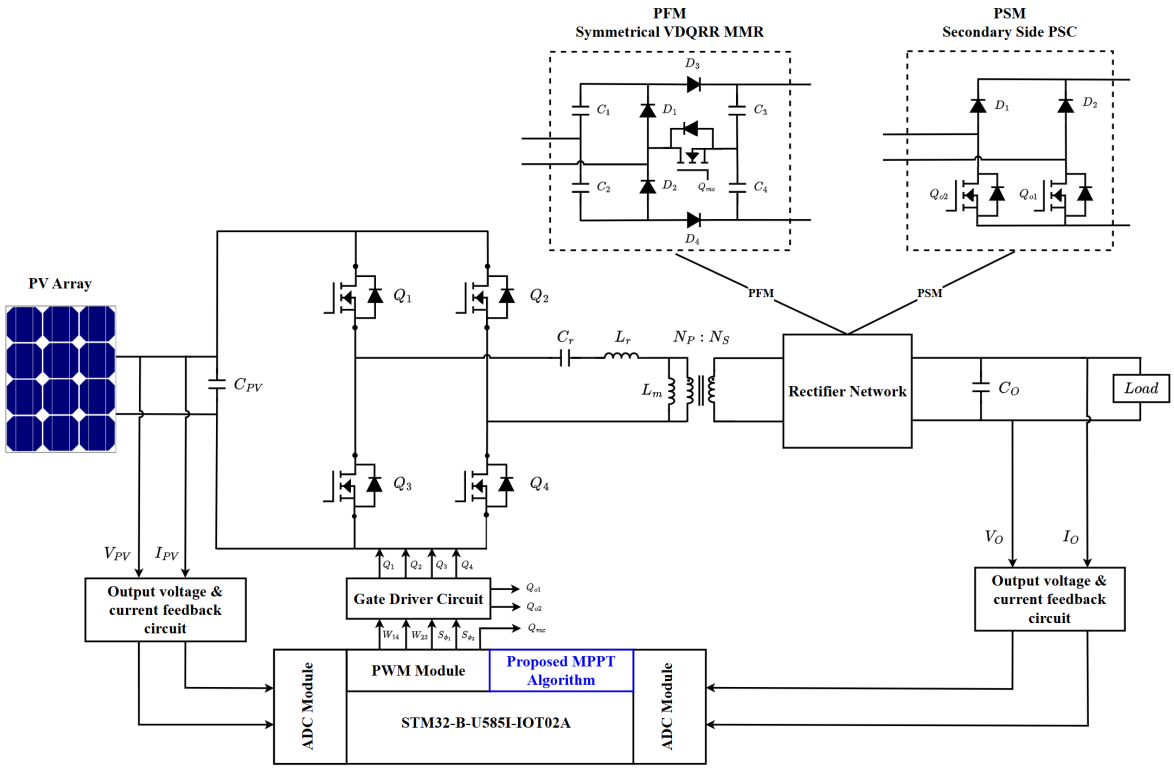


Figure 3.1. System diagram for two stage GMPPT method.

Pulse Frequency Modulation (PFM) or Pulse Phase Shift Modulation (PSM) method modulation techniques are explored in this project, with their respective advantages and disadvantages discussed in a later section. For the PFM method, the secondary board is configurable between a voltage-doubler and a voltage-quadrupler circuit to help extend the effective input voltage range. However, due to the limited gain range of PFM method, the primary full-bridge stage will also be reconfigured into a half-bridge by opening switch $Q2$ and closing switch $Q4$. For the SS-PSM method, the switching signal of gate Q_{o2} is delayed by a phase angle φ relative to Q_1. Finally, the

output voltage of the converter will be clamped to $120V$ using an electronic load, mimicking how the output of a PV system is typically clamped by a DC grid bus.

Therefore, the implementation of the smart MPPT system using an LLC converter is structured into three key phases: algorithm development, hardware prototype setup and analysis, and microcontroller integration. The MPPT algorithm using PFM and PSM method is first developed and validated through simulations in the MATLAB SIMULINK environment. This step ensures that the control logic performs reliably under varying input conditions before deployment. For the prototype testing, all components of the system are assembled in accordance with the system diagram shown in Figure 3.1. The constructed LLC resonant converter consists of various passive and active components. As this project primarily focuses on developing the GMPPT algorithm where analysis will mainly center on evaluating whether the designed hardware prototype can meet the algorithm's operational requirements. Finally, the microcontroller configuration involves initializing the PWM and ADC modules, as well as deploying the trained AI-based MPPT algorithm onto the microcontroller for real-time operation.

3.2 Development of Proposed Two Stage GMPPT Method

3.2.1 Photovoltaic Array

The algorithm is primarily developed for a photovoltaic (PV) system configured in a three-series, one-parallel (3S1P) arrangement. In this setup, three module connected in series, combining their voltages while maintaining the same current, with no additional parallel module. The PV panel used in this study is modeled based on the specifications of the 250Watts BLD250-290 60M mono crystalline array [63], which comprises three module of 20 solar cells each, forming the complete panel structure. Table 3.1 describes the specifications of the BLD250-290 60M module.

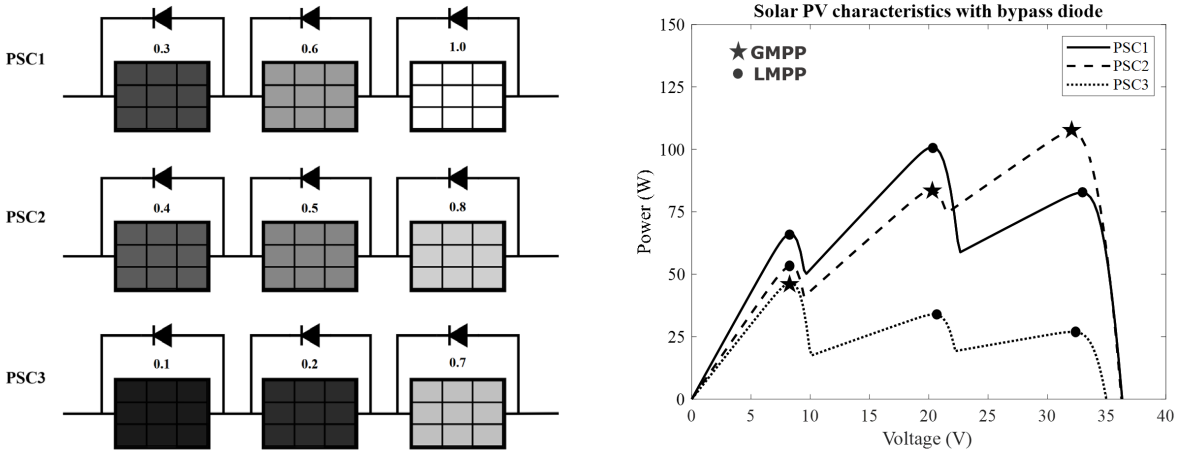
Table 3.1. Specifications of the BLD250-290 60M photovoltaic array.

Description	Values
Maximum Power (P_{max})	250W
Maximum Power Voltage (V_{mpp})	$30.79V$
Maximum Power Current (I_{mpp})	$8.12A$
Open Circuit Voltage (V_{oc})	$36.9V$
Short Circuit Current (I_{sc})	$8.59A$
Cell Efficiency (η)	17.96%
Module Efficiency (η)	15.37%
Temperature coefficient of V_{oc}	-0.36%
Temperature coefficient of I_{sc}	0.07%
Temperature coefficient of P_{max}	-0.38%
Nominal Operating Cell Temperature	$45 \pm 2°C$

The BLD250-290 60M monocrystalline module, rated at 250W, has an open-circuit voltage (V_{oc}) of 36.9V and a short-circuit current (I_{sc}) of 8.12 A. Based on these parameters, the fill factor (FF) of the array can be calculated as follows.

$$FF(\%) = \frac{V_{mpp} \cdot I_{mpp}}{V_{oc} \cdot I_{sc}} \times 100\% \qquad \textit{Eq 3.1}$$
$$= \frac{(30.79)(8.12)}{(36.9)(8.59)} \times 100\% = 78.88\%$$

The internal structure of BLD250-290 60M monocrystalline array consists of three modules connected in series and each substring is protected by its own bypass diode. Due to the varying irradiance levels across the series-connected modules, the resulting PV curve can exhibit up to three distinct peaks. These peaks correspond to multiple LMPPs, among which the highest is referred to as the GMPP. Figure 3.2 shows the P-V curve of the BLD250-290 60M array under different PSCs.



(a) *(b)*

Figure 3.2. Three tested PSCs. (a) Settings of each PV module, PSC1 (Top): 1000 W/m², 600 W/m², 300W/m²; PSC2 (Middle): 800 W/m², 500 W/m², 300 W/m²; and PSC3 (Bottom): 700 W/m², 200 W/m², and 100 W/m². (b) Corresponding PV curves.

3.2.2 Two Stage GMPPT Method

The aim of this project is to develop a GMPPT algorithm capable of accurately tracking the GMPP under various PSCs. Hence, a function $f(x) \rightarrow Y$ to estimate the GMPP and its corresponding voltage under a given shading pattern (SP) as shown in Figure 3.2(b) is needed. Due to the complexity and nonlinearity of the relationship between shading patterns and their resulting electrical characteristics, manually defining the function f using conventional rule-based approaches is impractical. To address this, this project will employs artificial neural networks (ANN) and 1D-convolutional neural networks (1D-CNN) by using to estimate the power and voltage values corresponding to the LMPPs in the first stage.

The overall flow of the proposed two-stage smart MPPT algorithm is illustrated in Figure 3.3. The algorithm operates with an update rate of 100Hz, a choice that effectively balances dynamic response and steady-state accuracy under rapidly changing irradiance conditions [64]. The process begins with measuring the open circuit voltage (V_{Moc}), from which the normalization parameter (I_{sc}) and the denormalization parameters (T, P_{max} and V_{oc}) are calculated. An automated sampling mechanism then captures three current values (I_1, I_2, I_3), each representing specific regions of the I–V curve. These sampled currents are normalized by dividing them by the calculated $I_{sc}$and are

subsequently processed by a neural network, either an ANN or a 1D-CNN, to estimate the LMPPs. The predicted LMPPs are denormalized using P_{max} and V_{oc}, and the highest value is selected through a maximum function to determine the GMPP. This GMPP is then converter into a initial modulating value, enabling the converter to operate at the optimal point using either pulse frequency modulation (PFM) or secondary side phase shift modulation (SS-PSM). The selected modulation strategy governs both the current sampling and the initialization of P&O algorithm, which is employed in the second stage to fine tune tracking accuracy. In PFM, modulation is achieved by adjusting the switching frequency, while SS-PSM uses phase shift modulation. These operational differences, along with the detailed implementation of the P&O algorithm, will be discussed in subsequent sections.

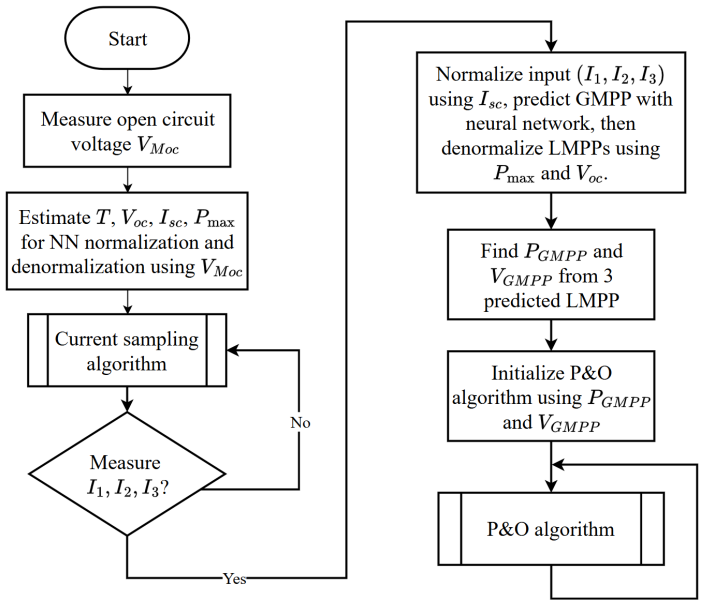


Figure 3.3. Flowchart for the proposed two stage GMPPT method.

Furthermore, taking the current as the input offers two key advantages.

- First, due to the resistive voltage drop across the wires connecting the PV array to the converter, using voltage and power as inputs may not accurately represent the true characteristics of the PV array. By using current, the impact of these voltage drops is minimized, leading to a more reliable input for modeling.
- Second, at the proposed sampling point, the PV current remains nearly constant within a specific voltage range. This means it is not necessary to precisely reach a specific voltage value to obtain an accurate current reading. As long as the voltage falls within an acceptable threshold, the current can be effectively sampled. This approach significantly reduces the required sampling time, enhancing the overall efficiency of the tracking process.

3.2.3 Current Sampling Method

The neural network architecture in this study employs three key input parameters: the current values I_{1x}, I_{2x}, and I_{3x} (x represent the start, middle or end current value at the flat region of I-V curve) which are sampled from the flat regions of the I–V curve. These flat regions, highlighted by the orange boxes in Figure 3.4, define the acceptable voltage ranges that the neural network can use as input. Each region is centered around a relatively constant current value and bounded by an

upper and lower limit with a voltage tolerance of $\Delta V = \pm 2.1\Delta$ from the predefined voltage points. The defined boundaries ensure that the sampled input currents fall within a stable and tolerant range, thereby enabling a faster and more reliable sampling process.

To ensure consistent sampling across these flat regions, the current measurements need to be taken at specific voltage points, V_{target}: $0.4V_{oc}/3$, $0.4 + V_{oc}/3$, and $0.4 + 2V_{oc}/3$ where V_{oc} represents the open-circuit voltage of the PV panel. In Region 1, I_{11} denotes the lower boundary of the target range, while I_{13} denotes the upper boundary.

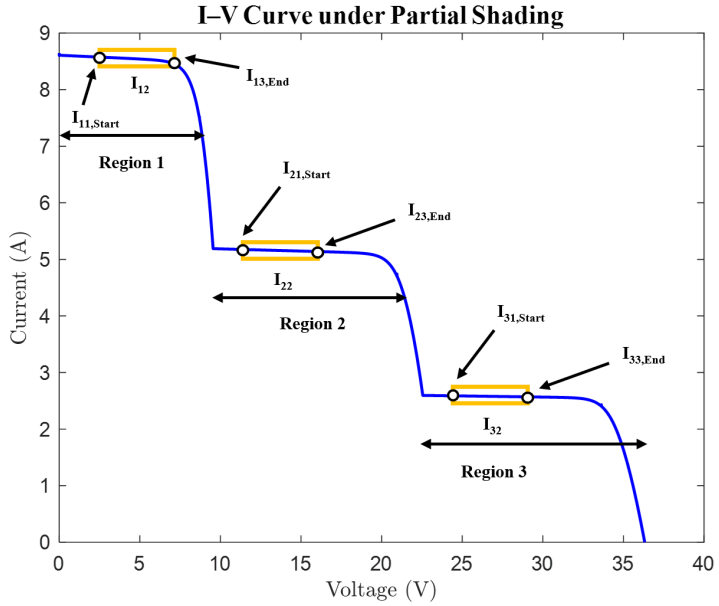


Figure 3.4. Illustration of the current sampling position of the proposed method.

However, as temperature increases, the I–V curves exhibit a noticeable leftward shift. This behavior is consistent with the negative temperature coefficient of the open-circuit voltage V_{oc}, which results in a reduction of voltage under constant irradiance conditions. Consequently, the originally defined flat regions become compressed or shifted along the voltage axis, increasing the likelihood that the sampled current may fall outside the intended flat region—particularly in Region 1, as illustrated in Figure 3.5. While the sampling windows (Represented by the orange boxes) remain within acceptable flat regions for regions 2 and 3, the upper boundary for Region 1 begins to drift outside the flat zone at higher temperatures. This limitation, however, is expected. Attempting to force the sampling voltage further toward the lower end of the I–V curve to recenter Region 1 would require the converter to operate at an extremely high voltage gain, which is often impractical in real applications. Despite this, the observed deviation of the sampled current from the center point to the upper boundary in Region 1 remains below 0.5A. This level of deviation is considered acceptable in practical scenarios, as it is within the typical error margin of current sensors, which often have accuracy limits that exceed this value. Hence, bounded for upper and lower limit with a voltage tolerance of $\Delta V = \pm 2.1\Delta$ will be used throughout the study.

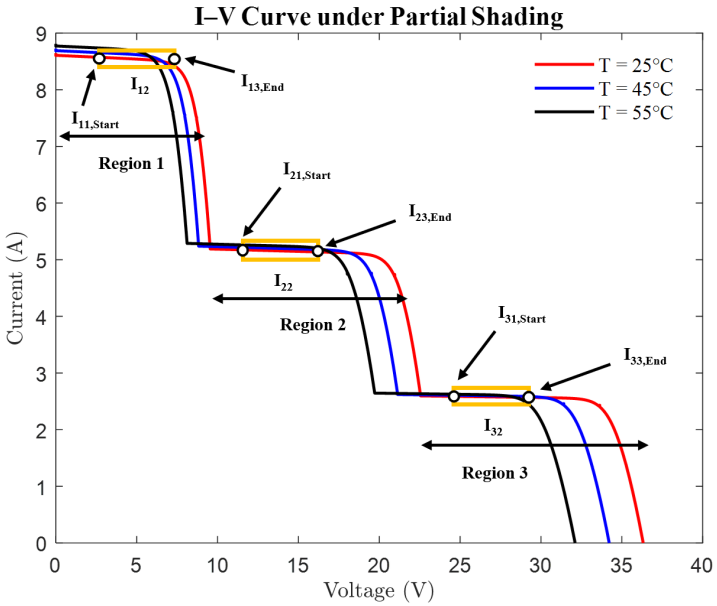


Figure 3.5. Illustration of the current sampling position under different operating temperature.

Finally, the flowchart for current sampling algorithm is shown in Figure 3.6 below. These regions are defined based on the open-circuit voltage ($V_{oc} = 37.92V$ for the BLD250-290 60M PV module) and are set at approximately $0.4V_{oc}/3, 0.4 + V_{oc}/3$, and $0.2 + 2V_{oc}/3$. The goal is to measure the current at these voltages to be used as inputs for the neural network.

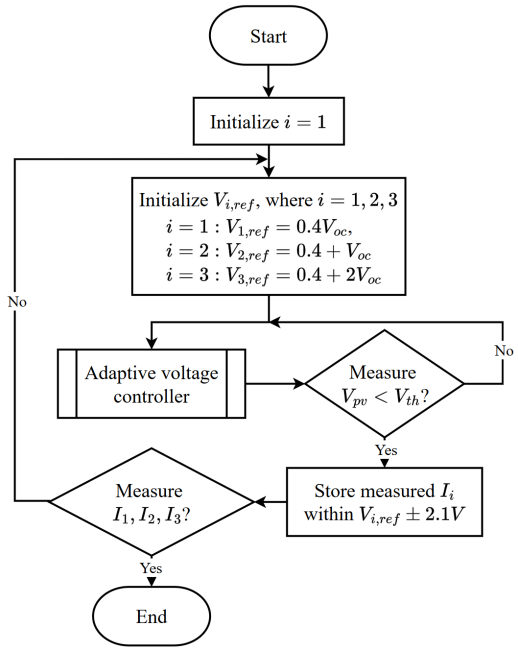


Figure 3.6. Flowchart of the current sampling algorithm.

The process of the current sampling algorithm begins by initializing an initial voltage reference, which will be given for the adaptive voltage controller to track. A counter is also initialized to keep track of how many current values have been successfully collected. In each step, the algorithm checks how close the current PV voltage is to the desired reference voltage. If the voltage is within

an acceptable range ($\pm 2.1\,V$), the current is recorded, and the counter increases by one. If not all three values have been collected, the control value is adjusted to target the next reference voltage. If the PV voltage is not close enough to the target, the control value is adjusted using the adaptive voltage controller (AVC) to gradually bring the voltage closer (explained in Section 3.2.4). This process repeats until all three current values are captured.

3.2.4 LLC Microgrid Adaptive Voltage Controller

A simple 1D-Look-Up Table (LUT) adaptive voltage controller is implemented to ensure the converter reaches the desired voltage threshold V_{th} efficiently. This controller dynamically adjusts the switching frequency based on the voltage error, calculated as $s = V_{pv} - V_{ref}$. For PFM, an initial frequency estimation using the first harmonic approximation (FHA), a frequency step α is selected according to predefined error ranges stored in the LUT. To achieve smooth and stable convergence, a corrective term $(\alpha \cdot s)/(2.1 + |s|)$ is applied, where α is adaptively chosen from the LUT based on the magnitude of s. The negative adjustment $f_{sw} = f_{sw} - (\alpha \cdot s)/(2.1 + |s|)$ ensures that when the voltage is above the target, the switching frequency decreases, thereby increasing the converter gain in line with PFM control principles. A larger voltage error results in a higher α value from the LUT, allowing faster response for significant deviations while maintaining stability as the system approaches the target voltage. The adaptive voltage control algorithm for pulse frequency modulation (PFM) is summarized in Algorithm 2 below.

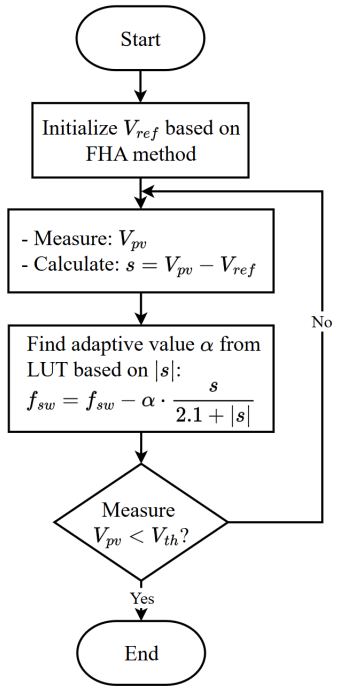


Figure 3.7. Flowchart of the current sampling algorithm.

Similar design flow also for SS-PSM method. Instead the initial phase shift is obtained by solving a quadratic equation derived in section 3.3.4 and a positive sign is used for $\varphi = \varphi + (\alpha \cdot s)/(2.1 + |s|)$ Equation. As increasing the phase shift will increase the effective gain of the circuit which is different compared to PFM method. The control flow is similar to PFM voltage controller, but phase shift angles are selected instead of frequency steps. Depending on the voltage error, adaptive phase shift are selected. A positive adjustment increases the phase shift when the voltage

is above the target, raising the converter gain to reach the desired lower voltage region on the PV curve. These modifications ensure effective control tailored for SS-PSM operation.

3.2.5 Synthtic Data Generation

To accurately predict the LMPPs, a data-generating distribution D is essential for producing a comprehensive dataset to train the neural network. For this purpose, the PV system specified in Table 3.1 was configured using MATLAB SIMULINK, as detailed in Section 3.4.1. The dataset collected from this setup consists of pairs $\{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\}$, where x represents three sampled current values $(x \rightarrow I_1, I_2, I_3)$, and y contains the corresponding power and voltage values $(y \rightarrow P_1, P_2, P_3, V_1, V_2, V_3)$. The primary objective is to train the neural network to minimize the expected loss across this dataset, thereby enhancing its predictive accuracy. To ensure the neural network is exposed to a wide range of scenarios, a large and diverse dataset was required to reflect various PSCs in a "3S1P" PV configuration. To achieve this, a MATLAB script was developed to generate all possible irradiance combinations based on 41 discrete values ranging from 1W/m² to 1000W/m², with increments of 25W/m². These combinations were generated with replacement, resulting in a total of 12341 combinations.

$$C(n+k-1,k) = \frac{(n+k-1)!}{k!\,(n-1)!} \qquad \textit{Eq 3.2}$$
$$= \frac{(41+3-1)!}{3!\,(41-1)!} = 12341$$

Where: $n = 21$ represents the number of discrete irradiance levels, $k = 3$ corresponds to the number of PV substrings.

For every generated pattern, the MATLAB script first updates the irradiance levels within the SIMULINK model, which includes both the PV system and a P&O based MPPT algorithm implemented using a MATLAB Function block. This process continues iteratively until the script has collected results for all 12341 combinations.

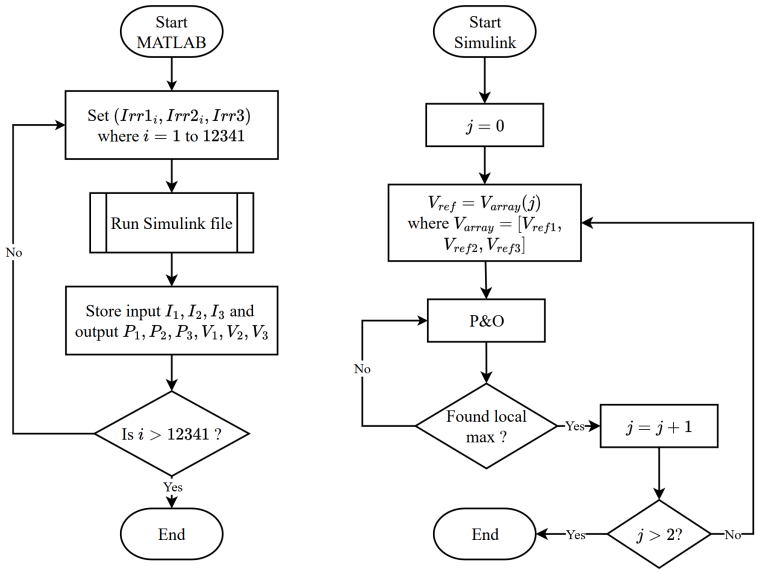


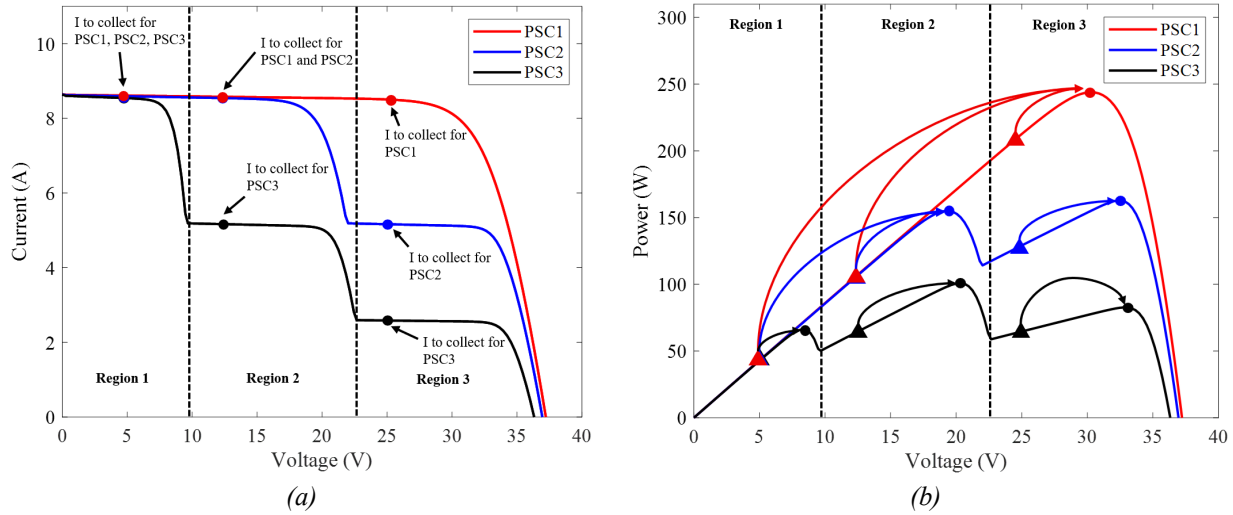
Figure 3.8. Flowchart for synthetic data generation using MATLAB and SIMULINK.

After setting the parameters the script will run the SIMULINK file with the algorithm decription as follows (and is illustrated in Figure 3.9):

- Starting at $0.4 \cdot V_{oc}/3$, then collect the respective current value.
- Perform P&O to find the local maximum.
- Continue at $0.4 + V_{oc}/3$, then collect the respective current value.
- Perform P&O to find the local maximum.
- Continue at $0.4 + 2V_{oc}/3$, then collect the respective current value.
- Perform P&O to find the local maximum.

After completing the simulation in SIMULINK, the MATLAB script automatically collects the three sampled current values $(x \rightarrow I_1, I_2, I_3)$, which serve as input features for the neural network, along with the corresponding power and voltage values at the local maximum points $(y \rightarrow P_1, P_2, P_3, V_1, V_2, V_3)$. As illustrated in Figure 3.9(b), the black curve representing PSC3, which contains three local maxima, identifies the local maximum points within each region. Once data collection is completed for all irradiance scenarios, the entire dataset is stored in a .csv file for subsequent training and evaluation of the neural network model. This automated data collection process ensures consistency and completeness, both of which are crucial for developing a reliable NN-based GMPPT algorithm.

For PSC1, which has only one LMPP, and PSC2, which has two LMPPs, the algorithm might search beyond the intended region to find other local maximum points. In the case of PSC1, the three calculated power values actually refer to the same LMPP but at different input voltages. To make this situation clearer and help the neural network during training, a simple method is used. If the detected LMPP is greater than a certain threshold, meaning it is outside the expected region, the values taken from SIMULINK will be set to zero. This indicates that there is no LMPP in that region for the given PSC condition. This method improves the clarity of the data and makes it easier for the neural network to learn correct patterns.



(a) *(b)*

Figure 3.9. The graphical illustration of data collection for three tested PSCs with 1 LMPP (red), 2 LMPPs (blue) and 3 LMPPs (black) with the I-V and P-V curves are divided into three distinct regions. (a) I-V curve with circle mark to indicate the current sampling point. (b) P-V curve with triangle mark to indicate initial P&O algorithm starting point and circle mark indicating LMPP found.

3.2.6 Data Generalization Technique and Data Pre-Processing and Normalisation

The normalization process is a critical step in machine learning, particularly in deep learning models, as unnormalized data can cause the model to become overly sensitive to small input variations, where even minor changes may lead to significant fluctuations in the output. Furthermore, the normalization process used in this project allows the model to accounts for temperature variations to enhance the prediction of LMPPs under different operating conditions, using a regression-based approach. By incorporating temperature effects into the normalization process, the input and output data of the neural network more accurately represent real PV behavior across varying thermal environments.

The datasheet typically specifies the open-circuit voltage V_{oc} under standard test conditions (STC), which assume a temperature of 25°C and standard irradiance levels. However, in real-world conditions, several factors such as voltage drops across cables, series resistance within the solar cells, and temperature-induced deviations cause the actual V_{oc} to be lower than the datasheet value. To account for these discrepancies, a voltage offset must be applied to the measured V_{oc} to better reflect practical operating conditions. In this project, the datasheet specifies a V_{oc} of 37.92V at STC (25°C and maximum irradiance), while the value obtained from the SIMULINK PV model under identical conditions is 35.38V, indicating an offset of 2.54V. Since both the simulation and datasheet operate at the same temperature and irradiance, this difference is primarily attributed to series resistance effects within the PV system. Therefore, an offset value of 2.54V is selected to compensate for these resistive losses. To obtain the offset value in practical applications with real solar panels, measuring the maximum V_{oc} under peak irradiance conditions is necessary to capture the effects of resistive losses, as temperature impacts are minimal when irradiance is at its highest.

$$V_{oc_{calc}} = V_{oc,m} + offset \qquad \textit{Eq 3.3}$$

Where $V_{oc,m}$ is the measured V_{oc} and $V_{oc_{calc}}$ is the one used for normalization.

The method used throughout the project begins by estimating the operating temperature T_{calc} of the PV module using the calculated open-circuit voltage $V_{oc_{calc}}$. The estimation is based on the temperature coefficient of open-circuit voltage obtained from datasheet and is expressed as:

$$T_{calc} = 25°C + \frac{V_{oc,calc} - V_{oc}}{TK_{V_{oc}} \cdot V_{oc}} \qquad \textit{Eq 3.4}$$

Where, V_{oc} represents the standard open-circuit voltage of a PV panel (from the datasheet). This formula helps infer temperature changes based on voltage shifts, assuming a linear relationship, which is a widely accepted approximation in PV modeling.

Furthermore, key PV parameters like short-circuit current I_{sc} and maximum power P_{mpp} are calculated as they are temperature-dependent, these are also recalculated using the estimated temperature. Datasheets typically provide their reference values at 25°C along with temperature coefficients. As temperature rises, I_{sc} increases slightly, while P_{mpp} decreases due to thermal losses. These effects are calculated using the following equations:

$$I_{sc} = I_{sc@25°C}\left[1 + TK_{I_{sc}}(T_{calc} - 25°C)\right] \qquad \textit{Eq 3.5}$$

$$P_{mpp} = P_{mpp@25°C}\left[1 + TK_{P_{mpp}}(T_{calc} - 25°C)\right] \qquad \textit{Eq 3.6}$$

Once the temperature and related parameters are computed, the sampled input currents I_1, I_2, I_3 are normalized by dividing them by the estimated I_{sc}. This scaling ensures that the ANN inputs fall within a consistent range, which improves training performance and model stability. On the output side, the ANN's predicted voltage and power values are denormalized using the estimated V_{oc} and P_{mpp}, allowing meaningful interpretation of the results.

$$I_{norm(1,2,3)} = \frac{I_{1,2,3}}{I_{sc}} \quad \textit{Eq 3.7}$$

$$P_{denorm(LMP1,LMP2,LMP3)} = P_{mpp} \cdot P_{LMPP1,LMPP2,LMPP3} \quad \textit{Eq 3.8}$$

$$V_{denorm(LMP1,LMP2,LMP3)} = V_{oc} \cdot V_{LMPP1,LMPP2,LMPP3} \quad \textit{Eq 3.9}$$

Although this method may seem simple, it provides a useful way for the neural network to get a rough idea of the temperature effect. The estimation does not need to be exact, as the neural network will adjust and learn the relationship during training. Since the connection between temperature, irradiance, and open-circuit voltage V_{oc} is highly non-linear, this approach is practical and cost-effective, especially for systems using microcontrollers with limited processing power. By using this basic adjustment, the neural network can still make reliable predictions under changing environmental conditions. The next section will explain the training process for two types of neural network architectures.

3.2.7 Training and Validation of Artificial Neural Network

To train and evaluate the ANN, the dataset was divided into three subsets: 70% for training, 15% for validation, and 15% for testing. This approach ensures that the model learns effectively while its performance is validated and tested on unseen data. In this case, the training set contains 8639 samples, while both the validation and testing sets contain 1851 samples each. As mentioned earlier, the ANN architecture adopted in this project is inspired by [10] with architecture comprises three main components: an input layer, two hidden layers that apply normalization and regularization techniques, and an output layer.

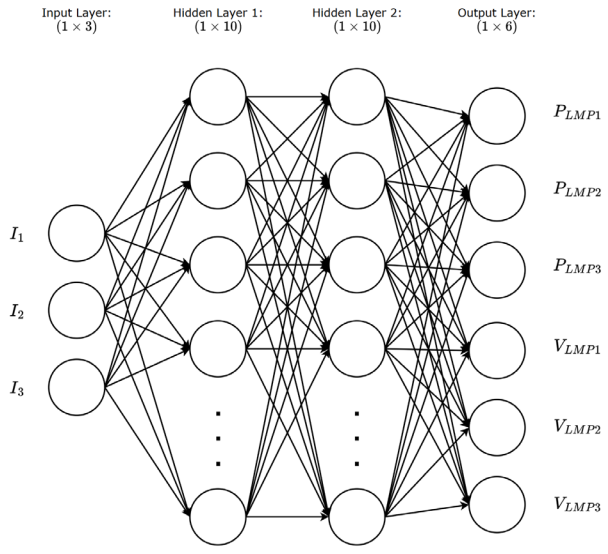


Figure 3.10. Detail architecture of ANN model.

The network shown in Figure 3.10 is built using a sequential model within a deep learning framework by referring to steps from [65], with the key design strategies summarized below.

1. Input Layer
   - The input layer accepts three normalized current values (I_1, I_2, I_3), which are sampled from three defined flat intervals of the PV module's I–V curve. These currents represent the system's electrical behavior under a given partial shading condition.
   - The input layer is fully connected to the first hidden layer and consists of 10 neurons with *ReLU* (Rectified Linear Unit) activation.
   - The weights are initialized using the Glorot Uniform initializer method (also known as Xavier initialization), which helps maintain stable gradients during training.

2. Hidden Layers
   - The model features two main hidden blocks, each comprising: a fully connected dense layer with 10 neurons and initialize using Glorot Uniform initialization method.
   - Batch normalization layer is applied to stabilize and accelerate training by normalizing the activations.
   - Dropout layer with a dropout rate of 0.1 are used to prevent overfitting by randomly deactivating 10% of the neurons during training.
   - Dense layer with 10 neurons and *ReLU* activation to introduce non-linearity and enable the network to learn complex patterns where this layered structure enhances the model's ability to generalize and handle nonlinear relationships between input currents and output parameters.

3. Output Layer
   - The output layer contains 6 neurons with linear activation (No activation function). These neurons correspond to the six target values: voltages and powers at three local maximum power points, specifically: V_{LMP1}, V_{LMP2}, V_{LMP3} and P_{LMP1}, P_{LMP2}, P_{LMP3}.

The architecture of ANN model with input layer, 2 hidden layers and output layer is shown below.

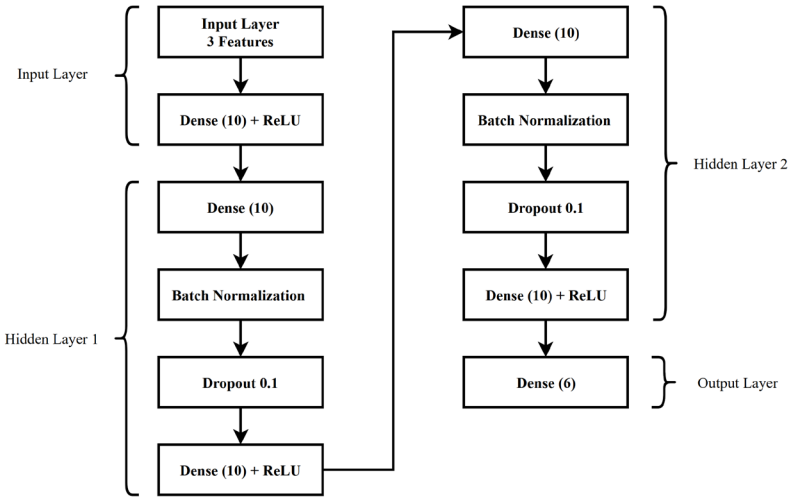


Figure 3.11. Detail architecture of ANN model.

Mean absolute error (MAE) is selected as the primary loss function for training, validation, and testing due to its robustness against outliers and random noise, which are common in photovoltaic (PV) measurements and simulation data. Unlike mean squared error (MSE), MAE provides a stable evaluation by treating all errors linearly, making it well-suited for this application. Throughout the training process, both training and validation MAE are monitored across epochs. The model is trained for a maximum of 200 epochs with a batch size of 16. An epoch refers to one complete pass through the entire training dataset, allowing the neural network to adjust its weights based on the observed errors. The batch size defines the number of training samples processed before the model's internal parameters are updated; in this case, the model updates its weights after every 16 samples. This mini-batch approach balances computational efficiency and learning stability, making it suitable for handling larger datasets without requiring excessive memory.

To improve training efficiency and prevent overfitting, two key callback functions are implemented. The *ReduceLROnPlateau* callback monitors the validation MAE and reduces the learning rate by a factor of 0.5 if no significant improvement (greater than 0.001) is observed over 20 consecutive epochs. This adaptive learning rate adjustment helps fine-tune the model during later stages of training, allowing for more precise convergence, while ensuring the learning rate does not fall below 1×10^{-6}. The *EarlyStopping* callback halts training if validation MAE does not improve over 50 consecutive epochs, preventing unnecessary computation and reducing the risk of overfitting. Additionally, it restores the model weights from the epoch where the best validation performance was recorded.

Validation is performed on a separate subset of the dataset to continuously assess generalization during training, while final performance evaluation is conducted using the test set where detail quantitative result is presented is Results section. This combination of MAE loss, adaptive learning rate scheduling, early stopping, and weight restoration ensures that the trained neural network achieves reliable accuracy and robustness when exposed to varying environmental conditions in real-world PV applications.

Table 3.2. Summary of specifications used when training ANN.

Design	Parameters
Architecture	ANN
Number of Input	3
Number of Ouput	6
Number of Dense Layers	4
Number of Nodes in each Hidden Layer	10
Number of training epochs	200
Initial Learning rate	0.001
Mini Batch Size	16
Training Dataset Size	8639
Validation Dataset Size	1851
Testing Dataset Size	1851

The model was trained using mini-batch gradient descent with a batch size of 16 for up to 200 epochs, where an epoch represents a full pass through the training data. Throughout training, both loss and MAE metrics were monitored for training and validation sets. Significant improvements

were observed in the early epochs, with the best validation MAE achieved at epoch 37 as shown in Figure 3.12. An adaptive learning rate strategy reduced the learning rate when progress plateaued, allowing finer adjustments during later stages. Training was stopped at epoch 87 by the EarlyStopping mechanism to prevent overfitting, and the model weights were restored from the best-performing epoch. Figure 3.12 shows the validation loss slightly above the training loss, indicating good generalization without overfitting or underfitting.

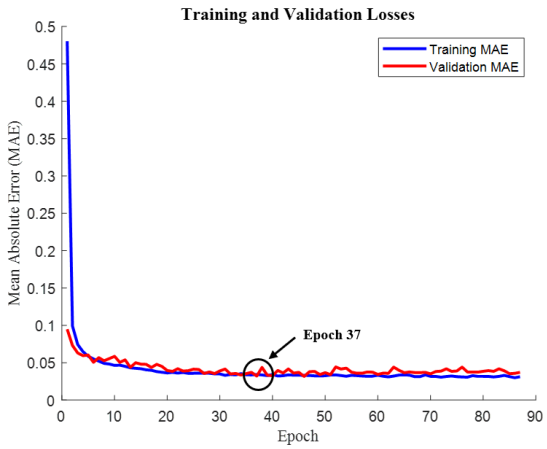


Figure 3.12. Training and validation losses of ANN using MAE evaluation metric.

3.2.8 Training and Validation of 1D-CNN

In the next stage of model development, a new sequential architecture incorporating a 1D-convolutional neural network (1D-CNN) was designed to enhance the model's ability to capture meaningful patterns within the input data. This structure enables the extraction of both low-level and high-level features, making it particularly effective for predicting local MPP voltages and powers under varying partial shading conditions. The architecture of the proposed 1D-CNN shown in Figure 3.13 comprises three main components: an input layer, two hidden layers and an output layer and is built using a sequential model within a deep learning framework by referring to steps from [65],

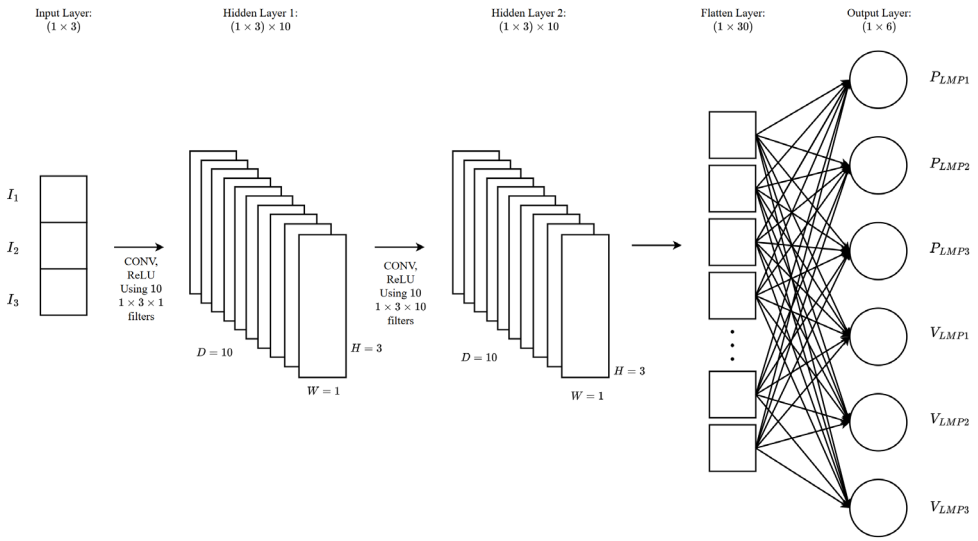


Figure 3.13. Detail architecture of ANN model.

The same dataset was used, divided into 70% for training, 15% for validation, and 15% for testing. In this case, the training set consists of 8639 samples, while the validation and testing sets each contain 1851 samples.

1. Input Layer
   - The input layer accepts three normalized current values $[I_1, I_2, I_3]$, which are sampled from three defined flat intervals of the PV module's I–V curve. These values capture the electrical behavior of the PV system under a specific partial shading condition.
   - Before being fed into the model, the 1D input vector is reshaped to a (3,1) format to comply with the input requirements of convolutional layers. This reshaping allows the use of 1D convolution operations, where filters slide across the input sequence to extract meaningful features.
   - The first layer is a Conv1D layer with 10 filters, a kernel size of 3, ReLU activation, and 'same' padding to maintain the original input length. The weights are initialized using the Glorot Uniform initializer to promote stable and efficient training.

2. Hidden Layers
   - The hidden section of the model consists of two convolutional blocks. Each block is designed to improve the model's depth, learning capability, and regularization
   - A Conv1D layer (without activation) with 10 filters and a kernel size of 3, using Glorot Uniform initialization and same padding.
   - A Batch Normalization layer, which helps accelerate training and maintain stable learning by normalizing layer inputs.
   - A Dropout layer with a dropout rate of 0.1, which helps prevent overfitting by randomly disabling 10% of the neurons during training.
   - Another Conv1D layer with 10 filters, kernel size 3, and ReLU activation to add non-linearity and enable the model to learn complex representations.
   - This layered convolutional structure enables the network to extract hierarchical patterns in the input currents, improving its capacity to generalize across different shading scenarios.
   - After passing through the convolutional blocks, the data is flattened using a Flatten layer, converting the multi-dimensional feature maps into a one-dimensional vector suitable for the output layer.

3. Output Layer
   - The output layer is a fully connected Dense layer with 6 neurons and linear activation (i.e., no activation function). These six outputs represent the predicted electrical characteristics of the PV system under the given input conditions: voltages and powers at three local maximum power points, specifically: V_{LMP1}, V_{LMP2}, V_{LMP3} and P_{LMP1}, P_{LMP2}, P_{LMP3}.
   - This design makes the CNN-based model more capable of capturing subtle correlations and variations in input patterns compared to standard dense networks, making it ideal for nonlinear regression tasks involving PV performance prediction under partial shading.

The architecture of 1D-CNN model with input layer, 2 hidden layers and output layer is shown below.

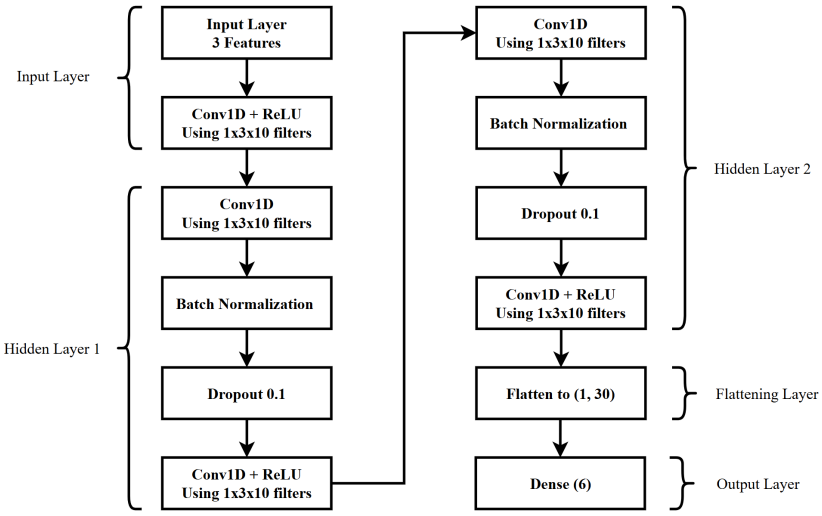


Figure 3.14. Detail architecture of 1D-CNN model.

The same training strategy applied to the ANN was also used for the 1D-CNN model to ensure consistency and fair performance comparison. Mean absolute error (MAE) was selected as the primary loss function due to its robustness against outliers and noise common in PV data. The 1D-CNN was trained for a maximum of 200 epochs with a batch size of 16, using mini-batch gradient descent for efficient learning. To enhance convergence and prevent overfitting, the *ReduceLROnPlateau* and *EarlyStopping* callbacks were implemented, dynamically adjusting the learning rate and halting training when improvements plateaued. Validation was performed on a separate subset, while final evaluation was conducted on the test set. This approach, combining MAE loss, adaptive learning rate scheduling, early stopping, and weight restoration, ensured that the 1D-CNN achieved reliable accuracy and robustness under varying environmental conditions, similar to the ANN setup.

Table 3.3. Summary of specifications used when training 1D-CNN.

Design	Parameters
Architecture	1-CNN
Number of Input	3
Number of Ouput	6
Number of Convolutional Layers	4
Filter Size	1×3
Number of Nodes in each Hidden Layer	10
Number of training epochs	200
Initial Learning rate	0.001
Mini Batch Size	16
Training Dataset Size	8639
Validation Dataset Size	1851
Testing Dataset Size	1851

The 1D-CNN model was also trained using mini-batch gradient descent with a batch size of 16 for up to 200 epochs with both loss and MAE metrics were monitored for training and validation sets. The 1D-CNN model demonstrated clear superiority over the ANN by achieving faster convergence and significantly lower errors. While the ANN reached its best validation MAE at epoch 37 and stopped at epoch 87, the 1D-CNN rapidly reduced both training and validation losses within the first 20 epochs, achieving optimal performance around epoch 70. Although the gap between training and validation losses in the 1D-CNN was slightly wider compared to the ANN, both losses remained consistently much lower than those of the ANN throughout the training process. EarlyStopping halted the 1D-CNN training at epoch 120, ensuring efficient convergence without overfitting. This highlights the 1D-CNN's enhanced capability to capture spatial and temporal patterns, delivering superior accuracy, faster learning, and stronger generalization under dynamic PV conditions. As illustrated in Figure 3.12, despite the noticeable gap, the validation loss stayed well-controlled, reflecting a balanced model that performs far better than the ANN in terms of overall error reduction.

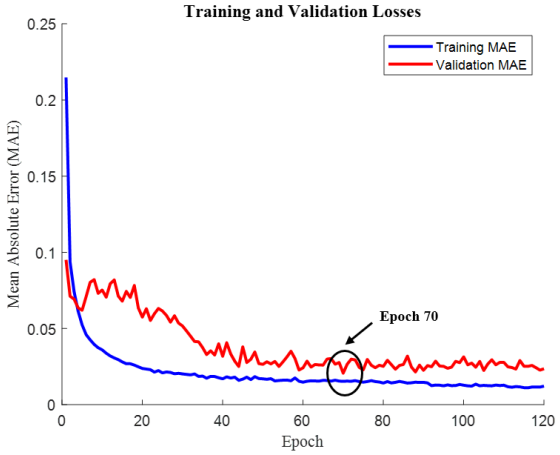


Figure 3.15. Training and validation losses of ANN using MAE evaluation metric.

3.2.9 Second Stage Hill Climbing Algorithm

The perturb and observe (P&O) algorithm used in this study incorporates several enhancements to improve adaptability and robustness under varying PV conditions. As illustrated in Figure 3.16, the algorithm begins by initializing three key parameters. These include the predicted GMPP values for both power P_{GMPP} and voltage V_{GMPP}, obtained from the neural network, along with the initial switching control variable, which corresponds to either the switching frequency in the case of the pulse frequency modulation (PFM) method or the phase shift for the single-stage phase shift modulation (SS-PSM) method. This initialization is performed through dedicated startup procedures specific to each modulation technique, ensuring that the controller starts operation near the expected optimal point. The initialization methods for PFM and SS-PSM are detailed in Section 3.3.3 and Section 3.3.6, respectively.

Following initialization in the PFM method, the P&O loop starts by calculating the PV power as the product of the measured voltage and current, expressed as $P_i = V_i I_i$. The algorithm then determines the change in power $\Delta P = P_i - P_{i-1}$ and the change in voltage $\Delta V = V_i - V_{i-1}$. Based on the power changes, an adaptive adjustment factor is selected from a lookup table to dynamically scale the perturbation magnitude according to system behavior. This adaptive strategy minimizes

oscillations around the maximum power point, promoting smoother and faster convergence. The control logic can be summarized as follows: if a perturbation leads to an increase in power, the algorithm maintains the same adjustment direction, indicating progress toward the maximum power point. Conversely, if the power decreases, it indicates that the maximum power point has been exceeded, prompting the algorithm to reverse the adjustment direction in the next cycle. Through this iterative process, the P&O algorithm efficiently tracks the power-voltage curve, ensuring optimal power extraction.

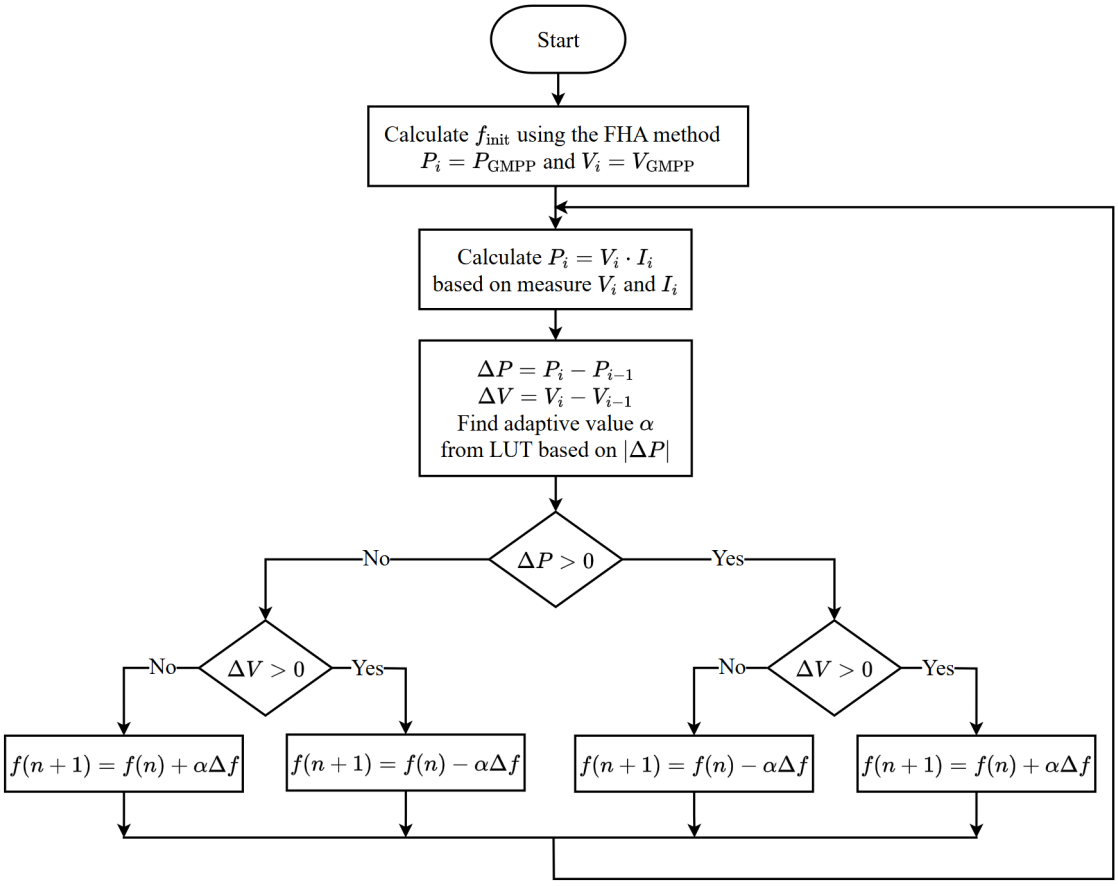


Figure 3.16. Flowchart for P&O algorithm for PFM modulation method.

A similar P&O approach is applied for the SS-PSM method, but instead of adjusting the switching frequency like in PFM, the SS-PSM method adjusts the phase shift angle. The control logic remains the same, where decisions are made based on changes in power, but the adjustment is applied to the phase shift. When there is a need to decrease the PV voltage, a positive adjustment increases the phase shift, which raises the converter gain and moves the operating point towards the desired region on the PV curve. One key difference is that the sign of the adjustment factor α is reversed compared to the PFM method because, in SS-PSM, increasing the phase shift increases the gain, while in PFM, increasing frequency decreases the gain.

3.3 LLC Converter for High Voltage Bus-Based PV

3.3.1 First Harmonic Approximation Method for PFM

This section will derived the gain Equation using frequency modulated method that will be used in this project. Many DC gain Equation has been shown in literature such as [16], [17], [18], [20], [22] but none of them provide detail derivation or explanation on the derivation part. Hence, derivation of this first harmonics appromination method DC gain Equation is essential as it will be used thoughout the project for PFM method and also throughout the derivation, some key assumption that is used while deriving the Equation will be pointed out as [24] says that it becomes less accurate when moving away from resonance frequency but detail explanation for this is not given.

Normally, there are two options in representing the the resonance tank. The first method is by using detailed analysis by considering all harmonics in the square waves while the seconds method is to approximate the resonant tank by only considering the fundamental components of the square waves. The second method will be used due to simplicity and the equivalent circuit of LLC resonant converter shown in Figure 3.17 consist of alternating input voltage V_{iF}. Correspondingly, the equivalent alternating output voltage is set as V_{oF}.

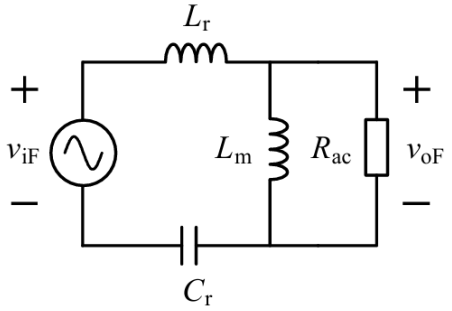


Figure 3.17. Equavalent circuit of resonant tank.

The resonant tank is a square wave due to the switching action of the full-bridge or half-bridge. This square wave can be applied to FHA, by decompose into its Fourier series.

$$v = \begin{cases} V_{in}, & 0 < t < \dfrac{T_s}{2} \\ -V_{in}, & \dfrac{T_s}{2} < t < T_s \end{cases}$$

The converter will have input voltage with 50% duty cycle where deadtime is ignored in this analysis. Hence, it can be represented as follows.

$$v_{in,i}(t) = \frac{4}{\pi} V_{in} \sum_{n=1,3,5,\ldots}^{\infty} \frac{1}{n} \sin(n\omega_{sw} t - \theta)$$

- V_{in} is the amplitude of the square wave (The DC input voltage).
- $\omega_{sw} = \frac{2\pi}{Ts}$ is the angular switching frequency.
- $n = 1,3,5, \ldots$ represents the odd harmonics (1st, 3rd, 5th, …) because a square wave with 50% duty cycle contains only odd harmonics.

- $\frac{4\pi}{V_{in}}$ comes from the fourier series of a square wave.
- θ is the phase shift.

Since the LLC resonant tank behaves as a narrowband filter, higher-order harmonics are significantly attenuated. Therefore, under the FHA, only the fundamental component ($n = 1$) is considered to dominate the system's behavior. The simplified input voltage becomes

$$v_{tank}(t) \approx v_{in,FHA}(t) = \frac{4}{\pi} V_{in} \sin(\omega_s t - \theta)$$

Hence, the root mean square voltage across transformer primary is $v_{tank,rms} = \frac{2\sqrt{2}}{\pi} V_{in}$. The transformer will step this voltage up to the secondary side, $v_{sec,rms} = n\frac{2\sqrt{2}}{\pi} V_{in}$.

Assuming that only purely resistive load is connected to the secondary side where the current is sinusoidal and in-phase with the voltage. Thus, the FHA current is approximated as follows.

$$i_{FHA}(t) = \sqrt{2} I_{rms} \sin(\omega_s t - \theta)$$

The rectifier converts the AC current into DC current. The DC output current, I_{out} is the average of the absolute value of the AC current over a half-cycle.

$$I_{out} = \frac{2}{T} \int_0^{\frac{T_s}{2}} |n i_{FHA}(t)| dt$$

Where $n = \frac{N_{sec}}{N_{pri}}$ is the turns ratio. Hence, $i_{FHA}(t) = \sqrt{2} I_{rms} \sin(\omega_s t - \theta)$ is substituted into the above Equation by assuming the phase shift, $\theta = 0°$.

$$I_{out} = \frac{2}{T_s} \int_0^{\frac{T_s}{2}} \frac{1}{n} \sqrt{2} I_{rms} \sin(\omega_s t)\, dt$$

$$= \frac{2}{T_s} \sqrt{2} \frac{1}{n} I_{rms} \left[-\frac{1}{\omega_s} \cos(\omega_s t) \right]_0^{\frac{T_s}{2}}$$

$$= \frac{2}{T_s} \sqrt{2} \frac{1}{n} I_{rms} \left(-\frac{1}{\omega_s} \cos(\pi) + \frac{1}{\omega_s} \cos(0) \right)$$

$$= \frac{2}{T_s} \sqrt{2} \frac{1}{n} I_{rms} \frac{2}{\omega_s} = \frac{4\sqrt{2} I_{rms}}{n \omega_s T_s} = \frac{4\sqrt{2} I_{rms}}{n 2\pi} = \frac{2\sqrt{2} I_{rms}}{n\pi}$$

Then, Ohm's law is used to related the output current to load.

$$I_{out} = \frac{V_{out}}{R_{load}}$$

$$\frac{2\sqrt{2} I_{rms}}{n\pi} = \frac{V_{out}}{R_{load}}$$

$$I_{rms} = \frac{n\pi V_{out}}{2\sqrt{2} R_{load}}$$

The equivalent resistance on the secondary side can be represented as shown below.

$$R_{ac} = \frac{v_{FHA}}{i_{FHA}} = \frac{\frac{2\sqrt{2}}{\pi}V_{in}}{\frac{\sqrt{2}I_{rms}}{\sqrt{2}}} = \frac{2\sqrt{2}V_{in}}{\pi I_{rms}}$$

$$= \frac{2\sqrt{2}V_{in}}{\pi\frac{n\pi V_{out}}{2\sqrt{2}R_{load}}} = \frac{8}{\pi^2} \times \frac{V_{in}}{nV_{out}} \times R_{load}$$

Where at resonance frequency, $nV_{in} \approx V_{out}$. Hence, the load resistance can be represented as follows (the factor of $1/4$ is obtained from the doubler rectifier circuit).

$$R_{ac} = \frac{8}{\pi^2}\frac{1}{n^2}\ R_{load} = \frac{8}{\pi^2}\frac{N_{pri}^2}{N_{sec}^2}\frac{1}{4}R_{load}$$

Next, the resonance tank transfer function is defined as the ratio of output voltage over input voltage in frequency domain by specifying the L_m (Magnetizing inductance), L_r (Resonant inductance), and C_r (Resonant capacitance). First, the impedance in parallel combination of L_m and R_{ac} is obtained as follows ($L_m||R_{ac}$).

$$Z_{tank} = \left[\frac{1}{sL_m} + \frac{1}{R_{ac}}\right]^{-1} = \frac{sL_mR_{ac}}{sL_m + R_{ac}}$$

The series impedance is obtained as follows.

$$Z_{series} = sL_r + \frac{1}{sC_r}$$

The total impedance seen by the input can be expressed as follows.

$$Z_{total} = j\omega L_r + \frac{1}{j\omega C_r} + \frac{j\omega L_m R_{ac}}{j\omega L_m + R_{ac}}$$

Using the complex transfer function, the gain for the resonant tank can be obtained by considering voltage divider rule.

$$V_{tank} = V_{in} \times \frac{Z_{tank}}{Z_{total}}$$

$$H(s) = \frac{V_{tank}}{V_{in}} = \frac{\frac{sL_mR_{ac}}{sL_m + R_{ac}}}{sL_r + \frac{1}{sC_r} + \frac{sL_mR_{ac}}{sL_m + R_{ac}}}$$

$$= \frac{sL_mR_{ac}}{sL_r(sL_m + R_{ac}) + \frac{1}{sC_r}(sL_m + R_{ac}) + sL_mR_{ac}}$$

$$= \frac{sR_{ac}L_m}{s^2L_mL_r + sR_{ac}(L_r + L_m) + \frac{L_m}{C_r} + \frac{R_{ac}}{sC_r}}$$

By multiplying sC_r to both numerator and denominator.

$$H(s) = \frac{s^2 R_{ac} C_r L_m}{s^3 C_r L_r L_m + s^2 C_r (L_r + L_m) R_{ac} + s L_m + R_{ac}}$$

The magnitude and phase angle can be calculated as follows.

$$H(j\omega) = \frac{-\omega^2 R_{ac} C_r L_m}{-j\omega^3 C_r L_m L_r - \omega^2 C_r (L_r + L_m) R_{ac} + j\omega L_m + R_{ac}}$$

Where.

- $s = j\omega$
- $s^2 = (j\omega)^2 = -\omega^2$
- $s^3 = (j\omega)^3 = -j\omega^3$

The denominator of the transfer function above can be separated into real and imaginary component.

$$-j\omega^3 C_r L_r L_m - \omega^2 C_r (L_r + L_m) R_{ac} + j\omega L_m + R_{ac}$$
$$= R_{ac}[1 - \omega^2 C_r (L_r + L_m)] + j\omega L_m [1 - \omega^2 C_r L_r]$$

The magnitude of the transfer function then can be calculated as follows.

$$|G(j\omega)| = \frac{\omega^2 R_{ac} C_r L_m}{\sqrt{\left[R_{ac}\left(1 - \omega^2 C_r (L_r + L_m)\right)\right]^2 + [\omega L_m (1 - \omega^2 C_r L_r)]^2}}$$

By considering the below variable, the transfer function can be simplified further,

- Resonance frequency, $f_0 = \frac{1}{2\pi\sqrt{L_r C_r}}$, $\omega_0 = 2\pi f_0 = \frac{1}{\sqrt{L_r C_r}}$
- Normalized frequency, $f_n = \frac{f_s}{f_0} = \frac{\omega}{\omega_0}$
- Inductance ratio, $k = \frac{L_m}{L_r}$
- Quality factor, $Q = \frac{\sqrt{L_r / C_r}}{R_{ac}} = \frac{\omega_0 L_r}{R_{ac}}$

Where the combination of some circuit parameter can be represented using the newly defined notation.

- $\omega^2 C_r L_r = \left(\frac{\omega}{\omega}\right)^2 \frac{L_r}{L_r C_r \omega_0^2} = f_n^2$
- $\omega^2 C_r L_r = 1 - f_n^2$
- $L_r + L_m = L_r(1 + k)$
- $\omega^2 C_r (L_r + L_m) = f_n^2 (1 + k)$
- $\omega^2 C_r (L_r + L_m) = 1 = f_n^2 (1 + k)$
- $\omega L_m = \omega k L_r = f_n \omega_0 k L_r$
- $\omega^2 C_r L_m = \omega^2 C_r k L_r = f_n^2 k$

A new representation for the gain equation for the resonance circuit can be obtained by substituting these expressions into the transfer function with magnitude above.

$$|G(j\omega)| = \frac{f_n^2 R_{eq} k}{\sqrt{\left[R_{eq}\left(1 - f_n^2(1+k)\right)\right]^2 + [f_n \omega_0 k L_r (1 - f_n^2)]^2}}$$

$$= \frac{f_n^2 k}{\sqrt{\left[\left(1 - f_n^2(1+k)\right)\right]^2 + \left[\frac{f_n \omega_0 k L_r}{R_{eq}}(1 - f_n^2)\right]^2}}$$

$$|G(j\omega)| = \frac{f_n^2 k}{\sqrt{\left[\left(1 - f_n^2(1+k)\right)\right]^2 + [f_n k Q (1 - f_n^2)]^2}} \qquad \textit{Eq 3.10}$$

Furthermore, derived gain Equation will can be simplified further to be represented as follows (Similar to the one presented in [16], [17], [18], [20], [22]).

$$|G(f_{sw}, K, Q)| = \frac{1}{\sqrt{\left(1 + \frac{1}{K} - \frac{f_r^2}{K f^2}\right)^2 + Q^2 \left(\frac{f}{f_r} - \frac{f_r}{f}\right)^2}}$$

Hence, the gain equation derived using FHA is most accurate at the resonant frequency, where the assumption that only the fundamental harmonic dominates holds true. As the switching frequency deviates from resonance, neglected higher harmonics and non-linear effects cause increasing discrepancies between the calculated and actual gain.

Based on the derived equations and findings from the literature, the LLC converter's frequency-dependent gain profile allows regulation of the output voltage simply by adjusting the switching frequency of the primary-side switches. However, as shown in Figure 2.17, the load conditions play a major role in shaping the gain curve and directly impact the converter's effective operating range. In PV applications, maintaining impedance matching between the converter and the PV array is critical for efficient MPPT. Since the P–V curve of a PV system changes with irradiance and temperature, the converter must support a wide switching frequency range to continuously adjust its input impedance in real time.

Another challenge comes from the fact that the load on the converter depends on the unknown power output of the PV array, which the MPPT is trying to find. This makes maintaining perfect matching even more complex. Moreover, at high PV power levels, the converter risks slipping into the capacitive region if the switching frequency drops below the resonant frequency. When this happens, the converter loses ZVS, leading to hard switching, increased switching losses, and extra thermal stress on the switching devices. If left unchecked, this could compromise system reliability or even cause permanent damage.

Furthermore, during the current sampling phase for the 1D-CNN prediction, the PV voltage VPV needs to be adjusted to around $5V$ and $25V$ to extract meaningful samples. To reach these points from a fixed output voltage of $120V$, the converter must achieve a voltage gain ranging from approximately 4.8 up to 24. Similarly, during normal MPPT operation, the required VPV can vary from around $8V$ up to $33V$, requiring a voltage gain between about 3.6 and 15. Altogether, this

means the converter must handle a gain range of approximately 3.6 to 24 to operate effectively across all conditions. However, after further analysis using FHA where the result is shown in Figure 3.20(b), it became clear that it is impossible to cover such a wide gain range using only a full-bridge primary-side configuration and a simple voltage doubler on the secondary side. Even assuming an ideal resonant tank that can vary its gain from 0.8 (buck) to 2.0 (boost), the achievable voltage range would only be from about $15V$ to $25V$ at the by using the analysis from Figure 3.20(b).

To overcome this limitation, a voltage doubler quadrupler reconfigurable rectifier (VDQRR), proposed by *Vinnikov et al.* [66] is introduced on the secondary side, allowing dynamic adjustment of the rectification ratio. Additionally, the primary-side full-bridge configuration is occasionally switched to a half-bridge configuration when necessary. These modifications enable the converter to achieve the broader voltage gain range required for reliable and flexible MPPT operation.

3.3.2 Extending Voltage Gain Range

To address the insufficient voltage gain provided by the PFM control method and to further demonstrate that the developed algorithm is functioning as intended, a voltage doubler quadrupler reconfigurable rectifier (VDQRR) proposed by *Vinnikov et al.* [66] is used. This approach provides the necessary gain to allow the converter to reach lower PV input voltages, particularly below $10V$.

The VDQRR topology enables the output stage to dynamically switch between voltage doubler and quadrupler configurations based on the system's voltage requirements. This reconfiguration is controlled by a mode-changing switch (S_{MC}), as illustrated in Figure 3.18. When S_{MC} is turned OFF, the circuit operates in voltage doubler (VDR) mode. When S_{MC} is turned ON, it shifts to voltage quadrupler (VQR) mode, providing a higher voltage gain to support wider MPPT operation.

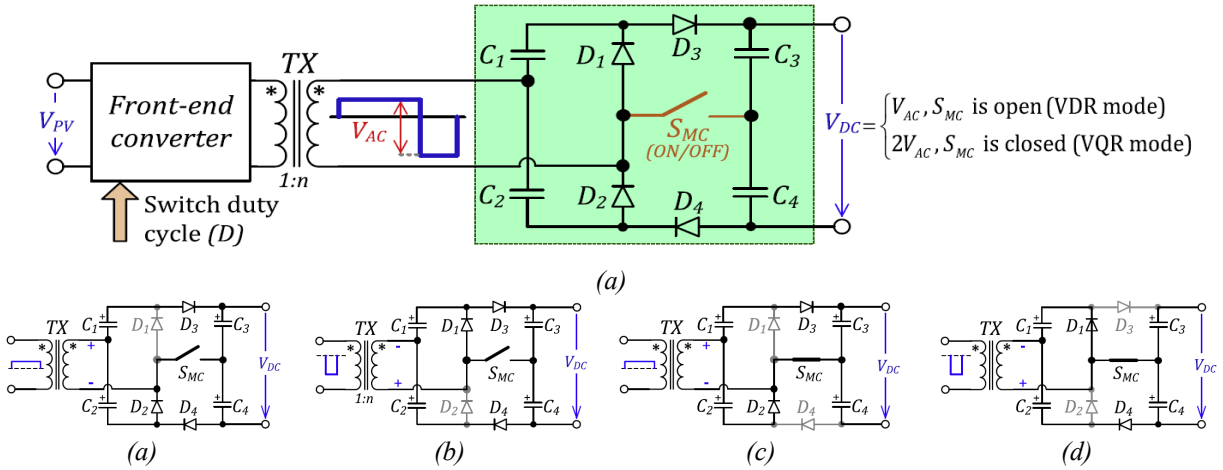


Figure 3.18. The equivalent circuits of the VDQRR operating with a balanced or asymmetric AC voltage. (a) and (b) illustrate the circuit in voltage doubler rectifier (VDR) mode, while (c) and (d) shows the circuit operating in voltage quadrupler rectifier (VQR) mode.

Reconfigurable rectifiers are commonly used in galvanically isolated step-up DC–DC converters to enhance performance across a wide output voltage regulation range. By integrating both voltage VDR and VQR modes in a single reconfigurable structure, the system can dynamically adapt its

DC gain. This approach provides a better trade-off between achieving high voltage gain and maintaining good power conversion efficiency. During operation, diode D_1 conducts during the negative half-cycle of the transformer's secondary voltage, allowing current to flow through D_1 and charge capacitor C_1 as shown in Figures 3.18(b) and Figures 3.18(d). Similarly, diode D_2 conducts during the positive half-cycle, allowing current through D_2 to charge capacitor C_2 as shown in Figures 3.18(a) and Figures 3.18(c). These actions occur in both VDR and VQR modes. Simply put, during each half-cycle, either C_1 or C_2 charges up to approximately the peak secondary voltage V, so together, C_1 and C_2 store a doubled voltage.

In VDR mode (when the mode-changing switch S_{MC} is OFF), diodes D_3 and D_4 transfer the charge from C_1 and C_2 to the output capacitors C_3 and C_4. This results in an output voltage that is approximately the same as the sum of C_1 and C_2, effectively a one-to-one voltage transfer relative to the peak secondary voltage.

In contrast, when operating in VQR mode (with S_{MC} ON), diodes D_3 and D_4 are reconfigured to support voltage multiplication. During the first half-cycle, the voltage from C_1 and C_2 are added to C_3, and during the second half-cycle, it is added to C_4. This stacking effect further increases the voltage, resulting in an output that is approximately four times the input. In other words, the VQR mode allows the converter to achieve higher output voltage levels while requiring only half the transformer's secondary voltage compared to the VDR mode. This makes VQR mode particularly useful when higher voltage gain is needed under low input PV voltage conditions. It allows the system to boost lower PV voltages, for example below $10V$, up to the desired output voltage of around $120V$ without pushing the switching frequency to extreme and inefficient levels.

On the other hand, when operating in a region where a lower voltage gain is needed, meaning the converter must reduce rather than increase the gain, the system needs to supply less voltage to the transformer. One effective method to achieve this is by shifting the primary side into a half bridge configuration [67], [68]. This can be done by keeping switch 2 open while closing switch 4 as shown in Figure 3.19 below [68]. As a result, the circuit operates as a half bridge, where the transformer is driven with a voltage swing from V_{DC} to 0, instead of from V_{DC} to $-V_{DC}$ as in full bridge operation. This approach effectively reduces the voltage stress on the transformer and lowers the overall voltage gain of the system, allowing better control when operating at higher output voltages.

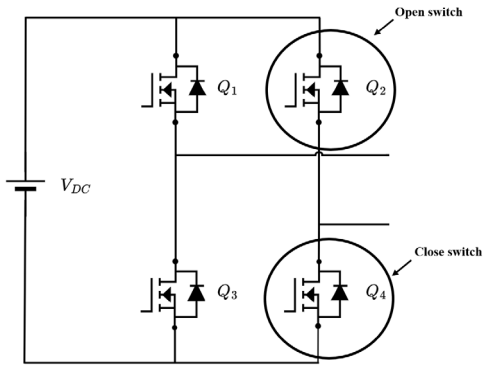


Figure 3.19. Reconfiguration of the primary-side switching circuit from full bridge to half bridge operation. By opening switch 2 and closing switch 4, the transformer voltage swing is reduced from Vdc to zero instead of Vdc to negative Vdc, effectively lowering the overall voltage gain to support high output voltage operation.

3.3.3 PFM Initialization Method

To select the initial switching frequency for the hill climbing algorithm or voltage scanning algorithm using the pulse frequency modulation technique, the first harmonic approximation method proposed by *Tang et al.* [18] is used, which involves solving a cubic Equation.

$$x^3 + ax^2 + bx + c = 0 \qquad \textit{Eq 3.11}$$

Where the coefficient for the cubic Equation using $K = \frac{L_m}{L_r}$.

- $f_n^2 = x$
- $a = \frac{K^{-2}(K+1)^2 - 2Q^2 - G^{-2}}{Q^2}$
- $b = \frac{Q^2 - 2K^{-2}(K+1)}{Q^2}$
- $c = \left(\frac{1}{KQ}\right)^2$

To determine the Q value, the variables R_{ac} and G must first be calculated to solve Equation 3.11. The reflected resistance on the primary side can be expressed.

$$R_{ac} = \frac{8}{\pi^2} N^2 \frac{1}{4} R_{load}$$

$$Q = \frac{\sqrt{\frac{L_c}{C_r}}}{R_{ac}}$$

Where the factor of $1/4$ is obtained from the voltage doubler circuit configuration.

By incorporating the predicted GMPP power and voltages, the Gain, G and the load resistance, R_{load} can be determine.

$$P_{out} = \frac{V_{out}^2}{R_{load}}$$

Where the output power can be approximated to be the same as GMPP power, $P_{GMPP} \approx P_{out}$ as the converter will have efficiency approximately 100%.

$$P_{GMPP} = \frac{V_{out}^2}{R_{load}}$$

$$R_{load} = \frac{V_{out}^2}{P_{GMPP}}$$

Hence, Q can be calculated as follows.

$$Q = \frac{4\pi^2}{8N^2} \cdot \frac{V_{out}^2}{P_{GMPP}} \cdot \sqrt{\frac{L_c}{C_r}}$$

Then, the gain is calculated using the V_{GMPP} estimated from the neural network.

$$G = \frac{V_{out}}{N_v \times V_{GMPP}}$$

The previous derivation applies specifically to the voltage doubler configuration. However, slight modifications can be made to the gain equation to suit different topologies:

For a half-bridge with voltage doubler (to access region 3), the gain can be calculated using:

$$G = \frac{V_{out}}{N_v \times V_{GMPP}} \cdot \frac{1}{2} \cdot 2$$

For a full-bridge with voltage doubler (to access region 2), the gain becomes:

$$G = \frac{V_{out}}{N_v \times V_{GMPP}} \cdot \frac{1}{2}$$

For a full-bridge with voltage quadrupler (to access region 1), the gain is given by:

$$G = \frac{V_{out}}{N_v \times V_{GMPP}} \cdot \frac{1}{2} \cdot \frac{1}{2}$$

According to *Tang et al.* [18], three possible roots for the variable x can be obtained when solving Equation 3.11. This results in six potential switching frequency solutions. Out of the six, four are either negative or imaginary and therefore not valid. Only a valid positive real root within the allowable LLC operating range can be used, making it the chosen switching frequency f_{sw}. Once this frequency is determined, the PV array's output current and power are initialized, and the algorithm begins operation using the PFM modulation technique.

3.3.4 PV Analysis Using Reconfigurable PFM Method

Next, similiar analysis done by [16] using the First Harmonic Approximation (FHA) equation is carried out to visualize the converter's operating region across the full PV power curve. The core idea is to determine the switching frequency f_{sw} required for the converter to match a specific power level. By assuming 100% efficiency the output power can be approximated to tbe the input power ($P_{PV} \approx P_{out}$) and hence the quality factor Q can be expressed in terms of the input power. Using this information, a gain curve can be plotted following process below.

The process is as follows:

- Input power is incrementally varied from $0W$ to $250W$, and for each value, the corresponding load resistance is calculated using the equation below, with the output voltage fixed at $120V$: $R_{load} = V_{out}^2/P_{PV}$.
- The graph is plotted the reflected resistance seen by the resonance tank is being calculated, $R_{ac} = \frac{8}{\pi^2}\frac{1}{n^2}\frac{1}{4}R_{load}$
- The quality factor can then be solve together with the $k = \frac{L_m}{L_r}$, $Q = \sqrt{\frac{L_r}{C_r}} / \frac{1}{R_{ac}}$
- Finally, the gain is then calculated using the derived Equation 3.11.

The above process is repeated for two additional configurations: the half-bridge with voltage doubler, which targets Region 3, and the full-bridge with voltage quadrupler, which targets Region

1. In each case, the gain equation is adjusted accordingly, as outlined in Section 3.3.3. The plotted graphs show that a lower quality factor Q, which occurs at lower power levels, leads to a higher voltage gain at the same switching frequency. This happens because lower power results in higher load resistance, increasing the reflected resistance and reducing Q. As a result, the gain curve shifts to the left. This trend is seen in all three graphs, where lower power conditions cause the gain curves to move leftward. In contrast, higher power leads to lower load resistance and a higher Q, which brings the curve towards right. These findings help explain how power levels affect converter performance and support better frequency planning for efficient power transfer.

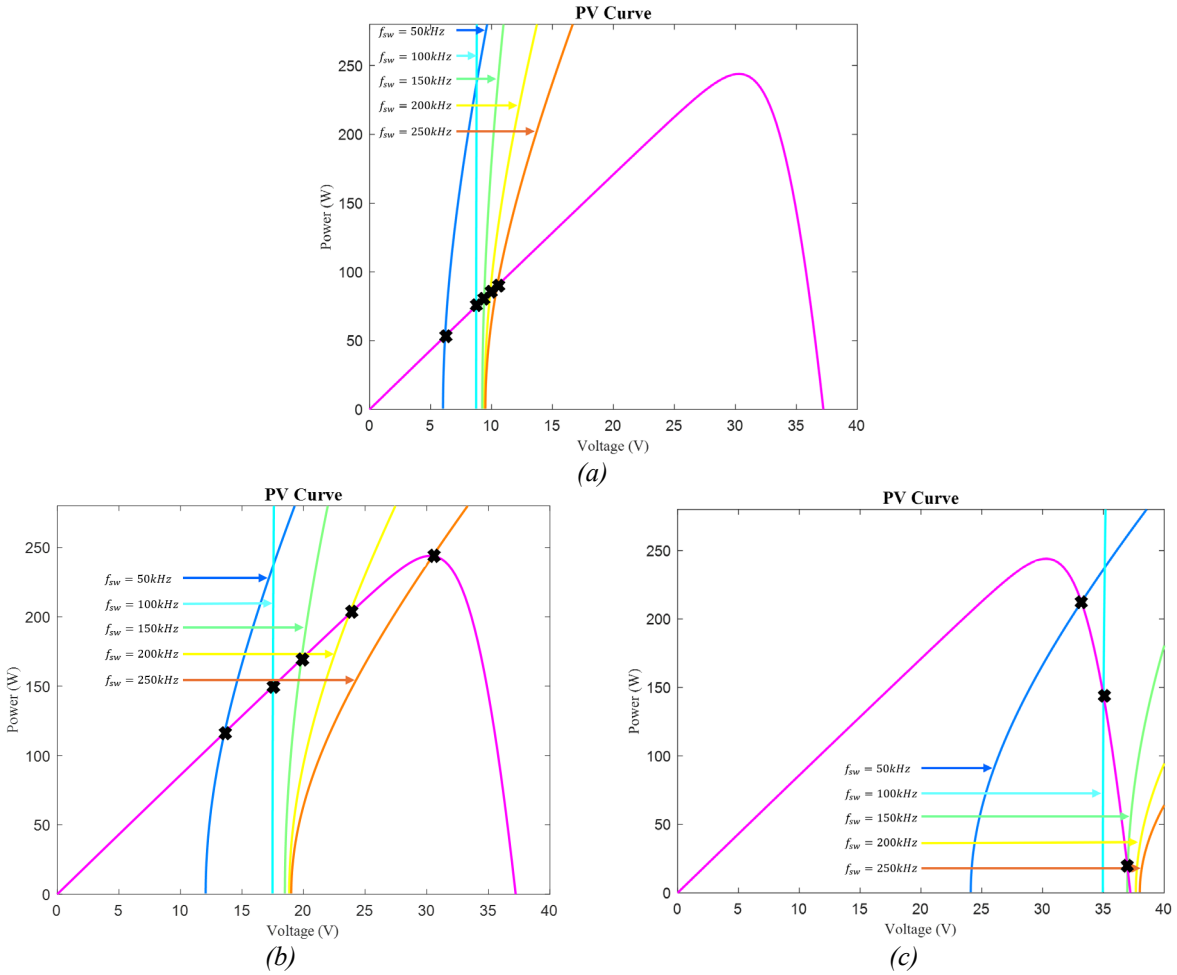


Figure 3.20. Comparison of input voltage V_{in} (blue) and sensor-measured voltage V_{sense} (purple) across varying switching frequencies, with V_{in} maintained at approximately 10V. (a) At f_{sw}=100kHz. (b) At f_{sw}=150kHz. (c) At f_{sw}=200kHz.

The black crosses on the gain-power plots indicate switching frequencies that allow the converter to operate at specific points on the PV curve. While the three methods reach their targeted regions, gaps remain where the converter cannot operate due to the limited gain range of the resonant tank. This range is insufficient for full MPPT coverage, especially given the non-optimized magnetic design of the prototype. A key observation is that the blue and cyan lines steepen and eventually reverse as power increases, signaling entry into the capacitive region. This leads to loss of zero-voltage switching, higher losses, and potential thermal stress. To overcome these limitations,

secondary side phase shift modulation (SS-PSM) proposed by *Wu et al.* [27] is a more robust method with extended gain range and preserved soft-switching operation.

3.3.5 Secondary Side Phase Shift Modulation

Using the secondary-side phase shift modulation (SS-PSM) technique introduced by *Wu et al.* [27], the output voltage can be controlled by adjusting the phase shift between the secondary-side switches and the primary-side switches, which operate with a fixed complementary PWM signal.

According to [27], the key advantage of this method is.

- Fixed Frequency Operation: Maintains soft-switching (ZVS) for all primary switches, minimizing switching losses.
- Effective Impedance Modulation: Phase shift indirectly controls the resonant tank's energy transfer efficiency, enabling voltage regulation without frequency variation.

To evaluate the accuracy and robustness of the gain Equation derived from by *Wu et al.* is tested.

$$G_{f_{sw}\approx f_r}(\phi, Q) = \frac{\sqrt{\pi Q} + \sqrt{-2(\cos\varphi)^2 + \pi Q + 2}}{\sqrt{\pi Q}(1 + \cos\varphi)}$$

Where

- $G = G_r(\varphi, Q)$ is the normalized voltage gain, treated as a constant.
- $Q = \frac{\sqrt{L_r/C_r}}{R_{ac}}$ is the normalized quality factor, also a constant.
- φ is the phase shift angle.

Then, the parameter of all the inductance and capacitance are put into MATLAB to calculate the obtained gain and the gain obtained is significantly higher compared to the one obtained for frequency modulation under the same loading condition when comparing to PFM method.

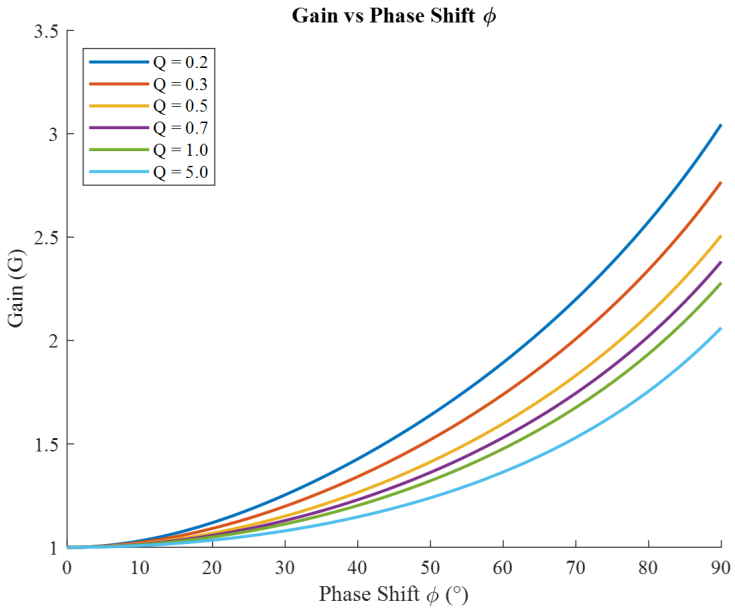


Figure 3.21. Gain curve under different loading factor with phase shift from 0 to 90 degree using SS-PSM modulation technique.

The same experiment is also carried out for SS-PSM method, following the procedure described in Section 3.3.4 for PV analysis using the reconfigurable PFM method. However, in this case, the gain equation is replaced with the one proposed by *Wu et al.* [27]. As shown in the plot, phase shift modulation allows the converter to achieve a wide gain range by adjusting the phase shift between 0 to 90 degrees. Therefore, next section will show the detail steps in solving this equation and be used as an initial approximation for hill-climbing algorithms or any other MPPT method.

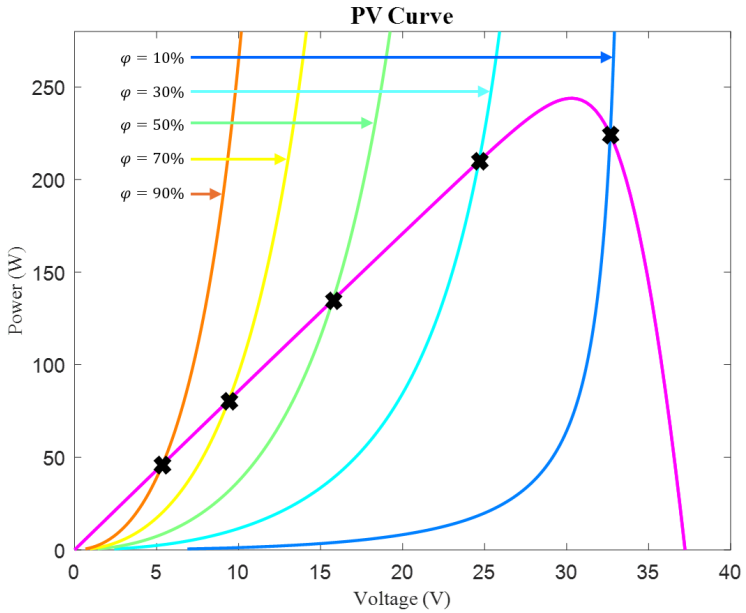


Figure 3.22. Gain curve under different loading factor with phase shift from 0 to 100 degree using SS-PSM modulation technique.

3.3.6 SS-PSM Initialization Method

Hence, this method is used and some derivation is being carried out to solve the Equation from [27] to obtain the initial phase shift required with the desired gain for the converter.

First the notation is simplified by setting, $c = \cos\varphi$. The Equation above is rearranged with the aim to solve c first, then only find the value of φ

$$G = \frac{\sqrt{\pi Q} + \sqrt{-2c^2 + \pi Q + 2}}{\sqrt{\pi Q}(1 + c)}$$

By multiplying $\sqrt{\pi Q}(1 + c)$ to both numerator and denominator.

$$G\sqrt{\pi Q}(1 + c) = \sqrt{\pi Q} + \sqrt{-2c^2 + \pi Q + 2}$$

By moving $\sqrt{\pi Q}$ to LHS of the Equation.

$$G\sqrt{\pi Q}(1 + c) - \sqrt{\pi Q} = \sqrt{-2c^2 + \pi Q + 2}$$

By factoring $\sqrt{\pi Q}$ out.

$$\sqrt{\pi Q}[G(1 + c) - 1] = \sqrt{-2c^2 + \pi Q + 2}$$

Then the Equation is squared and expand.

$$\pi Q[G(1 + c) - 1]^2 = -2c^2 + \pi Q + 2$$
$$\pi Q[G^2c^2 + 2G^2c - 2Gc + G^2 - 2G + 1] = -2c^2 + \pi Q + 2$$
$$\pi QG^2c^2 + 2\pi QG^2c - 2\pi QGc + \pi QG^2 - 2\pi QG + \pi Q + 2c^2 - \pi Q - 2 = 0$$

Then the equation can be solve using a quadratic equation.

$$Ac^2 + Bc + C = 0 \qquad \textit{Eq 3.12}$$

Where the coefficient for the quadratic equation used are.

- $A = \pi QG^2 + 2$
- $B = 2\pi Q(G^2 - G)$
- $C = \pi QG^2 - 2\pi QG - 2$

Then c is solved to get value within the range of -1 and 1.

$$c = \frac{-B \pm \sqrt{B^2 - 4AC}}{2A} \qquad \textit{Eq 3.13}$$

Finally, the phase angle can be calculated.

$$\varphi = \cos^{-1} c \qquad \textit{Eq 3.14}$$

There are two possible solutions for the phase angle. One of them is always $180°$, which falls outside the desired operating range and is therefore ignored and the second solution will be used for further analysis. Additionally, the phase shift of the converter is limited to a range between $0°$ and $90°$ to ensure safe and reliable operation.

3.4 Simulation Setup for Microgrid LLC Resonance Converter

This section describes the setup of the 3S1P PV configuration, the generation of PWM signals using both phase shift and PFM approaches, and the implementation of the LLC resonant converter. Two types of converter configurations are constructed. The first features a voltage doubler-quadrupler reconfigurable rectifier (VDQRR) on the secondary side, while the second integrates secondary-side phase shift modulation (SS-PSM). Both configurations are designed and the performance of both configurations will be compare and discuss in result section.

3.4.1 Setup of Photovoltaic System

The PV array is configured using a three-series, one-parallel (3S1P) arrangement, where each substring consists of 20 solar cells connected in series. To ensure protection under PSC, a bypass diode (D_1, D_2, D_3) is connected across each group of 20 cells. Additionally, a blocking diode is placed at the array output to prevent reverse current from flowing back into the modules. The complete electrical model of this 3×20 cell PV configuration, incorporating the series-connected cells, bypass diodes, and blocking diode, is illustrated below.

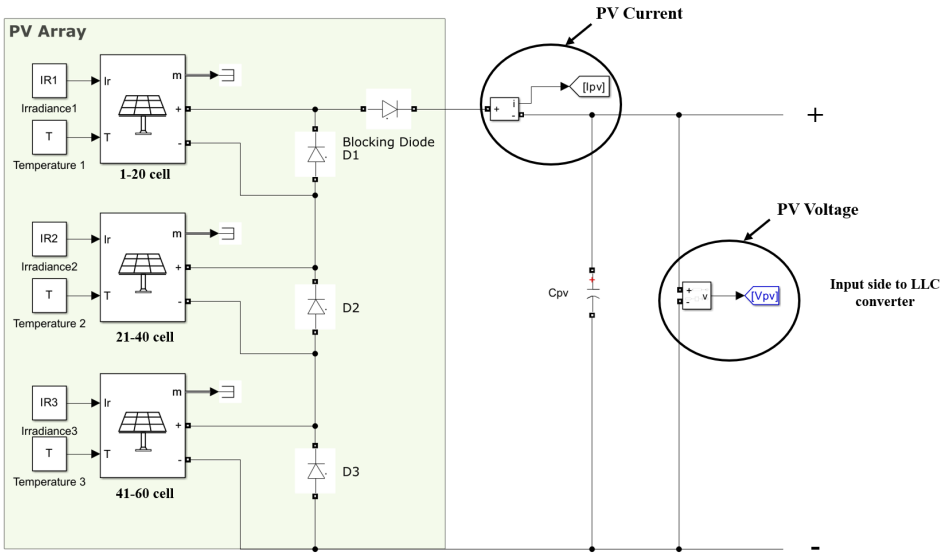


Figure 3.23. Simulink setup for the three-series, one-parallel (3S1P) PV configuration.

Each substring, as illustrated in Figure 3.23, is configured based on the specifications of the BLD250-290 60M photovoltaic array, detailed in Table 3.1. The PV panel used in this study is modeled after a $250W$ monocrystalline BLD250-290 60M array, which consists of three series substrings, each containing 20 solar cells. The open-circuit voltage (V_{oc}) is configured as $12.3V$, while the short-circuit current (I_{sc}) is $8.12A$. The maximum power point voltage (V_{mpp}) is set to $10.26V$, with a maximum power point current (I_{mpp}) of $8.07A$. The temperature coefficient for Voc is specified as $-0.36\%/°C$.

3.4.2 Generation of PWM signal

Figure 3.24 illustrates the SIMULINK setup used for generating PFM and SS-PSM signals, which control the switching behavior of the LLC resonant converter.

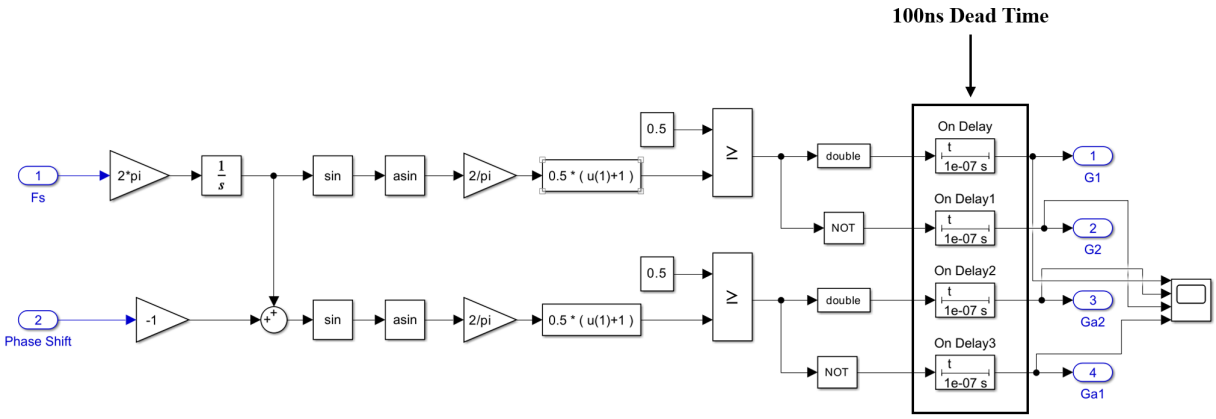


Figure 3.24. Simulink setup for the three-series, one-parallel (3S1P) PV configuration.

The first section of the setup focuses on PFM signal generation. A constant input F_S corresponding to the desired switching frequency f_{sw} is provided. This value remains fixed between sampling intervals, resulting in the creation of a ramp signal defined by the integration of $2\pi f_{sw}$.

$$\theta = \int 2\pi f_{sw} dt$$

This ramp signal is then processed through a sine function to produce a sinusoidal waveform, followed by an arcsine function to generate a periodic triangular-like waveform constrained between $-\frac{\pi}{2}$ and $\frac{\pi}{2}$. By scaling the output with $\frac{2}{\pi}$, the signal is normalized within the range of -1 to 1.

$$\sin^{-1}(\sin\theta)$$

Subsequently, the signal passes through a transformation block $0.5(u[1]+1)$, shifting the range to 0 and 1, effectively generating a 50% duty cycle PWM signal. Logical operations ensure proper waveform shaping, and a dead time of $100ns$ is introduced to prevent overlap during switching transitions, ensuring safe operation of the power switches.

For phase shift signal generation, the generation of referencing PWM signal are generated using the same approach as PFM. The only difference in generating SS-PSM signal is that a phase shift of φ is added to the integrated signal where input to the sine function will be out of phase, $\sin(2\pi f_{sw} + \varphi)$.

3.4.3 LLC Resonance Circuit using PFM

Figure 3.25 shows the complete SIMULINK setup of the LLC resonant converter integrated with a Voltage Doubler-Quadrupler Reconfigurable Rectifier (VDQRR) and output filter. The model includes a switching bridge with four MOSFETs (G_1 to G_4) driven by PWM signals with duty cycle of 50% for PFM, where the configuration allows selection between full-bridge and half-bridge operation by shorting Q_4 and opening Q_2. The AC output feeds into the LLC Resonance Tank consisting of L_r, C_r, and L_m. A transformer provides isolation, followed by the VDQRR, which switches between doubler and quadrupler modes. The output filter uses a capacitor (C_o) to smooth the DC voltage and the output voltage is clamped at $120V$, and the Output Load is modeled with R_{load} and R_{leak}. Measurement points are placed to monitor key voltages and currents for control and analysis.

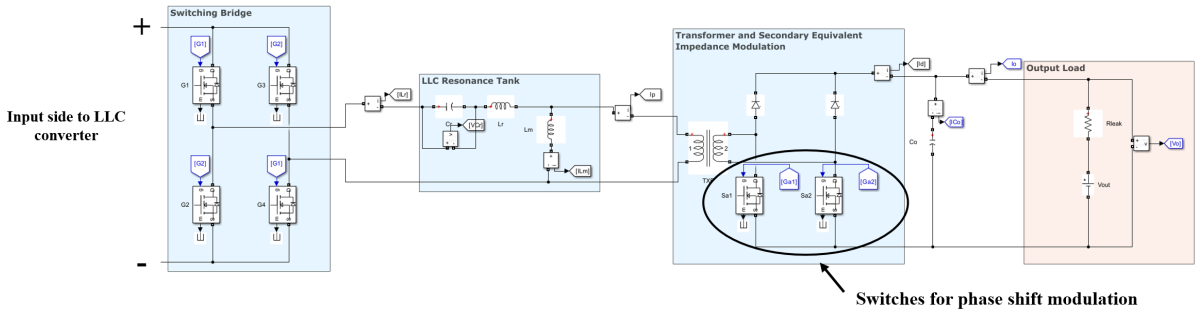


Figure 3.25. Simulink setup of LLC resonant converter using VDQRR circuit for PFM modulation method.

Figure 3.21 presents the block diagram of the proposed two-stage GMPP algorithm that will be updated every 10ms. The process begins with the green block, which measures the open-circuit voltage to calculate the normalization and denormalization parameters for the neural network. Next, the yellow block initiates the current sampling algorithm. Once sampling is complete, the neural network is called to predict the GMPP voltage and power. These predicted values are then used to initialize the purple block, which contains the P&O algorithm. The P&O algorithm refines the prediction by compensating for any errors and continues to track the GMPP under dynamic conditions.

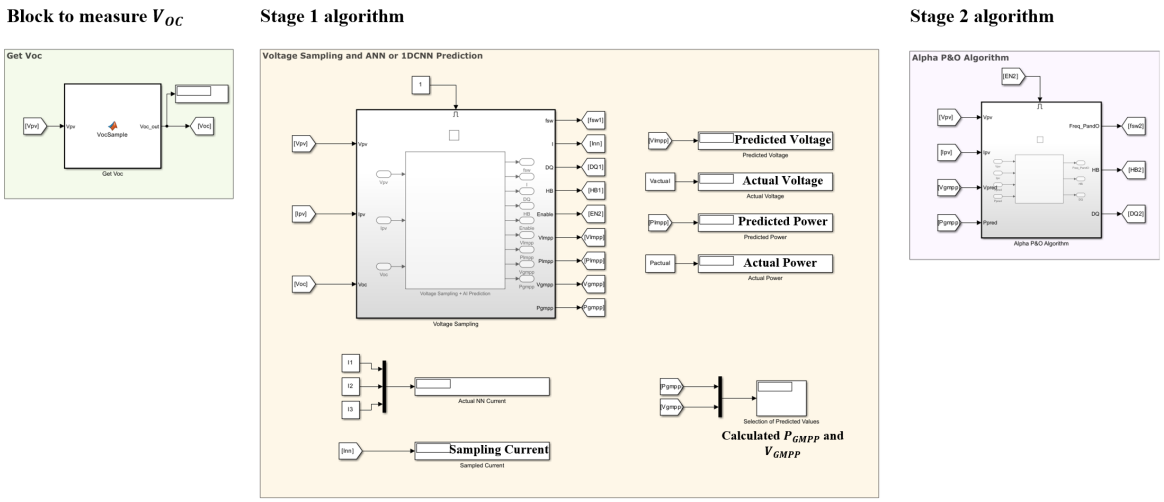


Figure 3.26. Block diagram for proposed two stage GMPP algorithm.

3.4.4 LLC Resonance Circuit using SS-PSM

Figure 3.27 illustrates the SIMULINK setup of the LLC resonant converter configured for SS-PSM. This setup, while similar to the PFM-based design, introduces two additional switches (G_{a1} and G_{a2}) on the secondary side to enable impedance modulation through adaptive phase shifting. The switching bridge on the primary side operates consistently at the resonant frequency, ensuring optimal soft-switching conditions. The AC output from the bridge feeds into the LLC resonance tank, consisting of L_r, C_r, and L_m, followed by a high-frequency transformer for isolation. Unlike the PFM configuration with a reconfigurable rectifier, this design adjusts the output by varying the phase shift between the primary and secondary switches. When the phase shift exceeds 1°, the effective gain increases, allowing dynamic control of the output voltage. The output is filtered by a capacitor and clamped at $120V$, with R_{load} and R_{leak} modeling the load characteristics. This SS-PSM approach provides an alternative modulation strategy, enhancing flexibility and efficiency in maximum power point tracking (MPPT) under varying operating conditions.

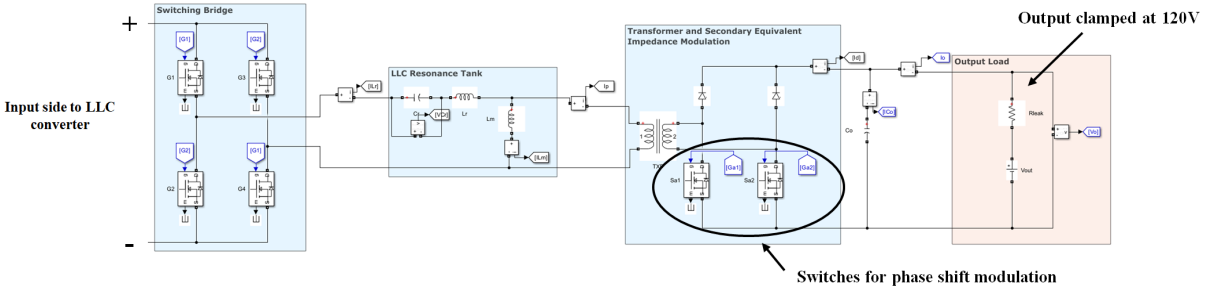


Figure 3.27. Simulink setup of LLC resonant converter for SS-PSM modulation method.

The overall block diagram of the proposed two-stage GMPP algorithm for phase shift modulation follows a similar flow to the PFM-based approach, with the primary differences being the modulation strategy and the initialization formula applied.

3.5 Setup of STM32-B-U585I-IOT02A Edge Computer

3.5.1 Connections to Edge Computer

First, the hardware connection of the STM32-B-U585I-IOT02A microcontroller shown in Figure 3.28 is summarized in Table 3.4 below by referring to the GPIO pinout connection from [69].

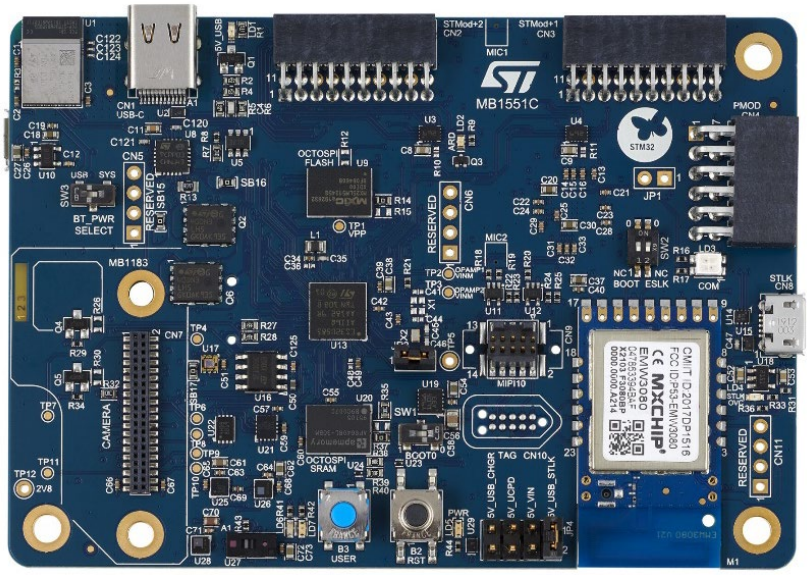


Figure 3.28. STM32-B-U585I-IOT02A microcontroller used for the two-stage GMPPT system.

Table 3.4. Summary of hardware connections of STM32-B-U585I-IOT02A microcontroller.

Peripheral	Description	Arduino Pin	STM32 Pin
ADC 1	To read analog signal from voltage feedback circuit range from $0 - 3.3V$.	$A0$	$PC0$
ADC 4	To read analog signal from current feedback circuit range from $0 - 3.3V$.	$A1$	$PC2$
Timer 1	Generated complementary PWM signal to control the gate driver circuit, enabling the switching of the GaN FET.	$A4, D9$	$PA7, PA8$

3.5.2 Configuration for USART STLINK

To transmit data to the computer, a serial communication protocol must be established. The STLink on the STM32 board facilitates data transmission via UART. Following the STLink MCU configuration from [70], the TX pin of STLink should be connected to the RX pin of STM32, while the RX pin of STLink should be connected to the TX pin of STM32. Additionally, pins PA9 and PA10 must be configured as TxD and RxD, respectively, with a baud rate of 115200 bits per second and a word length of 8 bits. The transmitted data can be viewed on a laptop or PC using the Serial Monitor in Arduino IDE or PuTTY by selecting the correct COM port and baud rate of 115200 bits per second.

3.5.3 Configuring ADC Pins

To measure real-time voltages from the voltage and current sensor board, the output pins must be read as ADC inputs. As shown in Figure 3.33, the V_{sense} and I_{sense} signals from the sensor board will be connected to pins PA0 and PA2, respectively. PA0 is configured as GPIO_Analog and ADC4_IN1, while PA2 is configured as GPIO_Analog and ADC1_IN3. Furthermore, the configuration involves setting both ADCs to asynchronous mode with a division factor of 1, a 12-bit resolution, and enabling regular scan conversion for ADC1.

The sampling frequency of both ADC1 and ADC4 can be determined using Equation 3.15, considering a sampling time of 1.5 cycles and a clock source of $16MHz$.

$$T_{conv} = (t_s + 12.5) \times T_{ADC_CLK} \qquad \textit{Eq 3.15}$$
$$= (1.5 + 12.5) \times \frac{1}{16 \times 10^6} = 0.875 \times 10^{-6}s$$

Where T_{conv} is the conversion time, t_s is the sampling time and T_ADC_{CLK} is the clock used by the ADC module. Hence, the sampling freqeuency for ADC1 and ADC4 is 1.15MHz.

3.5.4 Configuring Timer 1 for PFM PWM Generation

According to the schematic from [70], Timer 1 is an advanced-control timer that features complementary PWM outputs with programmable inserted dead time. It has a counter resolution of 16 bits and a prescaler factor that can be set to an integer value between 1 and 65536. Inside the project .ioc file, Timer 1 is enabled by selecting PWM generation for CH1 and CH1N under the Channel 1 toggle button. In the counter settings, Timer 1 is configured with a prescaler (PSC) value of 1 and a counter period (ARR) value of 1600, operating in count-up mode. The dead time can be adjusted under break and dead-time management by setting the BDTR register to 16. This configuration generates a complementary PWM signal at $100kHz$ with a dead time of $100ns$ where the choice of 100ns dead time is chosen by referring to [71], [72]. CH1 and CH1N correspond to the GPIO output pins PA7 and PA8, respectively. Hence, timer 1 will overflow when the CR value count until 1600 which corresponds to 10µs or 100kHz.

$$TMR1\ overflow = \frac{1}{f_{osc}} \times PSC \times (ARR - TMR1) \qquad \textit{Eq 3.16}$$
$$\frac{1}{T_{sw}} = \frac{1}{160 \times 10^6} \times 1 \times 1600 = \frac{1}{100 \times 10^3}$$

Two functions are used to dynamically adjust all parameters during runtime to generate PFM PWM waveform. Since the LLC converter operates using PFM, it must function within a frequency range of $70kHz$ to $200kHz$. The details of these boundary limits are explained in Section 3.3.4. The first function call check whether the switching frequency calculated by the GMPPT algorithm falls within the circuit's operating range of $70kHz$ to $200kHz$. If the frequency is within bounds, the second function is called to set the frequency. This second function calculates the ARR value using Equation 3.16, clears any previous update flags, and updates the period using the calculated ARR value. It then sets the duty cycle to 50% by configuring CCR1 to half of ARR and updates the dead time to 100ns using the BDTR_DTG register.

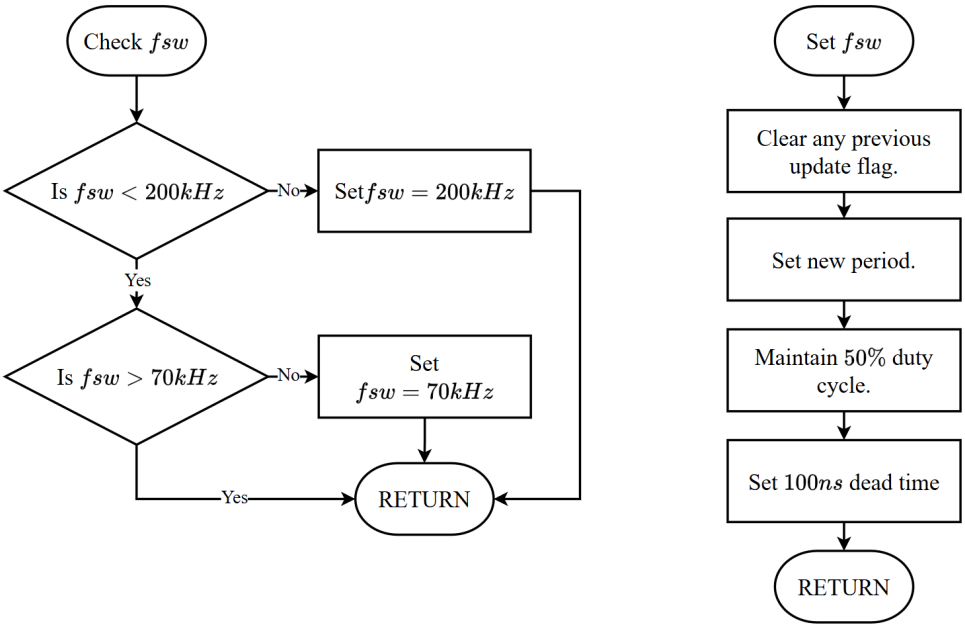


Figure 3.29. Flowchart for checking the lower and upper boundary switching frequency (Left) and setting switching frequency (Right).

3.5.5 Configuring Timer 16

Timer 16 is used to implement the sampling frequency of the GMPPT algorithm together with the collection of ADC data. Timer 16 will be set to overflow at $16kHz$, meaning a callback function (Interrupt) is triggered every $16kHz$. Within this function, five samples from both the voltage and current sensors will be collected. The median value of these samples is then calculated to filter out any outliers. Next, 160 data points from voltage and current sensors are collected and stored inside buffers with size of 160 at every $16kHz$ before executing the MPPT algorithm. This results in an effective MPPT sampling frequency of $f_{MPPT} = 16k/160 = 100Hz$. Before running the MPPT algorithm, the previously collected data is averaged, and the calculated V_{pv} and I_{pv} values are used as inputs for the algorithm.

Therefore, timer 16 will first need to be enable inside .ioc file. According to the documentation [73], Timer 16 is a general purpose timer. It has a counter resolution of 16 bits and a prescaler factor that can be set to an integer value between 1 and 65536. Inside the project .ioc file, timer 16 is enabled. In the counter settings, timer 16 is configured with a prescaler (PSC) value of 1 and a counter period (ARR) value of 10000, operating in count-up mode where it overflow every $16kHz$.

$$TMR16\ overflow = \frac{1}{f_{osc}} \times PSC \times (ARR - TMR16) \qquad \textit{Eq 3.17}$$

$$\frac{1}{T_{sw}} = \frac{1}{160 \times 10^6} \times 1 \times 10000 = \frac{1}{16000}$$

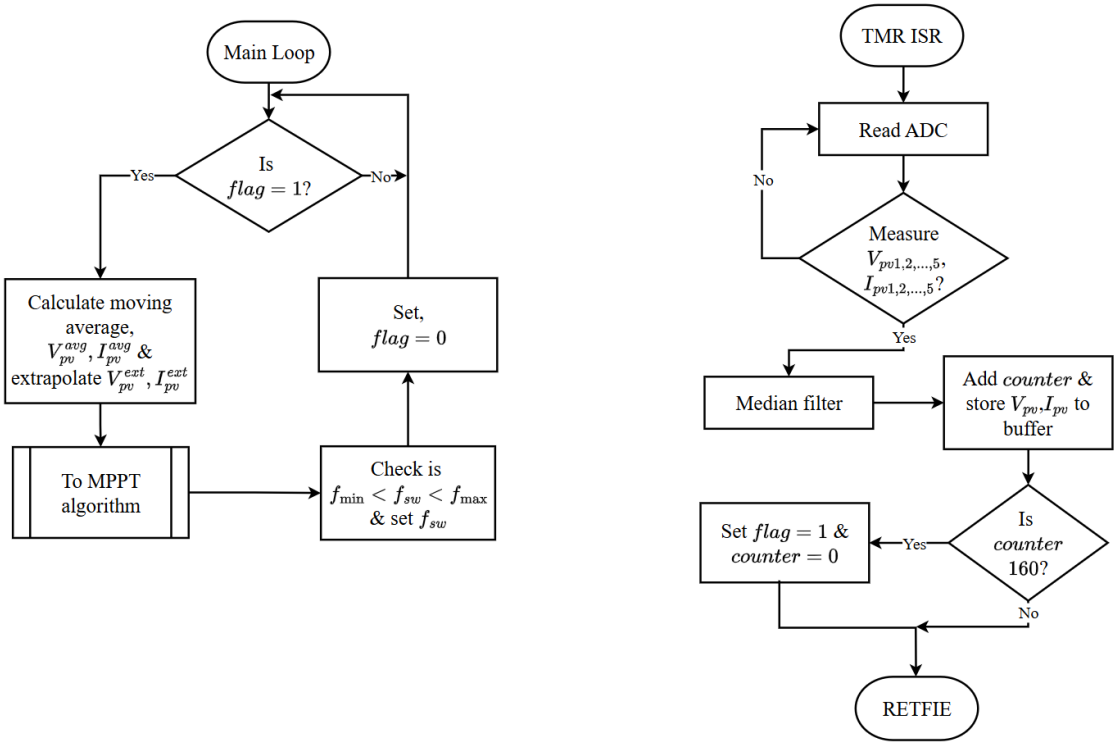


Figure 3.30. Flowchart for the main logic inside main function to execute the MPPT algorithm when the flag is set to 1 (Left) and callback function to collect 160 sensors data (Right).

3.5.6 X-CUBE-AI Installation and Setup

Before integrating the trained neural network model with real-time sensor data (I_1, I_2, I_3) into the firmware, X-CUBE-AI must be installed and configured within STM32CubeIDE. The STM32CubeIDE version 1.4.1 is first updated to the latest release to ensure compatibility, followed by the installation of X-CUBE-AI version 10.0.0 through the Embedded Software Packages Manager. Once installed, configuration begins by navigating to Middleware and Software Packages to enable the Artificial Intelligence core using the Software Packs Component Selector. A new network is then created where the TensorFlow Lite (TFLite) model trained as described in Section 3.2.7 and 3.2.8 is imported and assigned a custom name. Compression is set to "None" to prioritize accuracy since the model executes only once per cycle. The Analyze tool is used to verify that the model fits within the available Flash memory and RAM resources. After configuration, STM32CubeIDE automatically generates the required code and integrates the AI model into the *main.c* file. The generated C libraries include *ai_platform.h* along with the specific library for the AI model. Then, arrays are initialized for input and output, with three elements for the current values and six elements for the predicted outputs.

Two function is written where the first function *AI_Init* is used to initialize the AI model, handling any errors during initialization of the AI model. This function allocates activation buffers, creates the AI model instance, and assigns the input pointers. Another function *AI_Run* will then performs inference by processing the normalized current values and returning predicted voltages and powers for the local maximum points. If execution fails, appropriate error messages are displayed.

Within the main loop, real-time current values are stored, normalized, and passed to *AI_Run*. An output buffer is used to transmit the predicted voltages and powers via USART1 for analysis. Validation is performed by comparing these results with MATLAB SIMULINK outputs, confirming model accuracy as both environments produce identical predictions.

For resource usage, the ANN model occupies 12082 bytes of Flash and 2876 bytes of RAM. The CNN model requires more resources, using 19022 bytes of Flash and 4136 bytes of RAM, reflecting its higher complexity but improved performance.

3.6 Hardware Setup of LLC Resonance Converter

3.6.1 Setup of LLC Resonance Circuit

Figure 3.31 illustrates the complete setup of the two-stage GMPPT system. The LLC resonant converter is connected to a programmable DC power supply as the input source. An auxiliary $5V$ from a linear power supply is used to power the gate driver circuits and the voltage sensing circuit, while additional $\pm 15V$ supplies are dedicated to powering the current sensing circuits. A digital multimeter monitors the input voltage, and a programmable electronic load operates in constant voltage mode to clamp the output voltage at $120V$ during MPPT operation. Output power, voltage, and current are displayed directly on the electronic load's display. The STM32-U585I-IOT02A microcontroller will generate PWM signals with a 50% duty cycle, and during PFM operation, an FPGA is used to dynamically reconfigure the switching bridge into a half-bridge by opening switch $Q2$ and closing $Q4$. It is important to note that all hardware prototype validations are performed under PFM mode only.

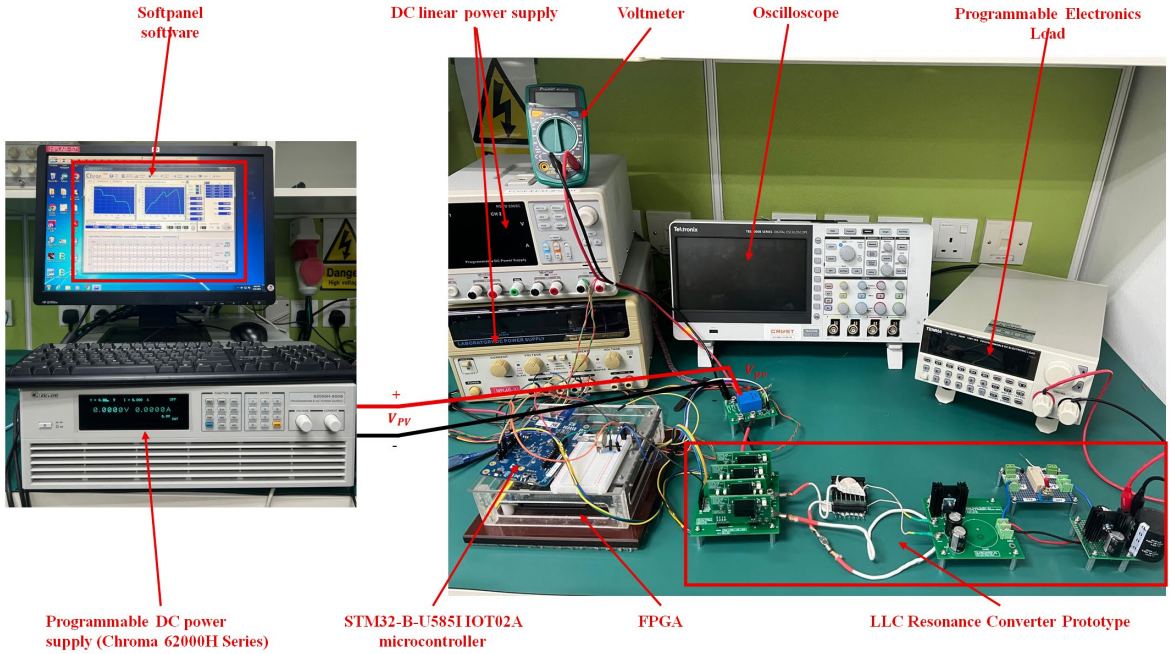


Figure 3.31. Hardware prototype setup of proposed two stage GMPPT method.

The detailed circuit diagram of the LLC resonant converter prototype is shown in Figure 3.32. The system consists of a switching network, a transformer, and a rectifier network. The input voltage for the converter is supplied by the programmable DC power supply, while an auxiliary $5V$ linear

power supply powers the gate driver circuits. The switching network is configured as a full-bridge using four GS61008T GaN FETs, each individually driven by its respective gate driver. The high-frequency transformer, with a turns ratio of 1: 5, steps up the waveform generated by the resonant tank, which consists of the resonant capacitor (C_r), resonant inductor (L_r), and magnetizing inductance (L_m). The stepped-up AC waveform is then rectified by the rectifier network. A symmetrical Voltage Doubler-Quadrupler Reconfigurable Rectifier (VDQRR) is employed. Furthermore, the VDQRR on the secondary side is equipped with a manual switch and an electronically controlled relay, enabling dynamic switching between doubler and quadrupler configurations through microcontroller commands.

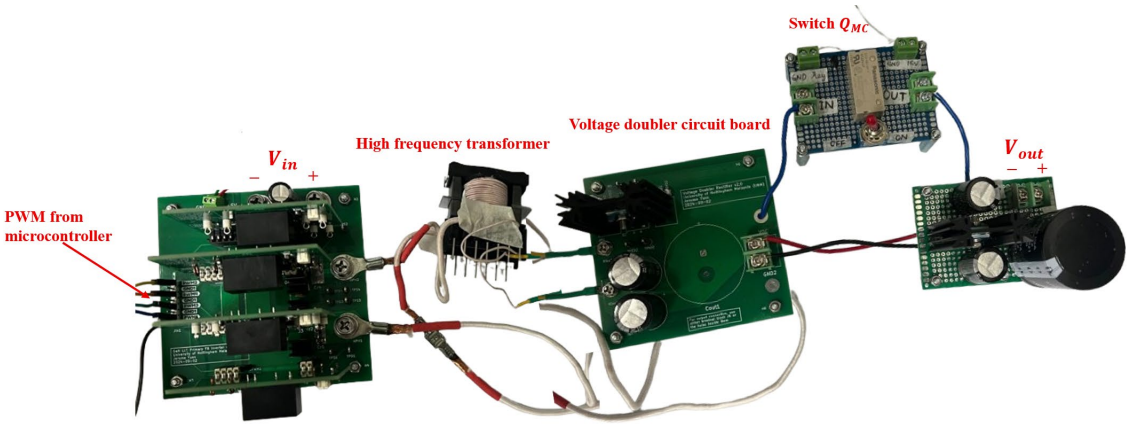


Figure 3.32. Circuit of LLC resonance converter prototype.

The detailed component values and circuit characteristics of the developed LLC resonant converter are summarized in Table 3.5. The prototype is designed for a maximum power rating of $250W$, supporting a maximum input voltage of $60V$ and a maximum input current of $10A$.

Table 3.5. Circuit operating conditions and parameters.

Description	Values
Rated power	$250W$
Maximum input voltage	$0 - 60V$
Maximum input current	$10A$
Resonant frequency, f_r	$100kHz$
Switching frequency, f_{sw}	$70kHz - 200kHz$
Resonance capacitance, C_r	$3\mu F$
Leakage inductance , L_r	$1\mu H$
Magnetizing inductance, L_m	$8\mu H$
Input capacitor, C_{pv}	$100\mu F$
Clamping capacitor, C_1, C_2, C_3, C_4	$33\mu F$
Output capacitance, C_o	$200\mu F$
Transformer turns ratio	1: 5
Switches	GS61008T GaN silicon power transistor

According to the given parameters (by the PhD student), the resonant tank components were the leakage inductance (L_r) and the magnetizing inductance (L_m) was measured to be $1\mu H$ and $8\mu H$ using the LCR meter. A resonant capacitance (C_r) of $3\mu F$ was selected based on the design requirements. Additionally, a $33\mu F$ input capacitor and a $200\mu F$ output capacitor were used as DC links for stabilizing the input and output stages, respectively. Based on the measured values, the calculated resonant frequency of the system is 91.89 kHz.

$$f_r = \frac{1}{2\pi\sqrt{L_r C_r}} \qquad \textit{Eq 3.18}$$

$$= \frac{1}{2\pi\sqrt{(1\times10^{-6})(3\times10^{-6})}} = 91888Hz \approx 100kHz$$

3.6.2 Calibration of Voltage and Current Sensor

The voltage and current sensors used in this setup as shown in Figure 3.33 below are designed to interface directly with the microcontroller's ADC inputs. The voltage sensor is capable of measuring input voltages from $0V$ to $60V$ with a sensitivity of $50mV/V$, while the current sensor measures currents from $0A$ to $10A$ with a sensitivity of $300mV/A$ The voltage sensor uses a differential amplifier based on the TLV2316IDR operational amplifier to scale the PV voltage from 0 to $60V$ into a 0 to $3V$ range suitable for ADC input. On the other hand, the current sensor is built with a LEM LA 25-NP transducer configured with three primary turns, converting 0 to $10A$ into a proportional voltage signal using a 100Ω resistor. This setup ensures accurate and stable measurements under varying PV operating conditions.

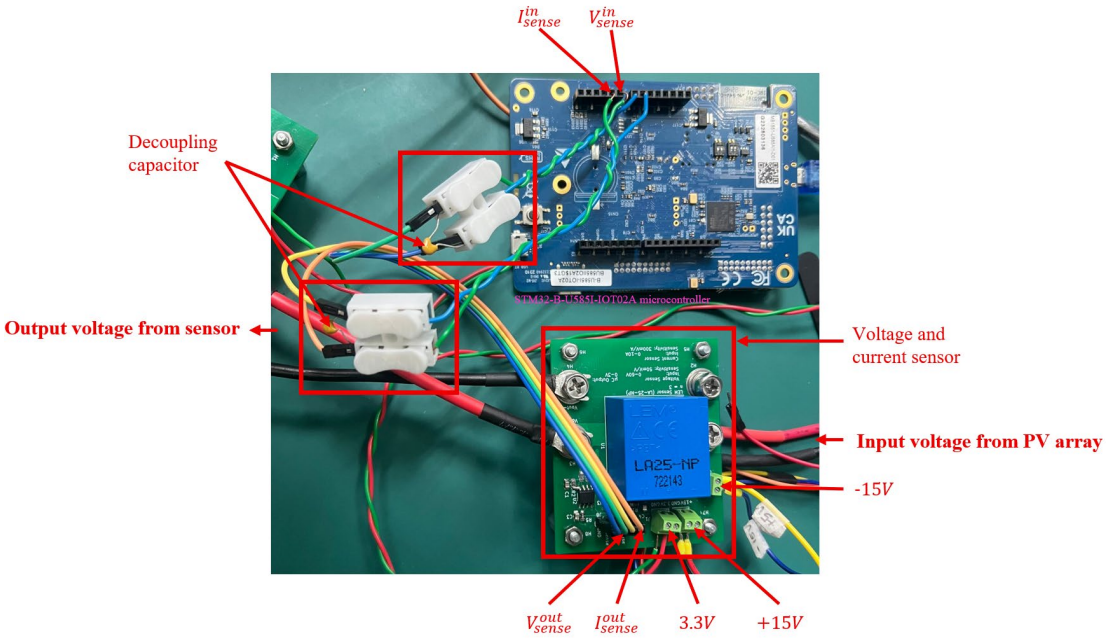
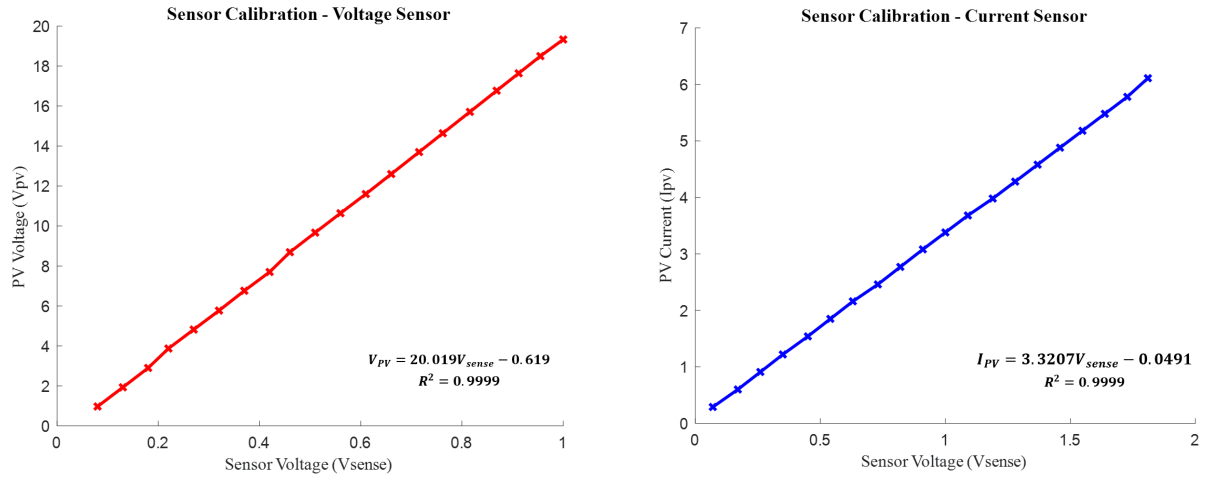


Figure 3.33. Circuit of gate driver circuit of primary board of LLC resonance converter prototype.

Due to practical non-idealities in the sensor circuits, slight deviations and offsets were observed in the readings. These inaccuracies arise from component tolerances as the sensors were custom-built

(as a summer research project) from discrete components. To improve measurement accuracy, a calibration process was performed.

The calibration was performed by gradually increasing the voltage of a linear power supply from $0V$ to $20V$. Corresponding sensor readings were collected from ADC1 for voltage sensing and ADC4 for current sensing. Since the sensor outputs exhibited ripple and were not purely DC, an oscilloscope was used to capture the waveform profiles for both input voltage and current. The root mean square (RMS) values obtained from the oscilloscope were considered the true reference values for calibration. Subsequently, the recorded sensor outputs and the corresponding oscilloscope RMS values were plotted, as illustrated in Figure 3.34. The plots reveal a strong linear relationship between the sensor readings and the true values, closely matching the theoretical sensitivities of the designed sensors. Linear regression was applied to derive correction equations for both the voltage and current sensors. After applying these correction factors, the adjusted sensor outputs showed significantly improved alignment with the true measurements, confirming the effectiveness of the calibration procedure.



(a) *(b)*

Figure 3.34. Sensor calibration results: (a) Voltage sensor calibration showing a sensitivity of approximately 50 mV/V with excellent linearity. (b) Current sensor calibration showing sensitivity of approximately 300 mV/A with excellent linearity.

Based on the collected data, the calibration equations were determined as $V_{PV} = 20.019\,V_{sense} - 0.619$ and $I_{PV} = 3.3207\,V_{sense} - 0.0491$. These results closely match the expected sensitivities of 50 mV/V (corresponding to a factor of 20) for voltage sensing and 300 mV/A (corresponding to a factor of 3.33) for current sensing, with small offsets of –0.619 V and –0.0491 V for voltage and current respectively. Both calibration curves achieved an R^2 value close to 1, confirming excellent linearity and measurement accuracy.

During calibration, it was observed that significant ripple appeared at the sensor outputs (explained in result Section 4.3.1) when the waveforms were probed and displayed on the oscilloscope. This ripple can be attributed to the finite bandwidth of the sensors, where the voltage sensor and current transducer bandwidths are lower than the high switching frequencies generated by the converter. To mitigate this issue, $33nF$ ceramic capacitors were added across the voltage and current sensor output terminals, placed as close as possible to the analog input pins [74]. This modification helped

suppress high-frequency noise components by providing a local low-pass filtering effect, improving measurement stability without introducing significant signal distortion as illustrated in Figure 3.33 [74].

3.6.3 Gate Driver Circuit

The gate driver circuit is required to turn the GS61008T GaN silicon power transistor [69] on and off. It steps up the control signal voltage from the microcontroller to match the desired gate-to-source voltage, V_{GS} level. Since a GaN FET behaves like a capacitive load, it requires significant current to rapidly charge and discharge quickly during switching events as the switching event need to occur between $70kHz$ to $200kHz$ as microcontroller alone may not supply enough current, leading to slower switching and increased switching losses. For GaN transistors, the optimal V_{GS} is typically between $5V$ to $6V$; operating below this range increases the drain-to-source on resistance, R_{DS} resulting in higher conduction losses.

An LED on the gate driver circuit lights up to indicate successful power-on, which occurs when a $5V$ supply is connected to power the gate driver circuits. Next, PWM signals generated by the STM32 microcontroller will be connected to the input of the gate driver circuit.

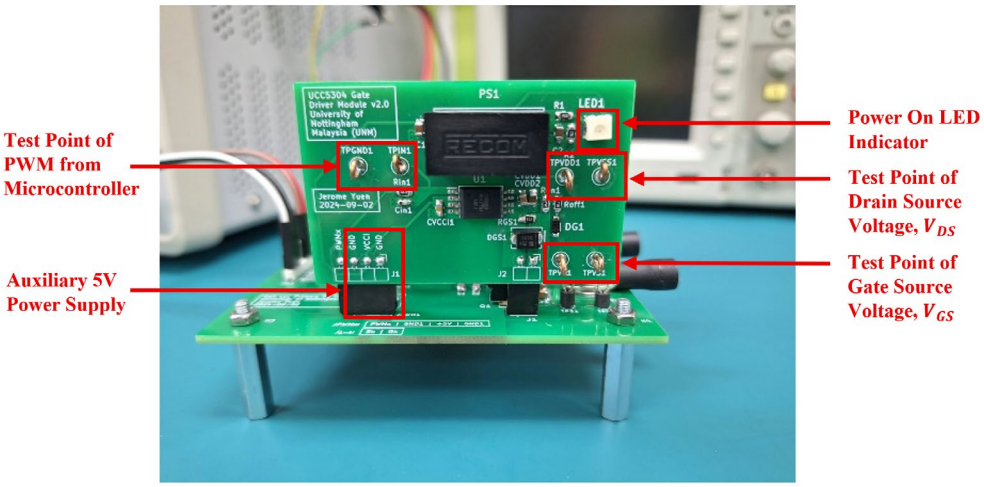


Figure 3.35. Circuit of gate driver circuit of primary board of LLC resonance converter prototype.

3.6.4 Chroma DC Power Supply

The 62000H-S Chroma DC power supply, a programmable power source, is used to emulate a photovoltaic (PV) panel under partial shading conditions by accurately replicating specific current-voltage (I-V) characteristics. The Chroma unit is controlled through its SoftPanel graphical user interface (GUI). In this project, predefined partial shading conditions (PSC) are generated using MATLAB Simulink to create specific I-V curves with multiple local maximum power points (LMPPs). These curves are then exported and uploaded into the SoftPanel software for use.

As shown in Figure 3.35, the uploaded PV characteristics contain three distinct LMPPs, clearly representing a realistic partial shading scenario. This allows the validation of the MPPT algorithm's capability to correctly identify the global maximum among multiple local peaks. The SoftPanel software interface also provides key measurement parameters such as open-circuit voltage (V_{oc}), short-circuit current (I_{sc}), maximum power point voltage (V_{mpp}), maximum power

point current (I_{mpp}), and maximum power (P_{mpp}). Furthermore, the interface displays real-time tracking values, including average voltage, current, and power, along with MPPT tracking efficiency. To initiate the simulation, the programmed PSC I-V curve is activated by pressing the "IV Run" button. This functionality enables flexible and repeatable testing of the MPPT algorithm under varying shading scenarios.

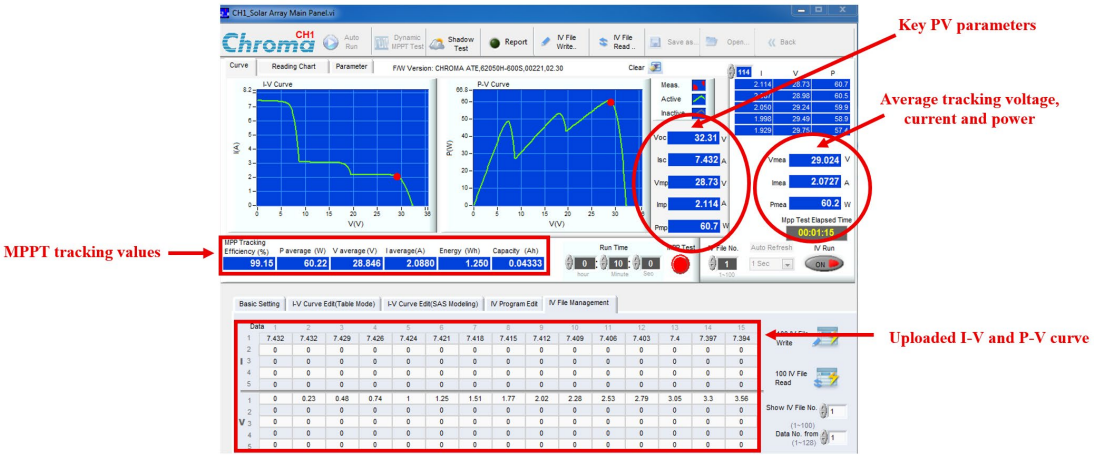


Figure 3.36. Softpanel GUI to control the 62000H-S Chroma DC power supply.

3.6.5 Electronics Load

The $300W$ TENMA 72-13210 programmable DC electronic load is used to clamp the output voltage at $120V$ by operating in constant voltage (CV) mode. In CV mode, the load emulates the behavior of a DC link in a grid-tied inverter, providing a stiff $120V$ bus that the MPPT stage expects in a solar power system. This setup enables testing of the MPPT algorithm without requiring an actual inverter or grid connection. Furthermore, the electronic load offers built-in protection by automatically increasing current draw to maintain $120V$, preventing overvoltage conditions that could damage the system. The setup using the chroma DC power supplt and electronics load allows realistic emulation of dynamic PV system operating conditions of the algorithm onto the hardware prototype without the need for an actual inverter or physical sunlight variations.

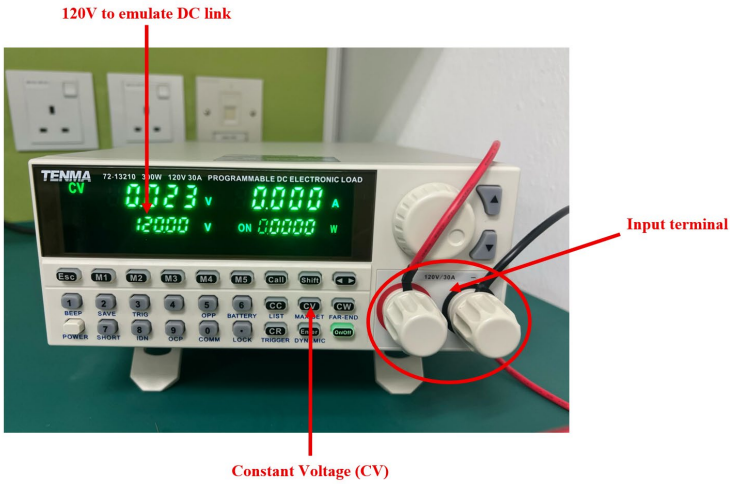


Figure 3.37. 300W TENMA 72-13210 programmable DC electronic load.

Chapter 4. Results and Discussion

This chapter begins with a quantitative analysis of the developed first-stage neural network algorithms, highlighting the strengths and trade-offs associated with each model. Subsequently, simulation results of the proposed two-stage GMPPT algorithm, utilizing both PFM and SS-PSM methods, are presented and compared. Experimental procedures, including open-loop tests, are conducted to evaluate the characteristics of the hardware prototype operating under the PFM method. Finally, the proposed algorithm is validated using the PFM method in conjunction with a programmable DC power supply, which simulates the behavior of a PV module.

4.1 Quantitative Analysis of Neural Network

4.1.1 Neural Network Accuracy Rate

The tracking accuracy of the two-stage GMPPT technique depends on accurately identifying the GMPP region from predicted LMPP candidates in the first stage. To evaluate this, the performance of the ANN and 1D-CNN models is assessed based on their ability to predict the GMPP. The Neural Network Accuracy Rate (NN_{AR}), defined as the percentage of correct predictions, is used as the evaluation metric. The models are tested using 1851 samples from a separate test set, which was not used during training. This ensures the results reflect the model's ability to generalize to unseen data.

$$NN_{AR} = \frac{Prediction_{correct}}{Prediction_{total}} \qquad \textit{Eq 4.1}$$

The results shown in Figure 4.1 are obtained by iterating through the 1851 test samples using the ANN and 1D-CNN models. During each iteration, the maximum among the predicted LMPPs is selected as the GMPP. The predicted GMPP from both models is then compared with the actual value (the labeled data). If the prediction GMPP is correct, a value of 1 is recorded; otherwise, a 0 is recorded. This process is repeated for all test samples. Once all predictions are completed, Equation 4.1 is used to compute the NN_{AR} value. As illustrated in Figure 4.1, the 1D-CNN model outperforms the ANN model by 3.6%, achieving an accuracy of 97.8% compared to 94.2% for the ANN. This suggests that the convolutional architecture of the 1D-CNN is better suited for identifying GMPP patterns.

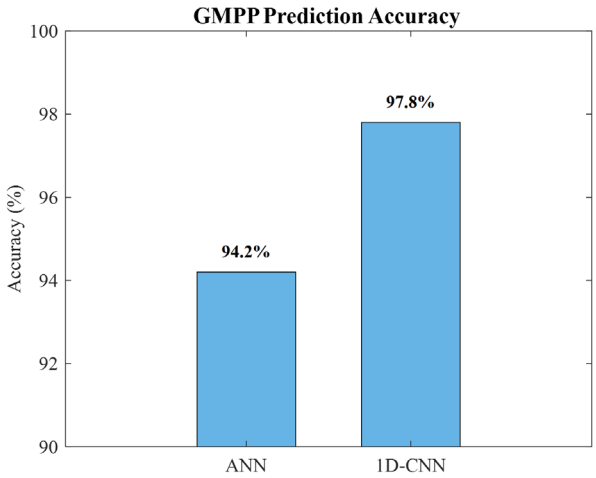
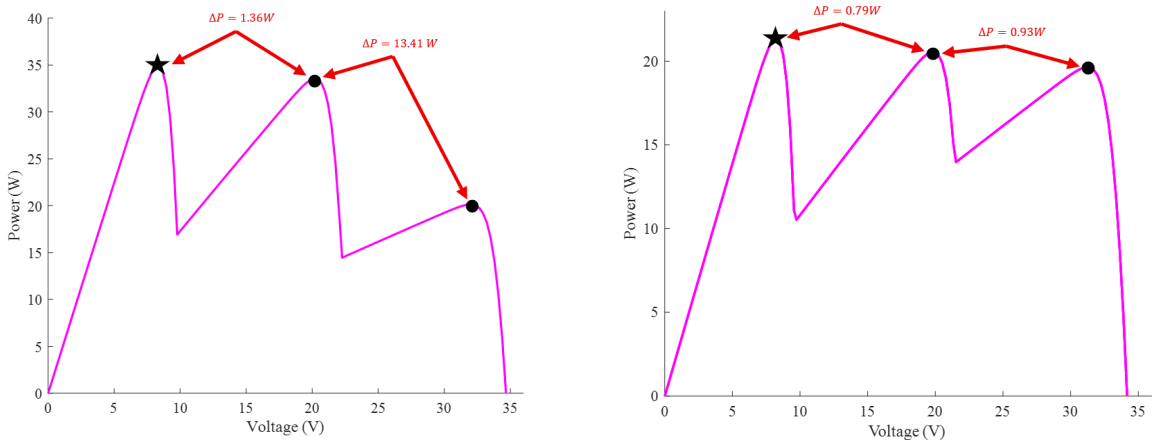


Figure 4.1. Comparison of GMPP prediction accuracy between ANN and 1D-CNN models.

After analysing the cases where the models failed to identify the GMPP, it was found that both the ANN and 1D-CNN models still exhibit occasional mispredictions under certain PSCs, as illustrated in the Figure below (an example). In case (a), the model fails to distinguish between the first and second peaks, which differ by only 1.36W (An error that is negligible in practical terms). Similarly, in case (b), all three peaks differ by less than 2W, making it difficult for the network to pinpoint the true GMPP.



(a) *(b)*

Figure 4.2. P-V curve under PSC condition with small ΔP between the local maximums, LMPPs where the neural network failed to predict the GMPP. (a) PSC condition with irradiance condition of 75, 200, 525W/m² at 25°C. (b) PSC condition with irradiance condition of 75, 125, 325W/m² at 25°C.

Both models performed well, with the ANN model achieving an NN_{AR} of 94.2% (1744/1851 correct predictions) and the 1D-CNN model showing higher accuracy at 97.8% (1810/1851 correct predictions). These results are comparable to [10], which reported 100% accuracy (252 out of 252 PSC samples), though the study used fewer samples with large irradiance changes and lacked comprehensive analysis. For high-accuracy needs, 1D-CNN is preferred, while ANN offers a balanced option between accuracy and computational efficiency.

4.1.2 Error Evaluation of LMPP Prediction

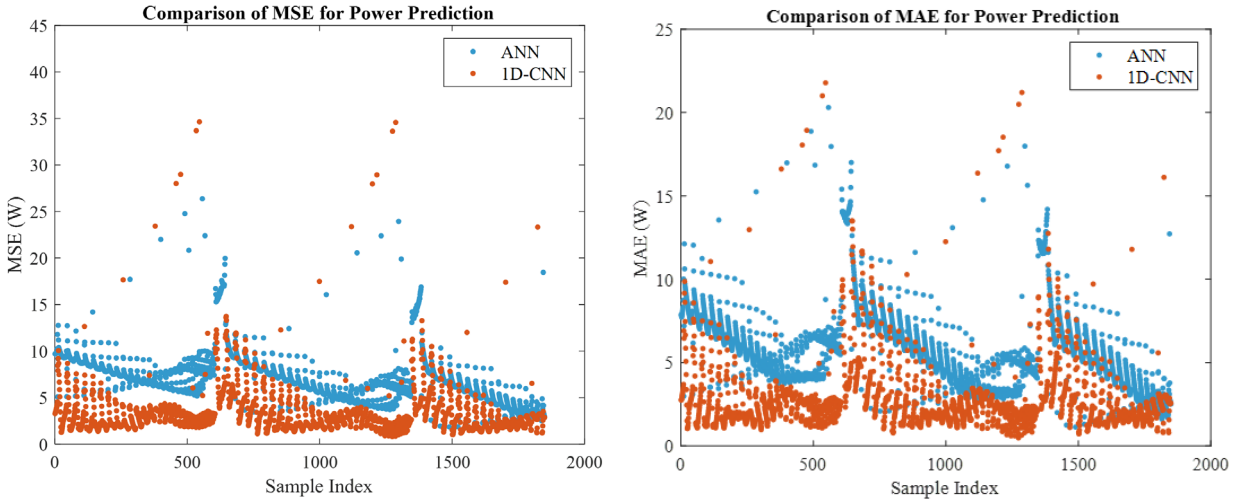
The second test evaluates the performance of the neural network architectures by comparing the predicted LMPPs with the actual values using mean squared error (MSE) and mean absolute error (MAE) as evaluation metrics. This test is conducted on the same 1851 test samples under various PSCs. A high MSE or MAE indicates significant deviation from the actual values, suggesting that the neural network may not be well-suited for accurately predicting LMPPs in such scenarios. MSE, which squares the errors, amplifies larger deviations, while MAE provides a more balanced view by treating all errors equally. Using both metrics offers complementary perspectives for assessing model performance.

$$MSE = \sqrt{\frac{\sum_{i=0}^{N=2}\left(P_{predict\ LMPP,i} - P_{real\ LMPP,i}\right)^2}{N}} \qquad \textit{Eq 4.2}$$

$$MAE = \frac{\sum_{i=0}^{N=2}\left|P_{predict\ LMPP,i} - P_{real\ LMPP,i}\right|}{N} \qquad \textit{Eq 4.3}$$

Figure 4.3 shows the result obtained by evaluating 1851 test samples using both the ANN and 1D-CNN models. For each sample, the MAE and MSE were calculated using Equations 4.2 and 4.3 by comparing the predicted LMPP values against the actual labeled data. It is important to note that the x-axis represents the test sample index; these samples are not sorted in any specific order, so no particular pattern along the x-axis should be interpreted.

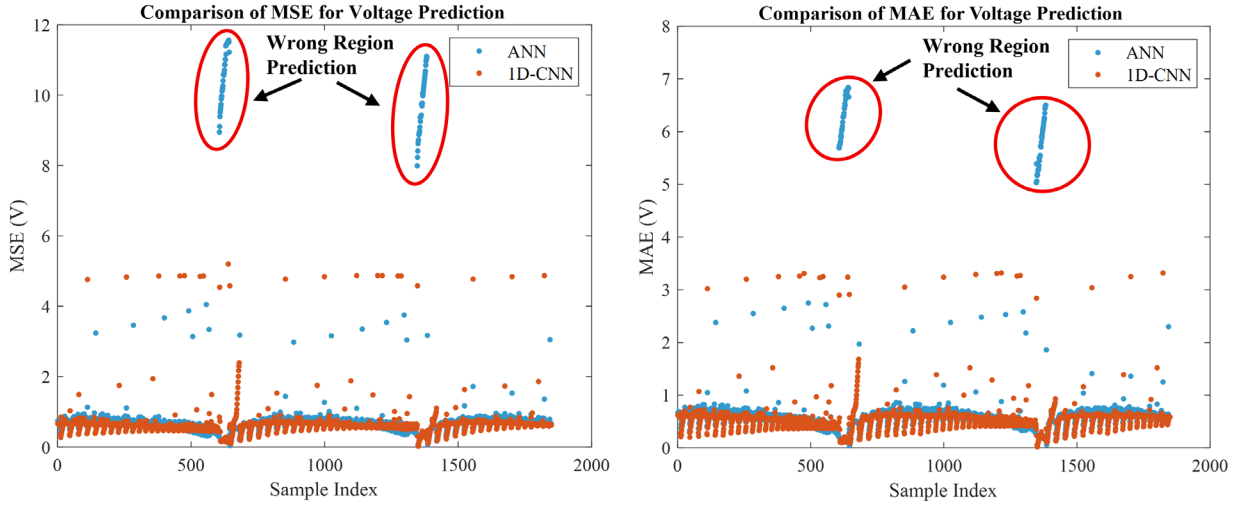
While both models show generally low error values, occasional spikes are observed where the network mispredicts one or more LMPPs. Overall, the 1D-CNN model demonstrates better performance particularly in power prediction compared to the ANN model.



(a) *(b)*

Figure 4.3. Comparison of power prediction errors between the ANN and 1D-CNN models. (a) Plot for MSE. (b) Plot for MAE.

As shown in Figure 4.4 below for voltage prediction, the 1D-CNN model achieves lower MAE and MSE values than the ANN model, consistent with the results observed in Experiment 1.



(a) *(b)*

Figure 4.4. Comparison of voltage prediction errors between the ANN and 1D-CNN models. (a) Plot for MSE. (b) Plot for MAE.

Notably, the graphs display sudden spikes labeled as “Wrong Region Prediction” for the ANN model. These spikes occur because the ANN incorrectly predicts PSC cases that contain only two peaks as shown in Figure 4.5. For instance, both the ANN and 1D-CNN correctly identify the GMPP, marked by a star. However, the ANN mistakenly predicts a second local peak in a region where only one local maximum power point (LMPP) actually exists. In contrast, the 1D-CNN accurately recognizes that no such local peak is present in that region.

Furthermore, the voltage predicted by the ANN in the third region lies very close to the first peak. This proximity increases the risk of misidentifying the GMPP if there are even slight fluctuations in the input current. This behavior indicates that the ANN’s predictions are more sensitive to minor input variations, making it less reliable in ambiguous scenarios. On the other hand, the 1D-CNN demonstrates better spatial understanding of the P–V curve, resulting in more stable and accurate predictions. This highlights the CNN’s superior ability to learn and interpret the overall shape of the P–V curve, rather than depending on memorized patterns. As a result, the CNN offers more robust and precise GMPP tracking, especially under varying partial shading conditions.

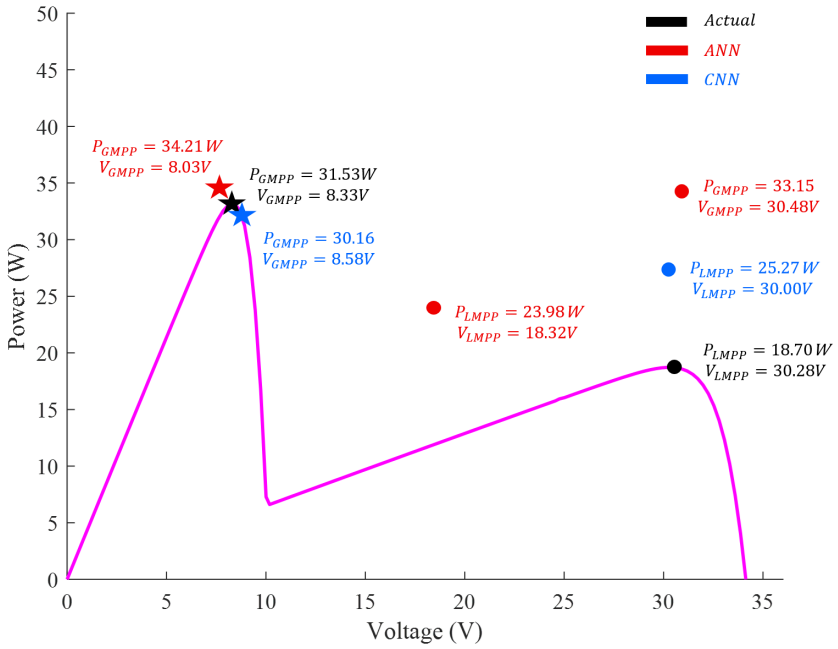


Figure 4.5. Comparison of GMPP and LMPP predictions by ANN and 1D-CNN under PSC under irradiance 75, 75, 500W/m² at 25°C. The actual P–V curve is shown in magenta. The GMPPs are marked with stars, while the LMPPs are indicated by circles. Actual values are represented in black, ANN predictions in red, and CNN predictions in blue.

4.2 Software Simulation for Two Stage GMPPT Algorithm

In the simulation phase, MATLAB SIMULINK is utilized to evaluate the performance of the proposed algorithm under varying temperature and irradiance condition using two modulation techniques, specifically PFM and SS-PSM with the MPPT algorithm that will be updated every 10ms. Therefore, the same testing conditions will be applied to both modulation techniques, with the results for PFM presented in Section 4.2.1 and SS-PSM method discussed in Section 4.2.2. The simulations were first conducted under constant temperature while varying the irradiance levels to 150, 200, and 850W/m², followed by 250, 700, and 1000W/m², and then 250, 350, and 850W/m² as shown in Figure 4.6.

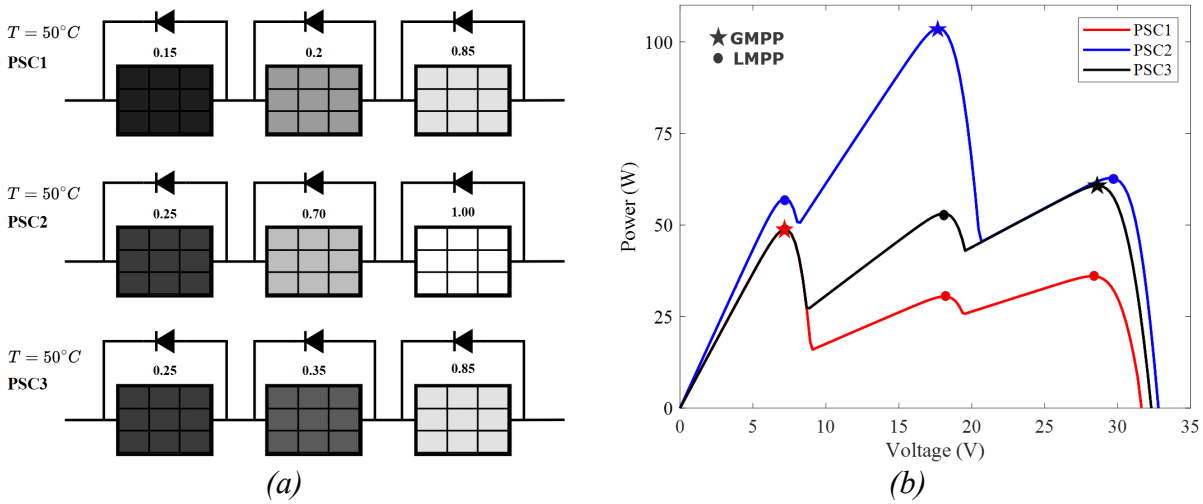


Figure 4.6. PSC with constant temperature where irradiance levels 150, 200, and 850W/m² (PSC1), 250, 700, and 1000W/m² (PSC2), and 250, 350, and 850W/m² (PSC3) (a) Setting for the 3S1P PV panel (b) P-V curve.

In the next stage, the irradiance was fixed at 200, 300, and 450W/m², while the temperature was varied to 25°C, 50°C, and 65°C as shown in Figure 4.7 to further evaluate the robustness of the algorithm under different thermal conditions.

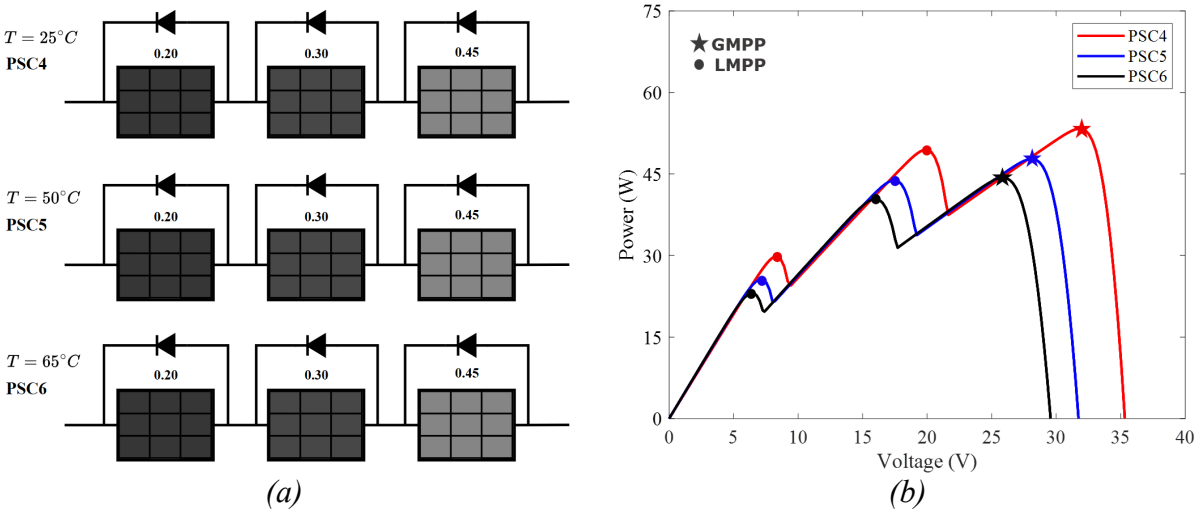


Figure 4.7. PSC with constant temperature where irradiance levels 150, 200, and 850W/m² (PSC4), 250, 700, and 1000W/m² (PSC5), and 250, 350, and 850W/m² (PSC6). (a) Setting for the 3S1P PV panel (b) P-V curve.

Figures 4.8 to 4.13(a) illustrate the system's dynamic response during the MPPT process. The first subplot shows the power waveform, followed by the voltage waveform, and finally the switching frequency variation. Figures 4.8 to 4.13(b) present the corresponding P–V curves under partial shading conditions, where black annotated points mark the actual local and global maximum power points (MPPs), while the local MPPs predicted by the 1D-CNN model are highlighted in a brownish-gold color. Circular markers indicate the local maxima, while the star symbol represents the GMPP.

The MPPT algorithm begins by measuring the open-circuit voltage V_{oc} (labelled with 1 in top plot of Figures 4.8 to 4.13(a)), followed by the collection of three current sampling points for neural network prediction. These points are sampled within a $\pm 2.1V$ threshold centered around $0.4 \times V_{oc}/3$, $0.4 + V_{oc}/3$, and $0.4 + 2V_{oc}/3$ respectively. Upon completing the first stage, the second stage is initiated by selecting the maximum predicted power and its corresponding voltage from the 1D-CNN model outputs to initialize the P&O algorithm.

Furthermore, the overall performance of the algorithm is evaluated using several key metrics, which are described below.

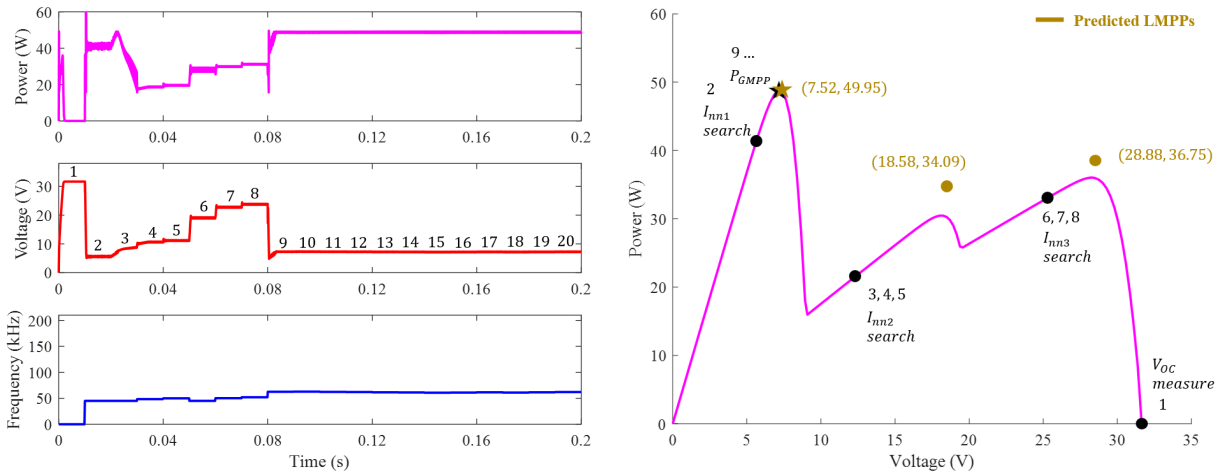
- The Total Current Sampling Points (TC_{SP}) metric represents the total time taken to complete the initial current sampling process during the first stage of the algorithm.
- The Tracking Time (TT) metric is defined as the total duration from the initial open-circuit voltage V_{oc} measurement until the P&O algorithm converges at the GMPP and begins to oscillate around it.
- The NN_{VP} and NN_{PP} metrics represent the predicted voltage and predicted power from the neural network, with their prediction accuracy evaluated using Equation 4.4 and 4.5.
- Tracking Average (TA) is evaluated to assess the P&O algorithm's ability to maintain the average harvested power relative to the true maximum power during steady-state operation.

$$NN_{PP} = 100\% - \frac{|P_{pred} - P_{actual}|}{P_{actual}} \times 100\% \qquad \textit{Eq 4.4}$$

$$NN_{VP} = 100\% - \frac{|V_{pred} - V_{actual}|}{V_{actual}} \times 100\% \qquad \textit{Eq 4.5}$$

4.2.1 PFM Modulation Techinique

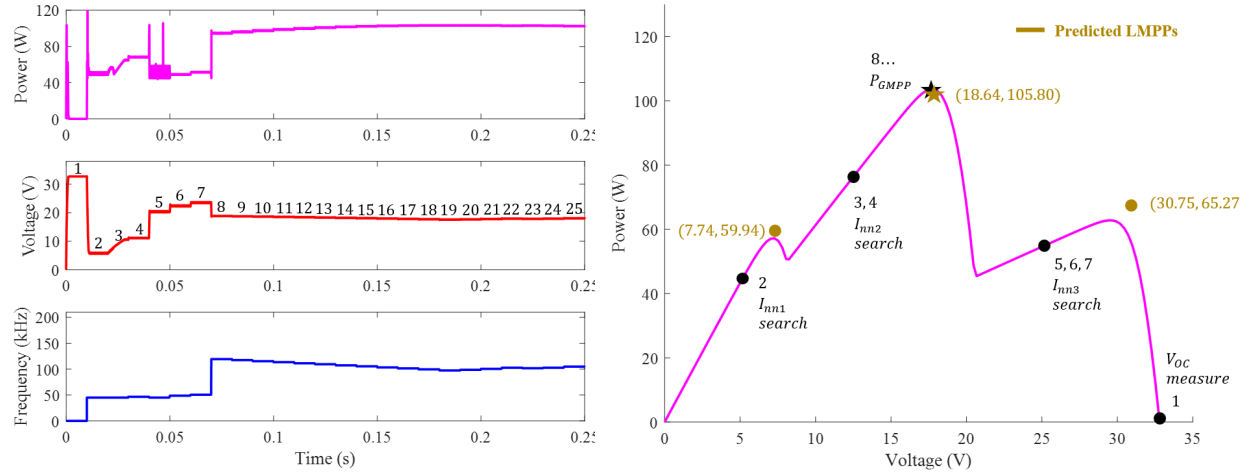
As illustrated in Figure 4.8, the proposed algorithm successfully tracks the GMPP under PSC1 with irradiance levels of 150, 200, and 850W/m² at a temperature of 50°C. The simulation results show that seven iterations are required to collect the current samples for the 1D-CNN input with Total Current Sampling Points (TC_{SP}) of 0.07s. The predicted GMPP is located at 7.52V and 49.95W (corresponding to region 1), closely matching the actual GMPP at 7.22V and 48.79W, achieving a neural network voltage prediction accuracy (NN_{VP}) of 95.84% and a neural network power prediction accuracy (NN_{PP}) of 97.62%. Finally, the system achieves an average output power of 48.70W compared to the theoretical maximum of 48.79W, resulting in tracking average (TA) of 99.82% and a total tracking time (TT) of 0.09s.



(a) *(b)*

Figure 4.8. Software simulation for proposed method under PSC1 with irradiance levels of 150, 200, and 850W/m² at a temperature of 50 °C. (a) The tracking waveform shows power (purple), voltage (red), and switching current (blue). (b) Corresponding P-V curve.

For PSC 2, where the irradiance levels are set at 250, 700, and 100W/m² at a temperature of 50°C, the algorithm requires six iterations to collect current samples for the 1D-CNN input, resulting in a TC_{SP} of 0.06s. The predicted GMPP is located at 18.64V and 105.80W (corresponding to region 2), closely matching the actual GMPP at 17.73V and 103.7W, achieving a NN_{VP} of 94.87% and a NN_{PP} of 97.97%. During steady-state operation, the system delivers an average power of 102.77W, yielding a TA of 99.10%, with a TT of 0.17s.

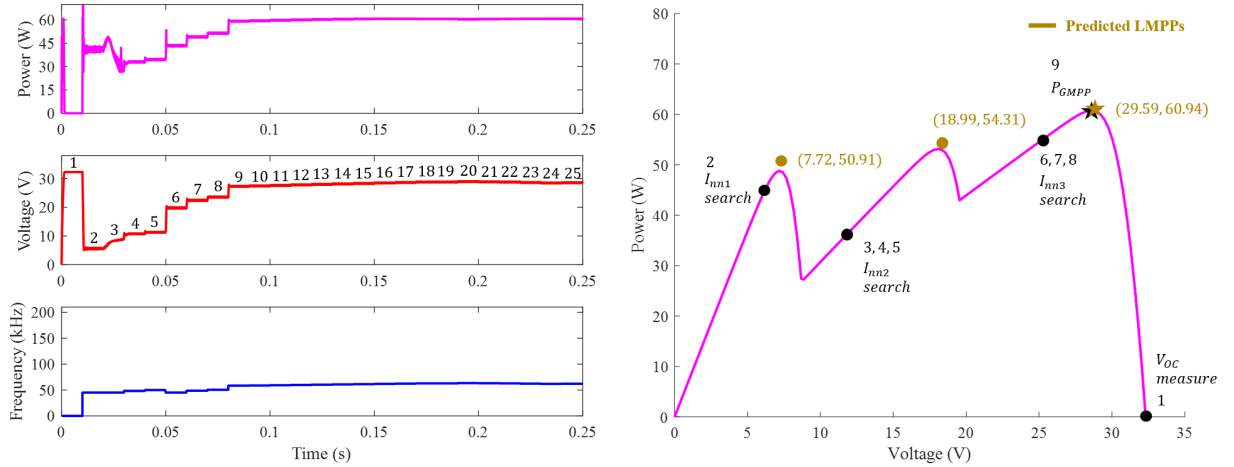


(a) *(b)*

Figure 4.9. Software simulation for proposed method under PSC2 with irradiance levels of 250, 700, and 1000W/m² at a temperature of 50 °C. (a) The tracking waveform shows power (purple), voltage (red), and switching current (blue). (b) Corresponding P-V curve.

For PSC 3, where the irradiance levels are set at 250, 350, and 850W/m² at a temperature of 50°C, the algorithm requires seven iterations to collect current samples for the 1D-CNN input, resulting in a TC_{SP} of 0.07s. The predicted GMPP is located at 29.59V and 60.94W (corresponding to region 3), closely matching the actual GMPP at 28.62V and 60.76W, achieving a NN_{VP} of 96.61% and a

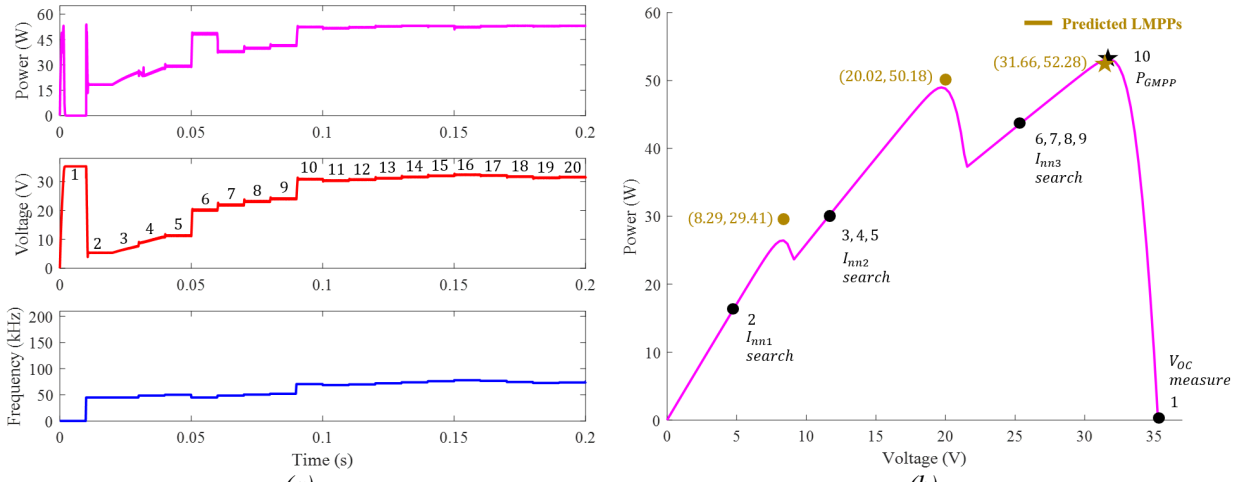
NNPP of 99.70%. During steady-state operation, the system delivers an average power of 60.64W, yielding a TA of 99.80%, with a TT of 0.15s.



(a) *(b)*

Figure 4.10. Software simulation for proposed method under PSC3 with irradiance levels of 250, 350, and 850W/m² at a temperature of 50 °C. (a) The tracking waveform shows power (purple), voltage (red), and switching current (blue). (b) Corresponding P-V curve.

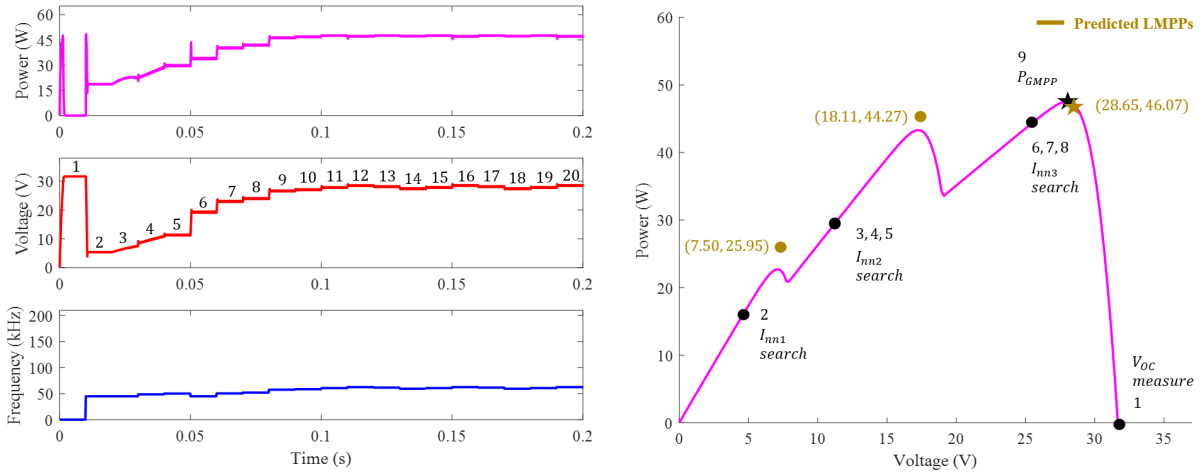
To evaluate the algorithm's robustness under different temperatures, Figures 4.11 to 4.13 present its performance at irradiance levels of 200, 300, and 400W/m², tested at 25°C, 50°C, and 65°C. For PSC 4 at 25°C, the algorithm requires eight iterations to collect current samples for the 1D-CNN input, resulting in a TCSP of 0.08s. The predicted GMPP is 31.66V and 52.28W (region 3), closely matching the actual GMPP of 31.72V and 53.14W, achieving NNVP of 99.81% voltage and NNPP of 98.38%. In steady-state, the system delivers an average power of 53.08W with TA of 99.89% and TT of 0.13s.



(a) *(b)*

Figure 4.11. Software simulation for proposed method under PSC4 with irradiance levels of 200, 300, and 450W/m² at a temperature of 25°C. (a) The tracking waveform shows power (purple), voltage (red), and switching current (blue). (b) Corresponding P-V curve.

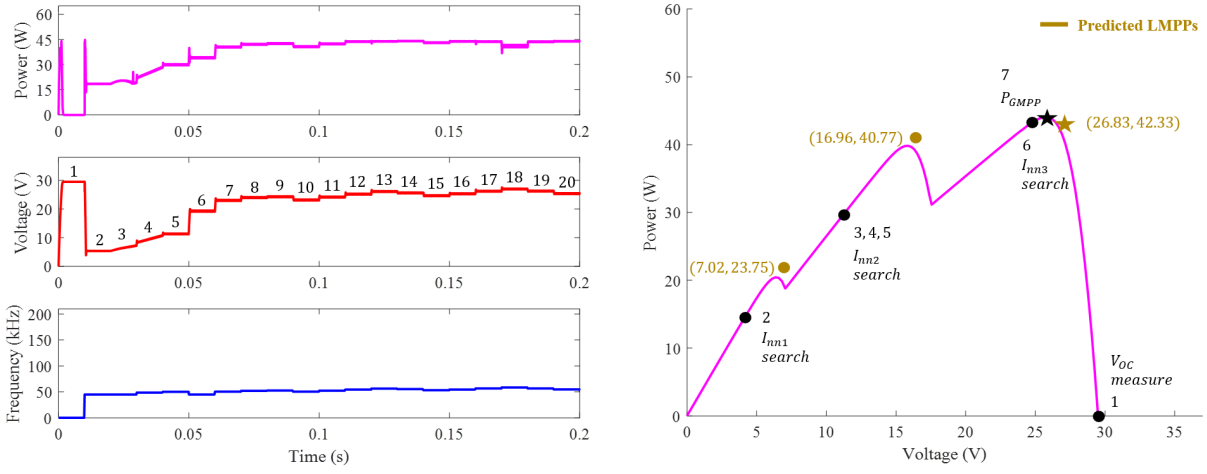
For temperature of 50°C for PSC 5, algorithm requires seven iterations to collect current samples for the 1D-CNN input, resulting in a TC_{SP} of 0.07s. The predicted GMPP is 28.65V and 46.07W (region 3), closely matching the actual GMPP of 28V and 47.55W, achieving NN_{VP} of 97.68% and NN_{PP} of 96.89%. In steady-state, the system delivers an average power of 47.53W with TA of 99.96% and TT of 0.10s.



(a) *(b)*

Figure 4.12. Software simulation for proposed method under PSC5 with irradiance levels of 200, 300, and 400W/m² at a temperature of 50 °C. (a) The tracking waveform shows power (purple), voltage (red), and switching current (blue). (b) Corresponding P-V curve.

For temperature of 65°C for PSC 5, algorithm requires five iterations to collect current samples for the 1D-CNN input, resulting in a TC_{SP} of 0.05s The predicted GMPP is 26.83V and 42.33W (region 3), closely matching the actual GMPP of 25.76V and 44.08W, achieving NN_{VP} of 95.85% and NN_{PP} of 96.03%. In steady-state, the system delivers an average power of 43.72W with a TA of 99.18% and TT of 0.13s.



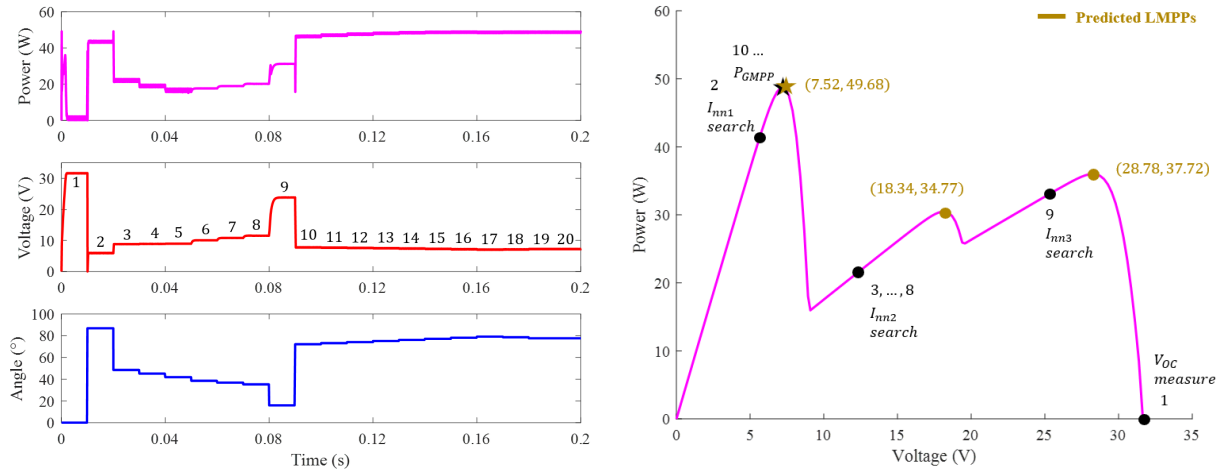
(a) *(b)*

Figure 4.13. Software simulation for proposed method under PSC6 with irradiance levels of 200, 300, and 400W/m² at a temperature of 65°C. (a) The tracking waveform shows power (purple), voltage (red), and switching current (blue). (b) Corresponding P-V curve.

4.2.2 SS-PSM Modulation Techinique

The same evaluation metrics and PSC conditions are applied to the SS-PSM method, with the only difference being its modulation strategy: a fixed switching frequency is used while the phase shift of the secondary switches is varied relative to the primary switches.

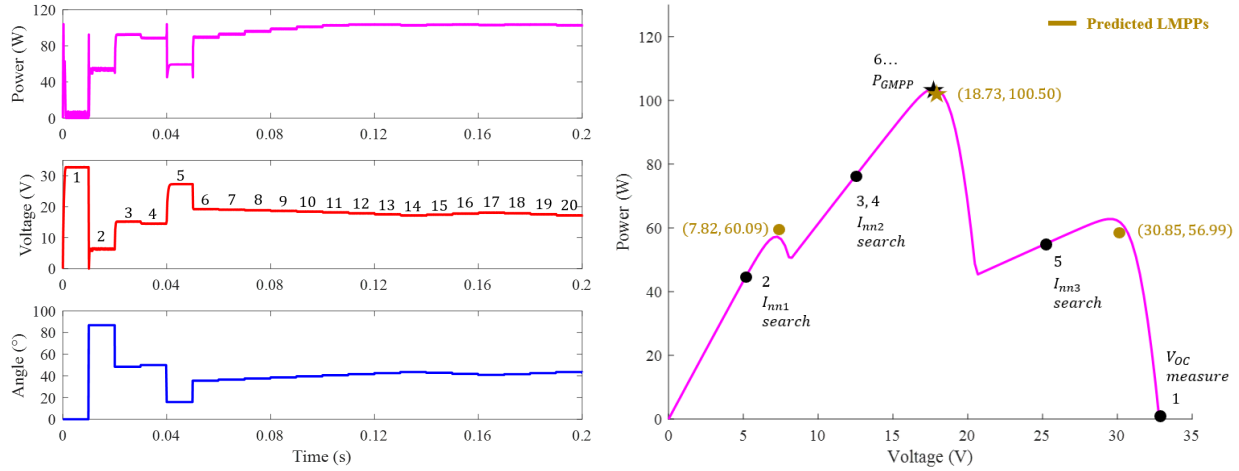
For irradiance levels of 150, 200, and 850W/m² at a temperature of 50°C, the algorithm requires eight iterations to collect current samples for the 1D-CNN input, resulting in a TC_{SP} of 0.08s. The predicted GMPP is located at 7.52V and 49.68W (corresponding to region 1), closely matching the actual GMPP at 7.22V and 48.79W, achieving NN_{VP} of 95.84% and NN_{PP} of 98.18%. Finally, the system achieves an average output power of 48.76W compared to the theoretical maximum of 48.79W, resulting in TA of 99.94% and TT of 0.15s.



(a) *(b)*

Figure 4.14. Software simulation for proposed method under PSC1 with irradiance levels of 150, 200, and 850W/m² at a temperature of 50 °C. (a) The tracking waveform shows power (purple), voltage (red), and switching current (blue). (b) Corresponding P-V curve.

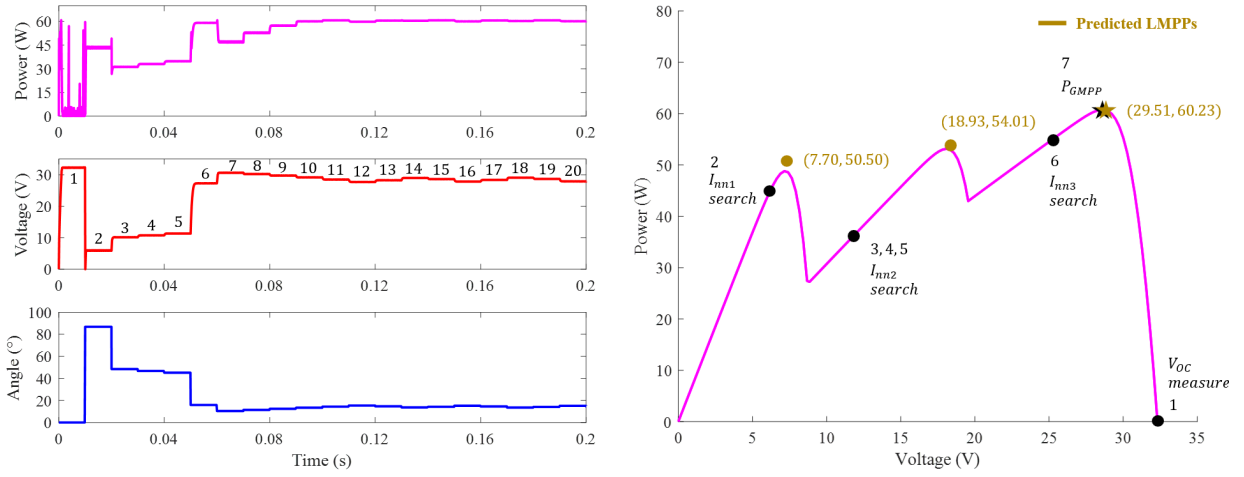
For PSC 2, where the irradiance levels are set at 250, 700, and 1000W/m² at a temperature of 50°C, the algorithm requires four iterations to collect current samples for the 1D-CNN input, resulting in a TC_{SP} of 0.04s. The predicted GMPP is located at 18.73V and 100.50W (corresponding to region 2), closely matching the actual GMPP at 17.73V and 103.7W, achieving NN_{VP} of 94.36% and NN_{PP} of 96.91%. During steady-state operation, the system delivers an average power of 103.50W, resulting in TA of 99.81%, with TT of 0.11s.



(a) *(b)*

Figure 4.15. Software simulation for proposed method under PSC2 with irradiance levels of 250, 700, and 1000W/m² at a temperature of 50 °C. (a) The tracking waveform shows power (purple), voltage (red), and switching current (blue). (b) Corresponding P-V curve.

For PSC 3, where the irradiance levels are set at 250, 350, and 850W/m² at a temperature of 50°C, the algorithm requires five iterations to collect current samples for the 1D-CNN input, resulting in a TC_{SP} of 0.05s. The predicted GMPP is located at 29.51V and 60.23W (corresponding to region 3), closely matching the actual GMPP at 28.62V and 60.76W, achieving NN_{VP} of 96.89% and NN_{PP} of 99.13%. During steady-state operation, the system delivers an average power of 60.69W, resulting in TA of 99.88%, with TT of 0.10s.

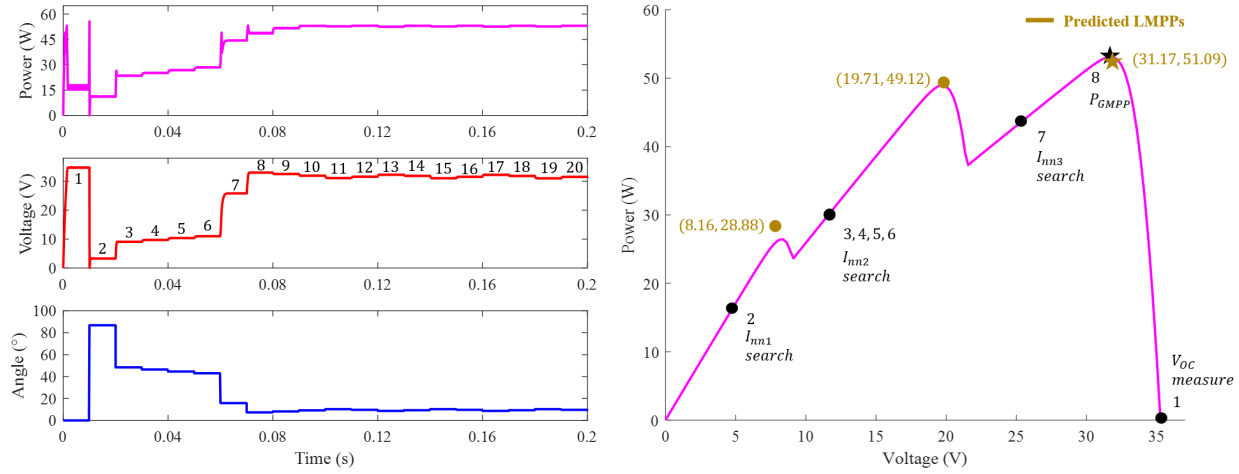


(a) *(b)*

Figure 4.16. Software simulation for proposed method under PSC3 with irradiance levels of 250, 350, and 850W/m² at a temperature of 50 °C. (a) The tracking waveform shows power (purple), voltage (red), and switching current (blue). (b) Corresponding P-V curve.

The same test is conducted to further evaluate the robustness of the algorithm under varying temperatures using the SS-PSM modulation method. Figures 4.17 to 4.19 show its performance at irradiance levels of 200, 300, and 400W/m² across temperatures of 25°C, 50°C, and 65°C.

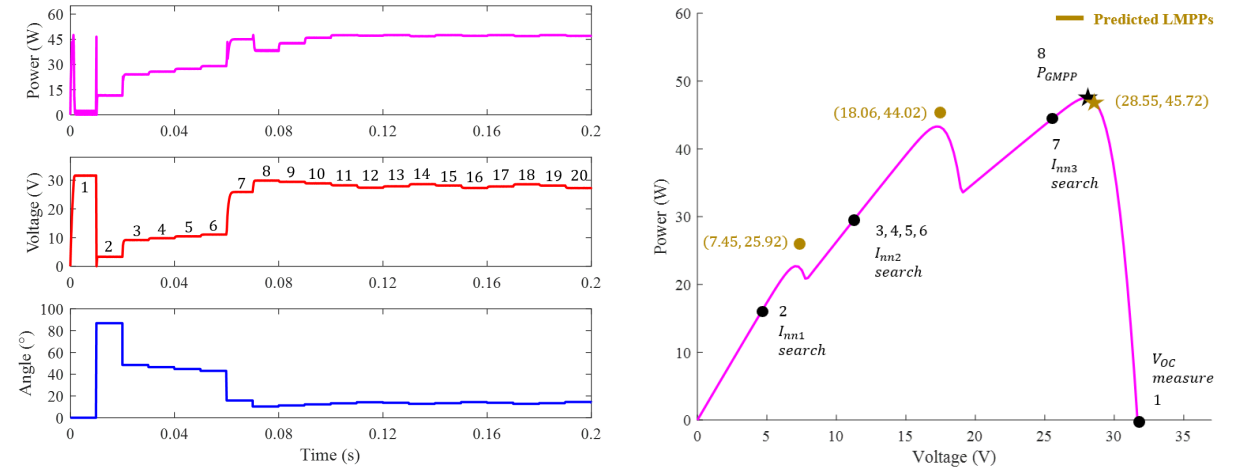
For PSC 4 at 25°C, the algorithm requires six iterations to collect current samples for the 1D-CNN input, resulting in a TC_{SP} of 0.06s. The predicted GMPP is 31.17V and 51.09W (region 3), closely matching the actual GMPP of 31.72V and 53.14W, achieving NN_{VP} of 98.27% and NN_{PP} of 96.14%. In steady-state, the system delivers an average power of 52.63W with a TA of 99.04% and TT of 0.09s.



(a) *(b)*

Figure 4.17. Software simulation for proposed method under PSC4 with irradiance levels of 200, 300, and 450W/m² at a temperature of 25°C. (a) The tracking waveform shows power (purple), voltage (red), and switching current (blue). (b) Corresponding P-V curve.

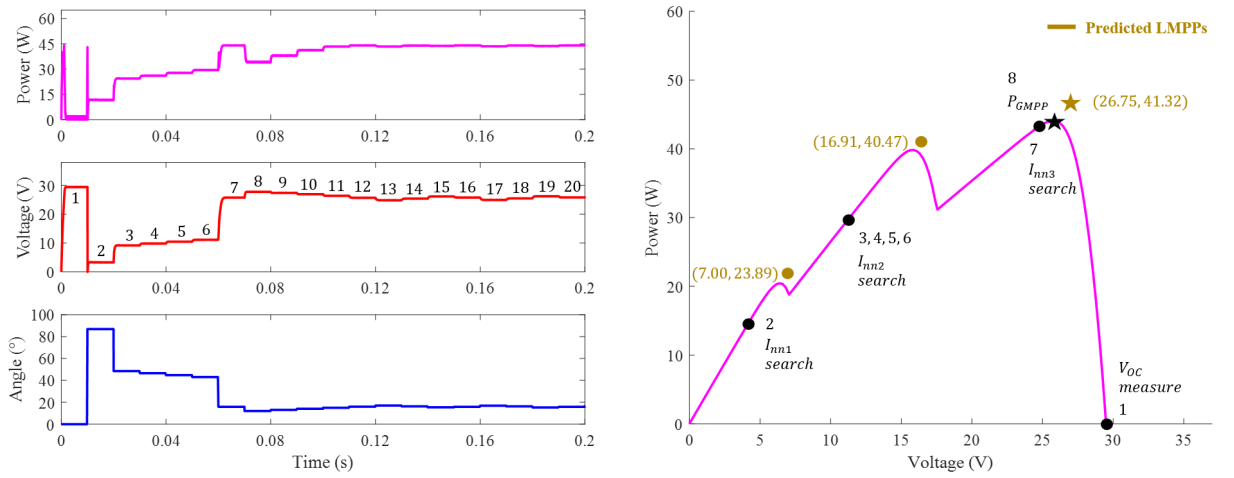
For temperature of 50°C for PSC 5, the algorithm the algorithm requires six iterations to collect current samples for the 1D-CNN input, resulting in a TC_{SP} of 0.06s. The predicted GMPP is 28.55V and 45.72W (region 3), closely matching the actual GMPP of 28V and 47.55W, achieving NN_{VP} of 98.04% and NN_{PP} of 96.15%. In steady-state, the system delivers an average power of 46.98W with TA of 99.80% and TT of 0.10s.



(a) *(b)*

Figure 4.18. Software simulation for proposed method under PSC5 with irradiance levels of 200, 300, and 400W/m² at a temperature of 50 °C. (a) The tracking waveform shows power (purple), voltage (red), and switching current (blue). (b) Corresponding P-V curve.

For temperature of 65°C for PSC 6, the algorithm requires six iterations to collect current samples for the 1D-CNN input, resulting in a TC_{SP} of 0.06s. The predicted GMPP is 26.75V and 41.32W (region 3), closely matching the actual GMPP of 25.76V and 44.08W, achieving NN_{VP} of 96.16% and NN_{PP} of 93.74%. In steady-state, the system delivers an average power of 44.02W with TA of 99.86% and TT of 0.10s.



(a) *(b)*

Figure 4.19. Software simulation for proposed method under PSC6 with irradiance levels of 200, 300, and 400W/m² at a temperature of 65°C. (a) The tracking waveform shows power (purple), voltage (red), and switching current (blue). (b) Corresponding P-V curve.

4.2.3 Comparison and Analysis for Simulation Result

The key performance indicators used in this evaluation are summarized in Table 4.1. For clarity, the following terms are used throughout the discussion: TC_{SP} stands for Total Current Sampling Period, TT refers to Tracking Time, NN_{PP} and NN_{VP} represent the prediction accuracy of power and voltage by the neural network, and TA refers to Tracking Accuracy during steady-state operation. These indicators were used to evaluate the performance of both PFM and SS-PSM methods under various PSCs.

The results show that both PFM and SS-PSM methods delivered high accuracy, with tracking accuracy values above 99% in all cases. The prediction of voltage and power was also consistently reliable, confirming the effectiveness of the 1D-CNN model in identifying the GMPP. The initialization approach based on the FHA proposed by [18], has proven to be an effective way to initiate the P&O algorithm. Building on this idea, the improved initialization strategy used in the SS-PSM method showed similar or even faster tracking performance compared to the PFM method. In most of the tested shading patterns, SS-PSM was able to track the MPP more quickly.

When compared with the results reported in [10], which achieved 99.99 percent accuracy and tracking times between 0.18 and 0.20s, the performance of both proposed methods is comparable. It is important to note that [10] used a boost converter, which is generally easier to control. In contrast, the use of an LLC resonant converter in this study adds more complexity to the control system. Matching the accuracy and speed of [10] using more complex circuit topology highlights the strength and reliability of the proposed algorithm.

While both methods are effective, SS-PSM is recommended. It provides faster tracking and better input-output efficiency because the switching frequency stays constant near the resonant point. This helps prevent circulating current flow, which can lead to extra power losses [27]. It also requires only a small hardware change, involving adjustment of the phase timing of the secondary switches. However, PFM still serves as a strong alternative, especially in scenarios where ease of implementation is more important than tracking speed. Another key highlight is that the proposed algorithm demonstrates consistent performance under varying temperature conditions, achieving results nearly identical to those obtained without temperature changes. This level of robustness is not addressed in [10], highlighting an advantage of the proposed method in this study.

Table 4.1. Summary of simulation results of the proposed two-stage GMPPT algorithm.

Pattern	Actual MPP (W)	PI	Proposed PFM	Proposed SS-PSM
I	48.79W	TC_{SP} (s)	0.07	0.08
		TT (s)	0.09	0.15
		NN_{PP} (%)	97.62	98.18
		NN_{VP} (%)	95.84	95.84
		TA (%)	99.82	99.94
II	103.70W	TC_{SP} (s)	0.06	0.04
		TT (s)	0.17	0.11
		NN_{PP} (%)	97.97	96.91
		NN_{VP} (%)	94.87	94.36
		TA (%)	99.10	99.81
III	60.76W	TC_{SP} (s)	0.07	0.05
		TT (s)	0.15	0.10
		NN_{PP} (%)	99.70	99.13
		NN_{VP} (%)	96.61	96.89
		TA (%)	99.80	99.88
IV	53.14W	TC_{SP} (s)	0.08	0.06
		TT (s)	0.13	0.09
		NN_{PP} (%)	98.38	96.14
		NN_{VP} (%)	99.81	98.27
		TA (%)	99.89	99.04
V	47.55W	TC_{SP} (s)	0.07	0.06
		TT (s)	0.10	0.10
		NN_{PP} (%)	96.89	96.15
		NN_{VP} (%)	97.68	98.04
		TA (%)	99.96	99.80
VI	44.08W	TC_{SP} (s)	0.05	0.06
		TT (s)	0.13	0.10
		NN_{PP} (%)	96.03	93.74
		NN_{VP} (%)	95.85	96.16
		TA (%)	99.18	99.86

4.3 Hardware Prototype Verification

4.3.1 Feedback Voltage Sensor

The waveforms in Figure 4.20 reveal the presence of high-frequency spikes superimposed on the sensor-measured voltage signal V_{sense} when the input voltage is maintain at approximately 5V. These spikes occur periodically at a frequency twice the switching frequency of the converter. At a switching frequency of $f_{sw} = 100kHz$, the noise period is about $5\mu s$, corresponding to a frequency of $200kHz$ as shown in Figure 4.20(a). Similarly, when the switching frequency is at $150kHz$ and $200kHz$, the noise period is about $3.3\mu s$ and $2.5\mu s$ as illustrated in Figures 4.20(b) and 4.20(c). These repetitive spikes observed in are mainly caused by parasitic inductance within the sensor circuitry, which becomes more prominent due to the high dv/dt and di/dt switching transitions in the converter. Since the sensor's cutoff frequency is lower than twice the converter's switching frequency, it cannot properly filter out this noise, leading to visible spikes that follow the switching behavior. Furthermore, similar voltage spiking behavior is observed when the input voltage is approximately 10V as shown in Figure 4.21, even though the input voltage changes while the switching frequency remains constant.

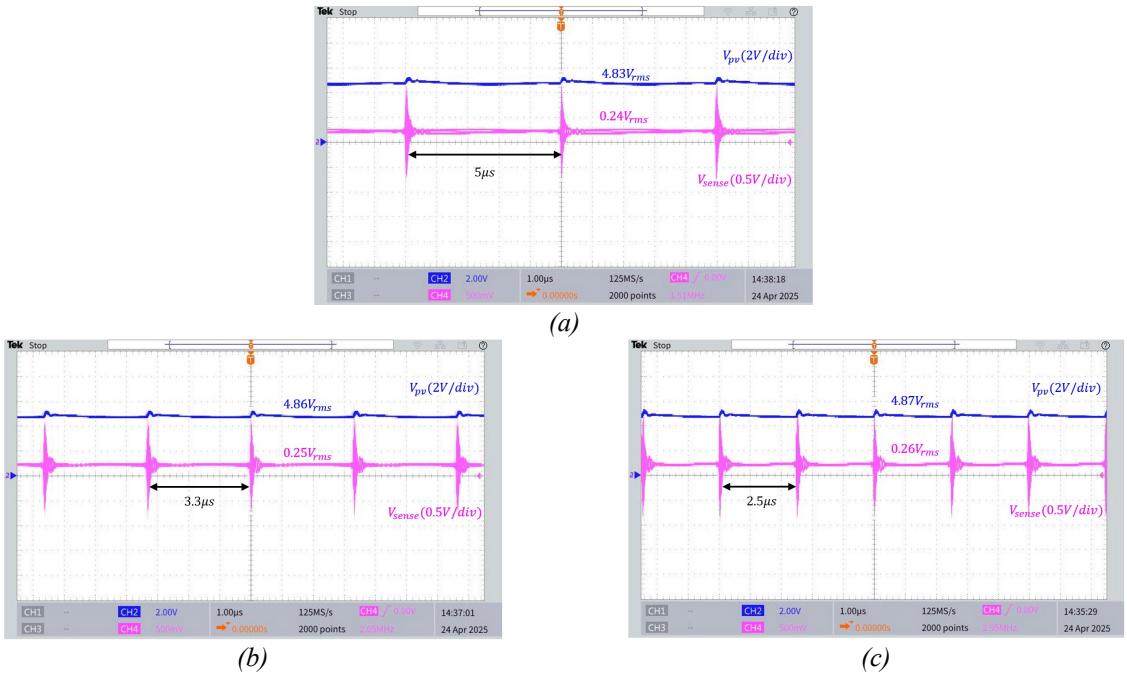


Figure 4.20. Comparison of input voltage V_{in} (blue) and sensor-measured voltage V_{sense} (purple) across varying switching frequencies, with V_{in} maintained at approximately 5V. (a) At f_{sw}=100kHz. (b) At f_{sw}=150kHz. (c) At f_{sw}=200kHz.

By applying the calibration equation obtained in Section 3.6.2, where $V_{PV} = 20.019V_{sense} - 0.619$, the extrapolated PV voltages are approximately 4.18V, 4.38V, and 4.59V, compared to the oscilloscope measurements in Figure 4.21(a), (b), and (c), which are 4.83V, 4.86V, and 4.87V respectively. Similarly, when the input voltage is around 10V, the extrapolated V_{PV} values are 8.79V, 9.19V, and 9.59V, while the measured RMS value from oscilloscope are 9.58V, 9.65V, and 9.69V, as shown in Figure 4.21(a), (b), and (c).

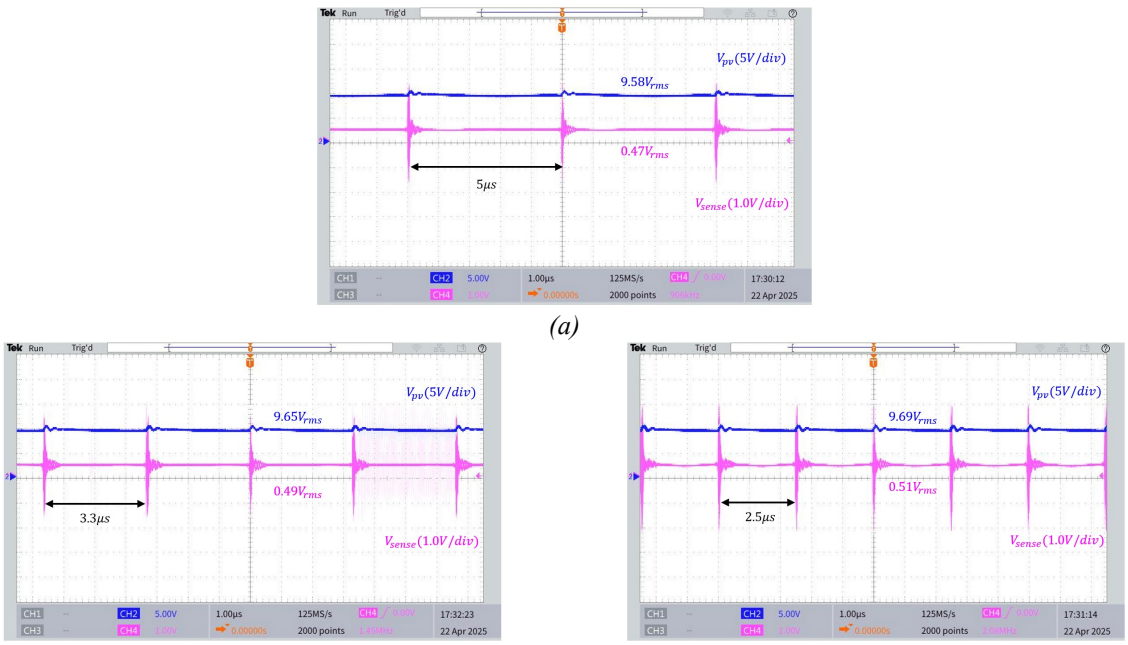


Figure 4.21. Comparison of input voltage V_{in} (blue) and sensor-measured voltage V_{sense} (purple) across varying switching frequencies, with V_{in} maintained at approximately 10V. (a) At f_{sw}=100kHz. (b) At f_{sw}=150kHz. (c) At f_{sw}=200kHz.

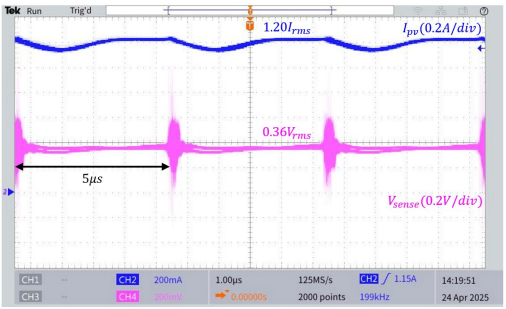
It is important to note that the measured RMS value from oscilloscope cannot be assumed to be 100% accurate and are mainly used here for comparison purposes. However, one clear observation is that an inaccuracy of approximately ±1 V exists, which could be attributed to the switching spikes and noise captured in the RMS measurement from oscilloscope. Despite these minor deviations, the calibration formula obtained previously is considered sufficiently accurate for this application.

As a temporary solution to mitigate the high-frequency disturbances, a decoupling capacitor is placed directly across the ADC input of the microcontroller to absorb switching noise. Additionally, digital filtering techniques are employed: a median filter is used to eliminate short-duration outliers, and a moving average filter smooths out longer-period fluctuations, ensuring more reliable sensor readings. Improving the measurement precision further would require the design of a dedicated analog filter, which is beyond the scope of this project.

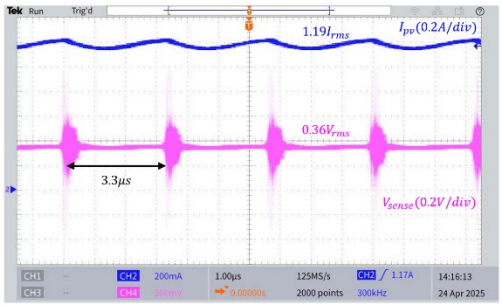
4.3.2 Feedback Current Sensor

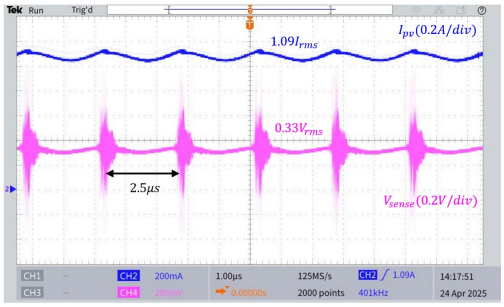
For the current sensing input, similar spiking behavior is observed in the sensor signal due to switching noise, even when the input voltage is approximately 5V and 10V, as shown in Figures 4.22 and 4.23.

By applying the calibration equation obtained in Section 3.6.2, where $I_{PV} = 3.3207V_{sense} - 0.0491$, the extrapolated I_{PV} values are approximately 1.15A, 1.15A, and 1.04A, compared to the oscilloscope measurements in Figures 4.22(a), (b), and (c), which are 1.20A, 1.19A, and 1.09A, respectively. Similarly, when the input voltage is around 10V, the extrapolated I_{PV} values are 2.61A, 2.28A, and 2.11A, while the measured RMS value from oscilloscope are 2.80A, 2.45A, and 2.25A, as shown in Figures 4.23(a), (b), and (c).



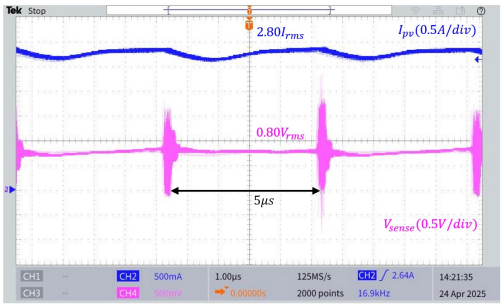
(a)



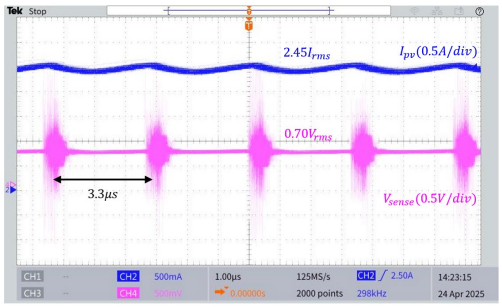


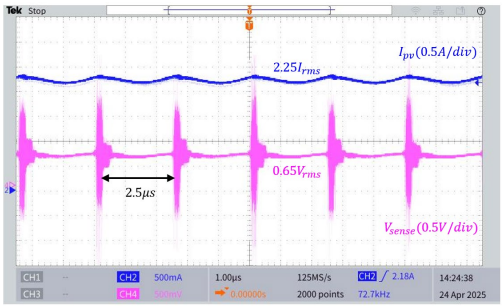
(b) (c)

Figure 4.22. Comparison of input current I_{in} (blue) and sensor-measured current I_{sense} (purple) across varying switching frequencies, with I_{in} range from 1.09A to 1.20A. (a) At f_{sw}=100kHz. (b) At f_{sw}=150kHz. (c) At f_{sw}=200kHz.



(a)





(b) (c)

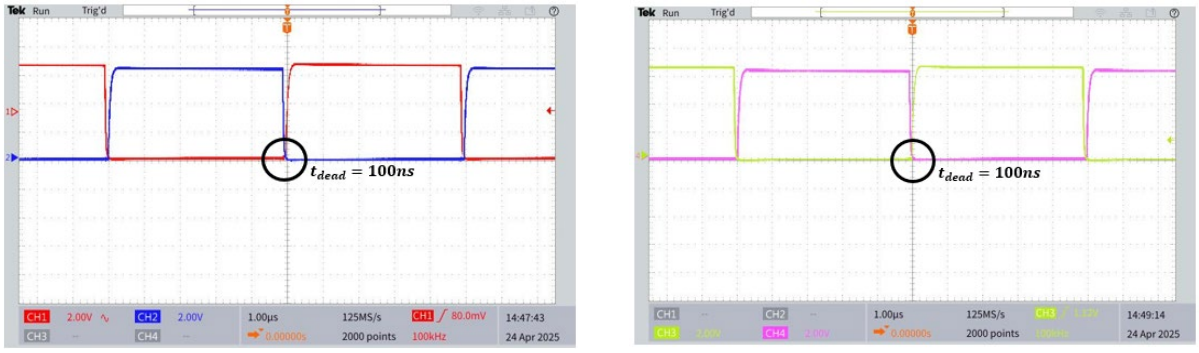
Figure 4.23. Comparison of input current I_{in} (blue) and sensor-measured current I_{sense} (purple) across varying switching frequencies, with I_{in} range from 2.25A to 2.80A. (a) At f_{sw}=100kHz. (b) At f_{sw}=150kHz. (c) At f_{sw}=200kHz.

It can be observed that the deviation between extrapolated and measured current values is generally smaller compared to the voltage readings. This is mainly because the current sensor has a smaller calibration gradient value, resulting in less amplification of deviations. Additionally, the current

sensor operates closer to its full dynamic range (uses about 0–8.5A of its full 0–10A range) compared to the voltage sensor, which only utilizes about 0–35V of its full 0–60V range. Consequently, the current sensor demonstrates better effective resolution than the voltage sensor. One way to improve the voltage sensor's accuracy would be to redesign it for a narrower range, making it more sensitive to small voltage changes. However, such modifications would require significant changes to the circuit design and passive component selection, which is beyond the scope of this project. To address the noise and improve measurement reliability, the same mitigation strategy is applied: a decoupling capacitor is placed directly across the ADC input of the microcontroller to absorb high-frequency switching noise. Additionally, digital filtering techniques are used, where a median filter removes short-duration outliers and a moving average filter smooths longer-period fluctuations, resulting in more stable and accurate sensor readings.

4.3.3 PWM Generation

Figure 4.24 illustrates how the gate-source voltages of the GaN FETs are driven in complementary pairs to ensure safe switching. In part (a), the red trace is V_{GS1} and the blue trace is V_{GS2}; in part (b), the green trace is V_{GS4} and the purple trace is V_{GS3}. Each switching waveforms are complementary and is delayed by dead time of $100ns$. This ensures that GaN FETs has fully turned off before its complementary pair turns on. This is needed as it introduce small delay prevents overlapping conduction (often called shoot-through) which could damage the circuit [72].

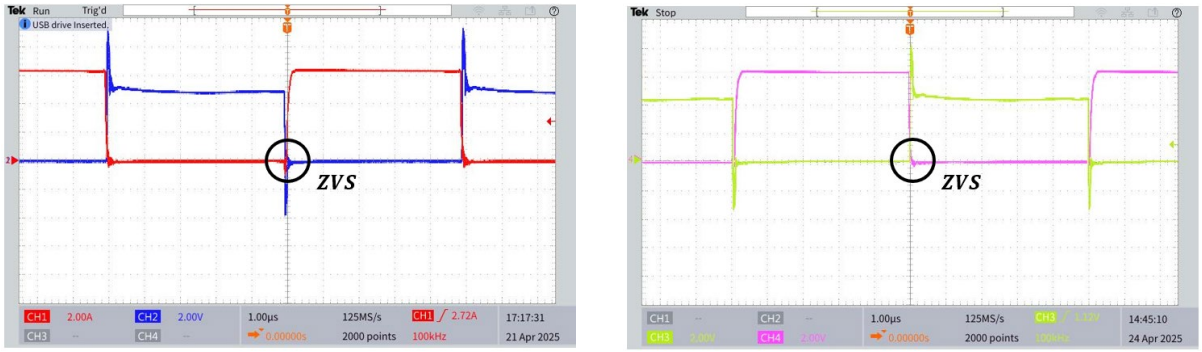


(a) *(b)*

Figure 4.24. Complementary gate-source voltage V_{GS} of GaN FET. (a) Waveforms of V_{GS1} (red) and V_{GS2} (blue). (b) Waveforms of V_{GS4} (green) and V_{GS3} (purple).

4.3.4 Open Loop Testing using PFM Modulation Technique

Before performing MPPT testing, an open-loop test is necessary to verify the correct operation of the LLC converter. In this preliminary test, the input voltage is fixed at approximately 5V, and the output is connected to a 330Ω resistive load. This step is important to confirm that the converter functions properly under controlled conditions. The waveforms shown in Figure 4.25 illustrate the ZVS behavior of the converter at a switching frequency close to $100\ kHz$. In both Figures 4.25(a) and (b), ZVS is observed when the drain-source voltage (V_{DS}) drops to zero before the corresponding gate-source voltage (V_{GS}) turns the switch on. This behavior confirms that the full-bridge LLC converter is operating correctly, and the results are consistent with the findings discussed in the literature review. Achieving ZVS is essential in this application, particularly for MPPT, as it enables soft switching, which reduces switching losses, limits voltage stress, and helps prevent potential damage to the switches. This contributes to improved efficiency and reliability of the overall system.



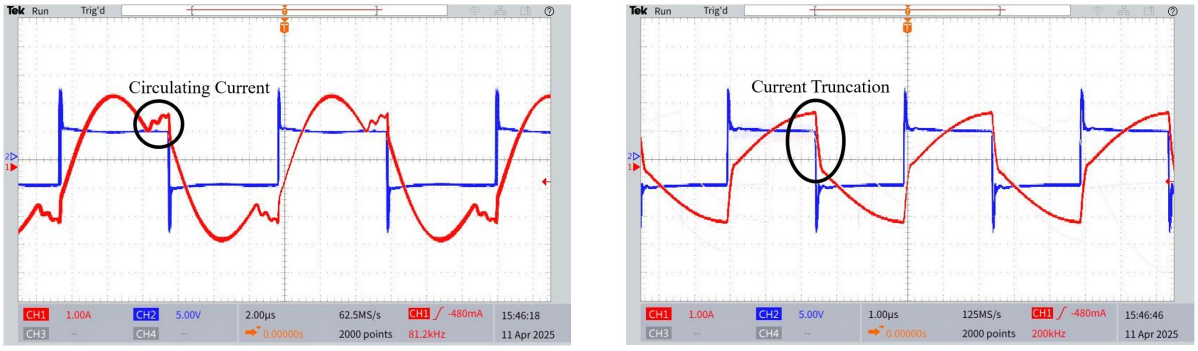
(a) (b)

Figure 4.25. Switching waveforms of LLC converter under steady-state operation. (a) Waveforms of V_{GS4} and V_{DS4}. (b) Waveforms of V_{GS3} and V_{DS4}.

The resonant inductor current I_{Lr} and drain-source voltage V_{GS4} (blue) are analyzed under various switching frequencies. As seen in Figure 4.26(c), when the switching frequency f_{sw} is above the resonant frequency f_r, switching interrupts the natural current cycle, causing truncation, hard commutation, and higher switching losses. In contrast, at $f_{sw} < f_r$ as shown in Figure 4.26(b), the current completes its cycle and enters a freewheeling mode, leading to higher conduction losses due to circulating current. At resonance, $f_{sw} = f_r$, current and voltage are naturally aligned, ensuring efficient energy transfer without forced commutation. This behavior occurs because I_{Lr} follows the LC tank dynamics, while V_{GS} is driven by fixed switching control. Tuning f_{sw} relative to f_r enables voltage gain control but distorts the ideal sinusoidal current profile. These findings are consistent with the theoretical analysis presented in the literature review.



(a)



(b) (c)

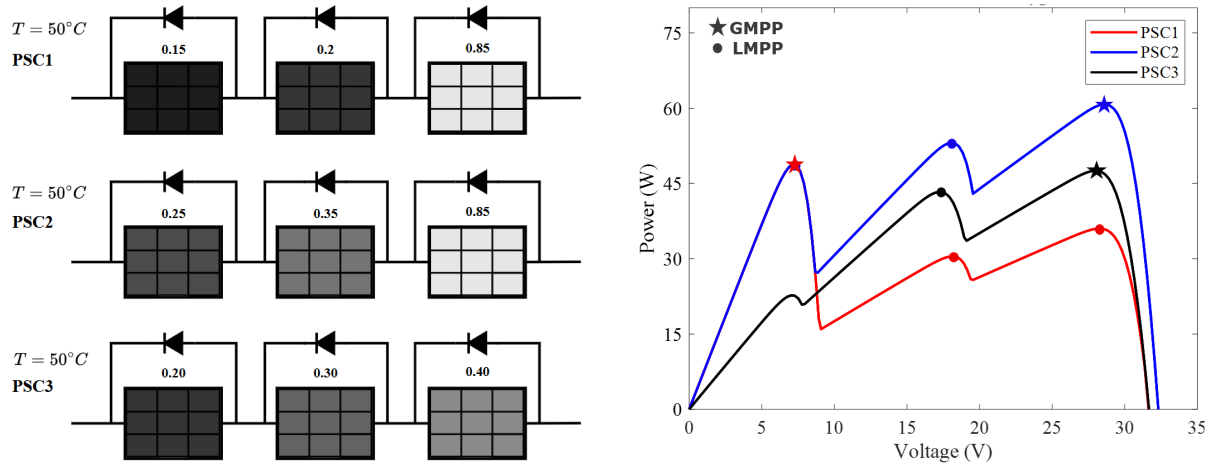
Figure 4.26. Resonant inductor current I_{Lr} (red) and drain-source voltage V_{GS4} (blue) at different switching frequency. (a) At resonance frequency. (b) Below resonance frequency. (c) Above resonance frequency.

4.4 Hardware Testing for Two Stage GMPPT Algorithm

In order to validate the advantages of the proposed method in real systems, experiments were conducted on five GMPPT methods. During the hardware prototype testing phase, predefined partial shading conditions (PSCs) were generated using MATLAB SIMULINK to create specific I-V curves containing multiple local maximum power points (LMPPs). These curves were then exported and uploaded into the SoftPanel software for practical use, with the full experimental setup following the configuration described in Section 3.6.4.

For this part of the project, only the PFM method was tested with six experiments, as the fabrication of the SS-PSM method was completed late in the semester and the hardware prototype was not ready in time. In the hardware MPPT testing, the algorithm begins by measuring the open-circuit voltage V_{oc}. Due to hardware constraints and limitations, the three current sampling points required for neural network prediction are collected only once. Each current value is initially sampled based on an estimated voltage determined using the First Harmonic Approximation (FHA) method. No further voltage adjustments are made to ensure the samples fall within the predefined ±2.1 V threshold, as the hardware prototype used for testing does not support dynamic voltage tuning. Hence, the algorithm will only perform three iterations to collect current samples for the 1D-CNN input, resulting in TC_{SP} of 0.03s for all test cases. Although this speeds up the process, it may cause the first and second sampled currents to fall outside the intended flat regions of the I-V curve.

The first three tests were performed under a constant temperature of 50°C, with varying irradiance levels of 150, 200, and 850W/m², followed by 250, 350, and 850W/m² under a constant temperature of 50°C, and then 200, 300, and 400W/m², as shown in Figure 4.6.

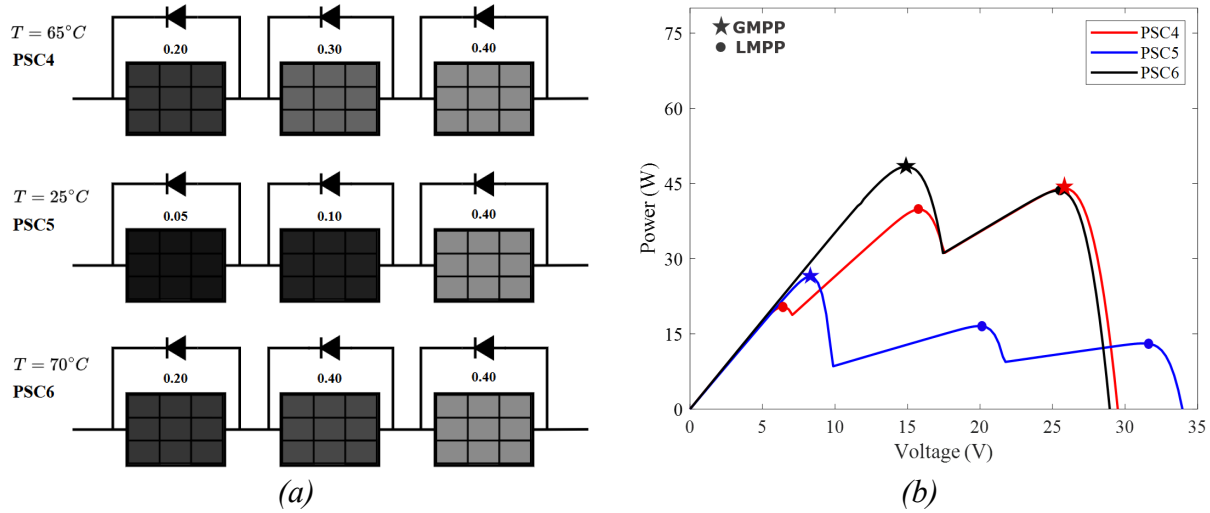


(a) *(b)*

Figure 4.27. PSC with constant temperature where irradiance levels 150, 200, and 850W/m² (PSC1), 250, 700, and 1000W/m² (PSC2), and 250, 350, and 850W/m² (PSC3) (a) Setting for the 3S1P PV panel (b) P-V curve.

The fourth test was carried out at a higher temperature of 65°C, while maintaining irradiance levels of 200, 300, and 400W/m². The fifth test was conducted at 25°C with irradiance levels of 50, 100, and 400W/m². Lastly, the sixth test evaluated the system under a temperature of 70°C with irradiance conditions of 200, 400, and 400 W/m². The setting from test 4 to 6 are shown in Figure

4.28 below. These tests allowed the algorithm to be evaluated under a range of realistic operating conditions with varying temperatures and irradiance levels.



(a) *(b)*

Figure 4.28. PSC with constant temperature where irradiance levels 150, 200, and 850W/m² (PSC4), 250, 700, and 1000W/m² (PSC5), and 250, 350, and 850W/m² (PSC6). (a) Setting for the 3S1P PV panel (b) P-V curve.

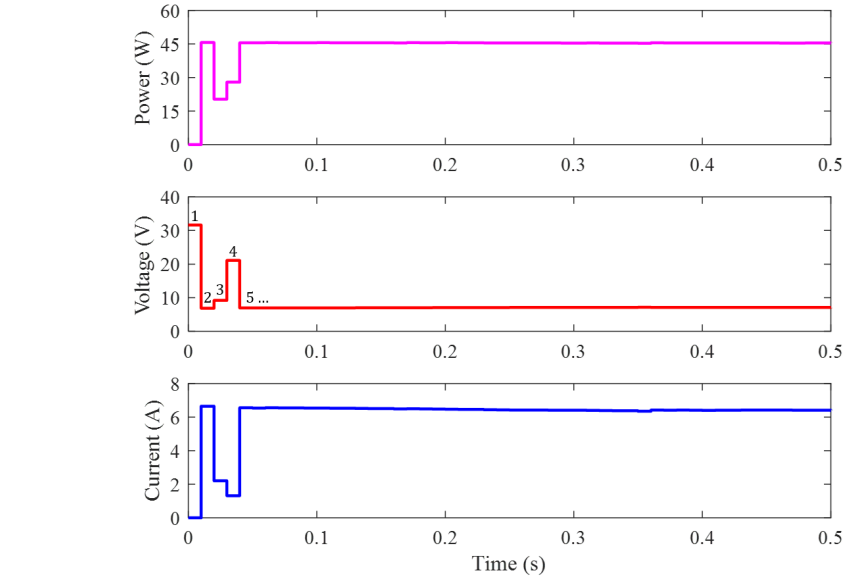
For data collection, the results were recorded and plotted using MATLAB. Data were transmitted via UART communication from the microcontroller to the laptop, where they were saved, organized, and processed for plotting. The SoftPanel software was used to display the instantaneous tracking efficiency of the algorithm, and snapshots of these results are captured and shown in Section 4.4.1.

Figures 4.29 to 4.34(a) show the system's dynamic response during the MPPT process, where the first subplot shows the power waveform, followed by the voltage waveform, and then the current waveform. Figures 4.29 to 4.34 (b) show the instantaneous tracking response captured from the SoftPanel software, displaying the corresponding P–V and I–V curves under partial shading conditions. In these plots, red annotated points mark the trajectory followed by the algorithm during tracking, while the local maximum power points (LMPPs) predicted by the 1D-CNN model are highlighted in orange. Circular markers are used to indicate the LMPPs, and the global maximum power point (GMPP) prediction is represented with a star symbol.

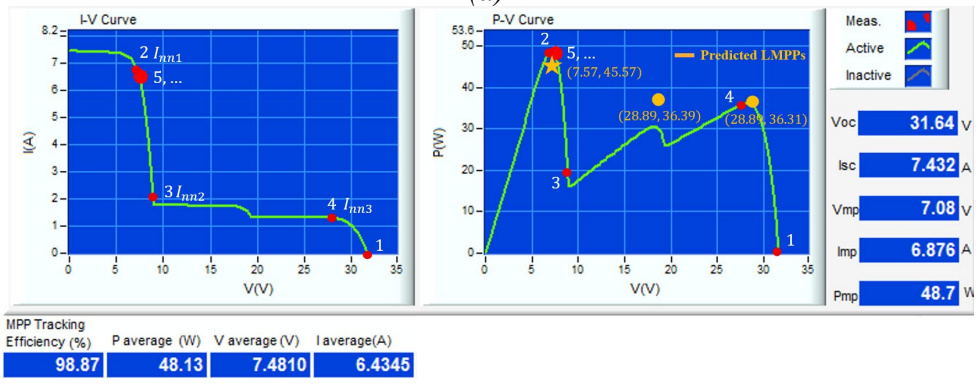
Furthermore, the overall performance of the MPPT algorithm is evaluated using the same set of metrics as in the simulation phase. These include the Total Current Sampling Points (TC_{SP}), which measures the time taken for initial current sampling, and the Tracking Time (TT), defined as the duration from the initial Voc measurement to the convergence of the P&O algorithm at the global maximum power point. Additionally, the neural network's predicted voltage (NN_{VP}) and predicted power (NN_{PP}) are assessed for prediction accuracy. Finally, the Tracking Average (TA) metric evaluates the P&O algorithm's ability to maintain the average harvested power relative to the true maximum during steady-state operation.

4.4.1 PFM Modulation Techinique

As shown in Figure 4.29, the proposed algorithm successfully tracks the GMPP under PSC1, with irradiance levels of 150, 200, and 850W/m² at 50 °C. The algorithm performs three iterations to collect current samples for the 1D-CNN input, resulting in TC_{SP} of 0.03s. However, as indicated in Figure 4.29(b), the first current sampling point falls outside the flat region of the I–V curve. The predicted GMPP was located at 7.57V and 45.75W, closely matching the actual GMPP at 7.08V and 48.7W. This results in NN_{VP} of 93.08% and NN_{PP} of 93.94%. The system achieved an average output power of 48.13W, compared to the theoretical maximum of 48.7W, yielding TA of 98.87% and TT of 0.04s. This fast tracking was partly coincidental, as the converter was already operating at its minimum predefined switching frequency of $70kHz$. At this lower limit boundary, the converter could not further increase its gain to reach a lower voltage, resulting in a naturally smoother and quicker convergence toward the GMPP.



(a)



(b)

Figure 4.29. GMPPT performance of the proposed method under PSC1 with irradiance levels of 150, 200, and 850W/m² at a temperature of 50 °C. (a) The tracking waveform shows power (purple), voltage (red), and switching current (blue). (b) Screenshot of the soft panel GUI.

For PSC2, with irradiance levels of 250, 350, and 850W/m² at 50°C, the algorithm performs three iterations to collect current samples for the 1D-CNN input, resulting in TC_{SP} of 0.03s. Similar to earlier observations, the first current sampling point fell outside the flat region of the I-V curve, as shown in Figure 4.30(b). The predicted GMPP was located at 29.48V and 60.15W (region 3), closely matching the actual GMPP at 28.73V and 60.7W. Despite the first current sampling point fell outside the flat region of the I-V curve, the neural network able to achiev NN_{VP} of 97.39% and NN_{PP} of 99.09%. During steady-state operation, the system maintained an average power of 60.22W, resulting in a TA of 99.15%, with TT of 0.16s.

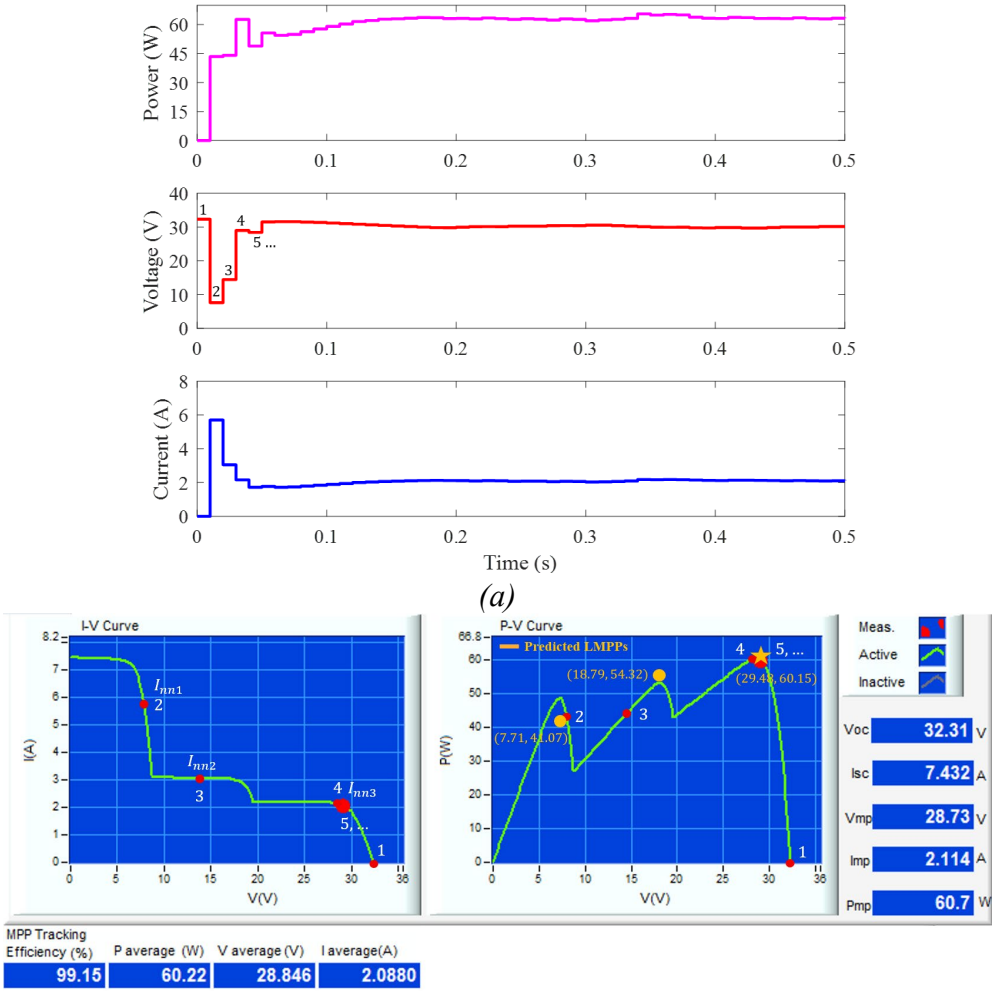


Figure 4.30. GMPPT performance of the proposed method under PSC2 with irradiance levels of 250, 350, and 850 W/m² at a temperature of 50 °C. (a) The tracking waveform shows power (purple), voltage (red), and switching current (blue). (b) Screenshot of the soft panel GUI.

For PSC 3, where the irradiance levels are set at 200, 300, and 400W/m² at a temperature of 50°C, the algorithm performs three iterations to collect current samples for the 1D-CNN input, resulting in TC_{SP} of 0.03s. The predicted GMPP is located at 28.65V and 45.28W (corresponding to region 3), closely matching the actual GMPP at 27.90V and 47.5W, achieving NN_{VP} of 97.31% and NN_{PP} of 95.33%. During steady-state operation, the system delivers an average power of 47.16W, yielding TA of 99.21%, and TT of 0.17s.

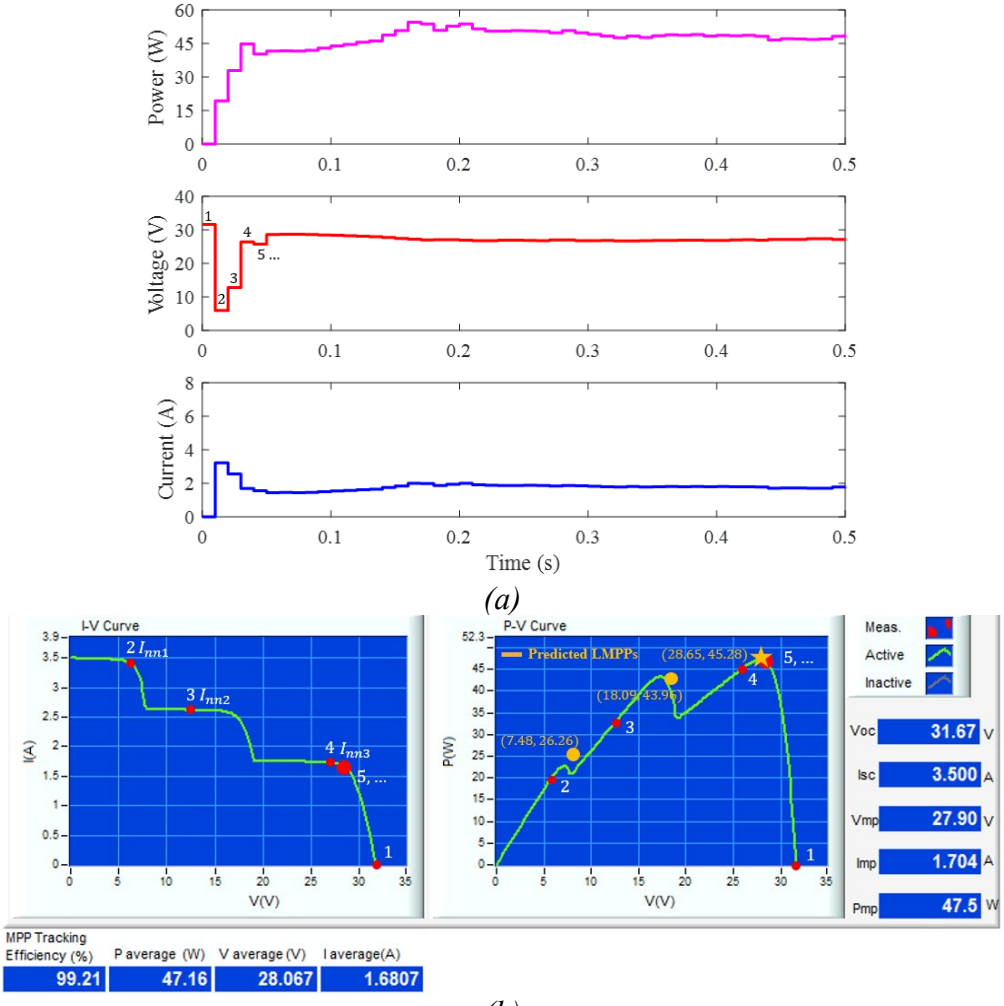


Figure 4.31. GMPPT performance of the proposed method under PSC3 with irradiance levels of 200, 300, and 400 W/m² at a temperature of 50 °C. (a) The tracking waveform shows power (purple), voltage (red), and switching current (blue). (b) Screenshot of the soft panel GUI.

Next, temperature variation was introduced to test the robustness of the algorithm. Figures 4.32 show the results for the same irradiance levels of 200, 300, and 400 W/m² but at temperature of 65°C, compared to PSC3. For PSC4 at 65°C, the algorithm performs three iterations to collect current samples for the 1D-CNN input, resulting in TC_{SP} of 0.03s. However, the first current sampling point is not at the flat region of the curve, causing the model to initially interpret the P-V curve as having only two LMPPs. Despite this, the model correctly predicted the GMPP at 26.8V and 41W (region 3), closely matching the actual GMPP of 25.73V and 44.1W. The neural network achieved NN_{VP} of 95.84% and a power prediction accuracy NN_{PP} of 92.97%. In steady-state operation, the system delivered an average power of 43.73W, resulting in TA of 99.22% and TT of 0.17s.

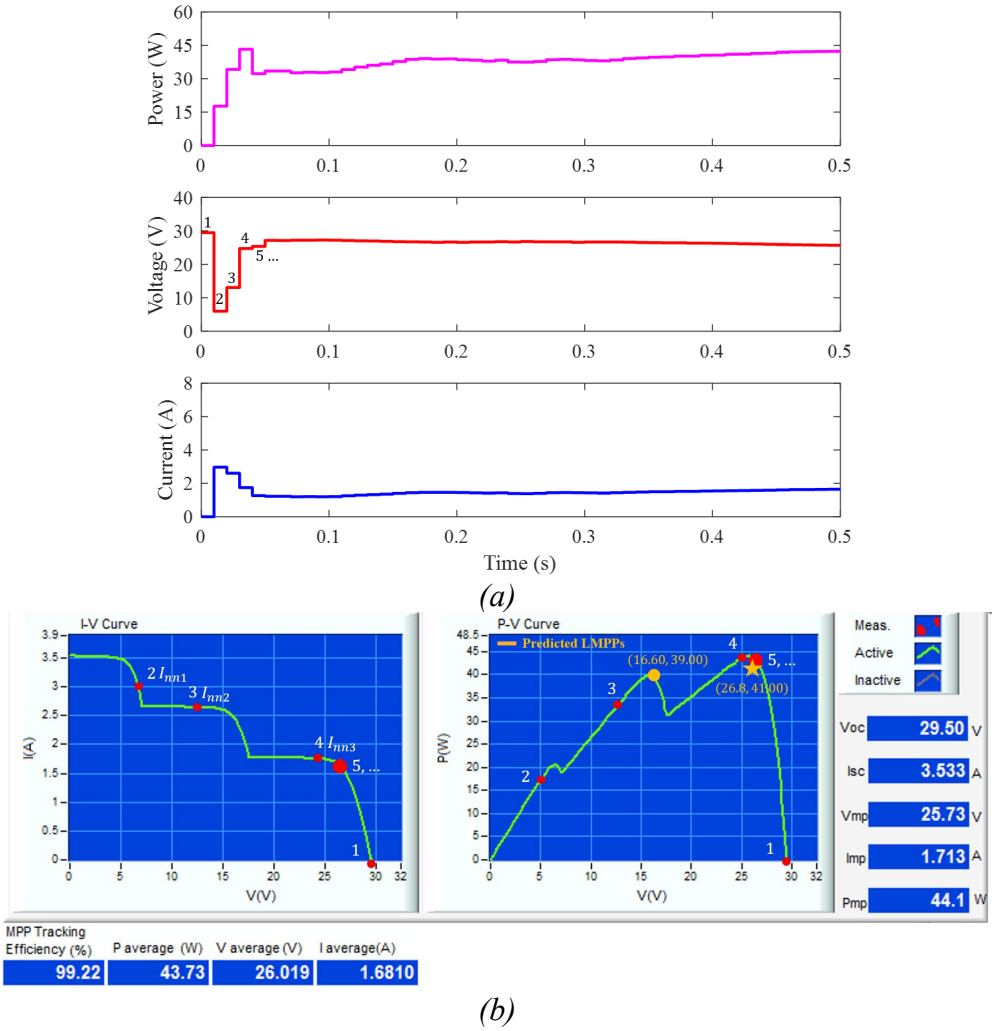


Figure 4.32. GMPPT performance of the proposed method under PSC4 with irradiance levels of 200, 300, and 400 W/m² at a temperature of 65°C. (a) The tracking waveform shows power (purple), voltage (red), and switching current (blue). (b) Screenshot of the soft panel GUI.

For PSC5 at a temperature of 25°C, the algorithm performs three iterations to collect current samples for the 1D-CNN input, resulting in TC_{SP} of 0.03s with the first current sample was relatively close to the flat region, allowing the model to detect three LMPPs, although with minor errors in the predicted locations. Despite this, the GMPP prediction remained accurate at 7.90V and 27.55W (region 1), closely matching the actual GMPP of 8.26V and 26.5W. The neural network achieved NN_{VP} of 95.64% and NNPP of 96.04%. In steady-state, the system delivered an average power of 26.15W, resulting in TA of 98.82% and TT of 0.04s.

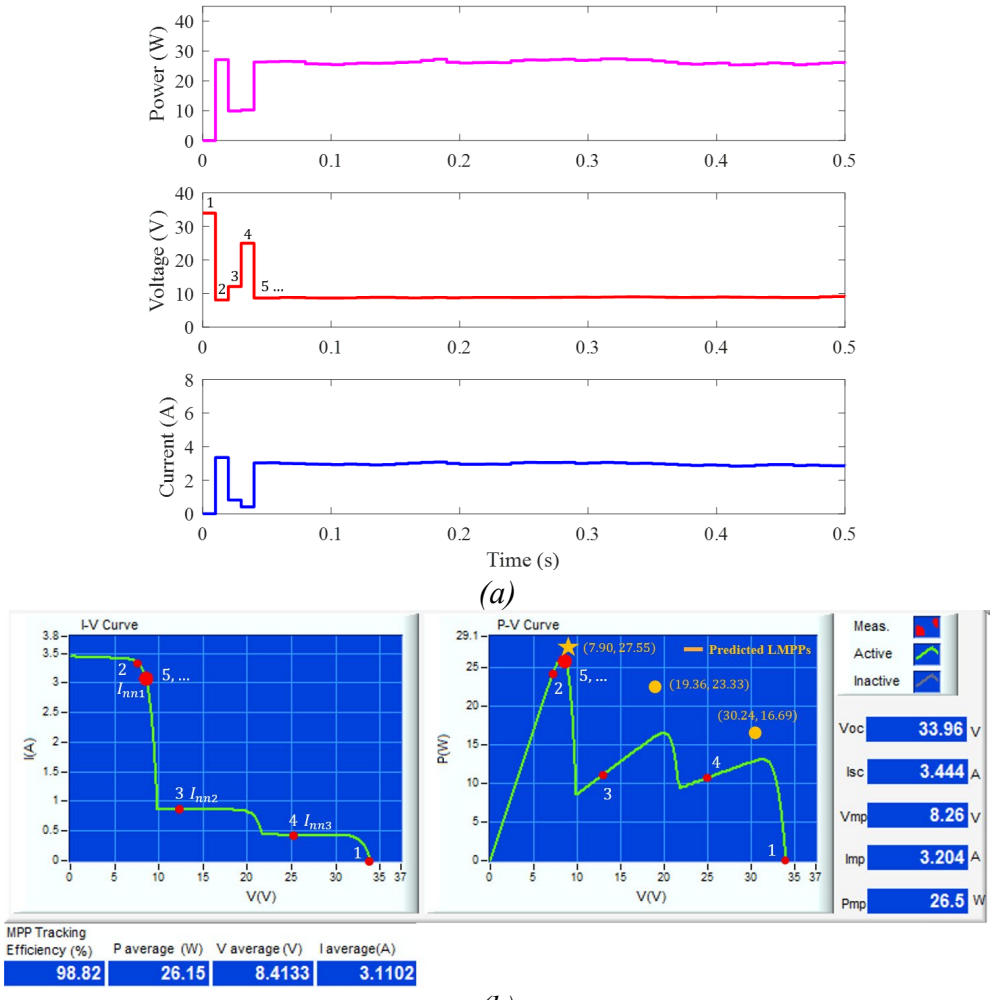


Figure 4.33. GMPPT performance of the proposed method under PSC5 with irradiance levels of 50, 100, and 400 W/m² at a temperature of 25°C. (a) The tracking waveform shows power (purple), voltage (red), and switching current (blue). (b) Screenshot of the soft panel GUI.

For PSC6 at a temperature of 70°C, the algorithm performs three iterations to collect current samples for the 1D-CNN input, resulting in TC_{SP} of 0.03s. The predicted GMPP was located at 16.23V and 46.23W (region 2), closely matching the actual GMPP of 14.94V and 48.4W. The neural network achieved NN_{VP} of 91.37% and NN_{PP} of 95.52%. Notably, the model correctly identified only two LMPPs without falsely predicting a third, demonstrating its strong ability to accurately interpret the P-V curve. In steady-state, the system delivered an average power of 48.19W, resulting in TA of 99.53% and TT of 0.05s.

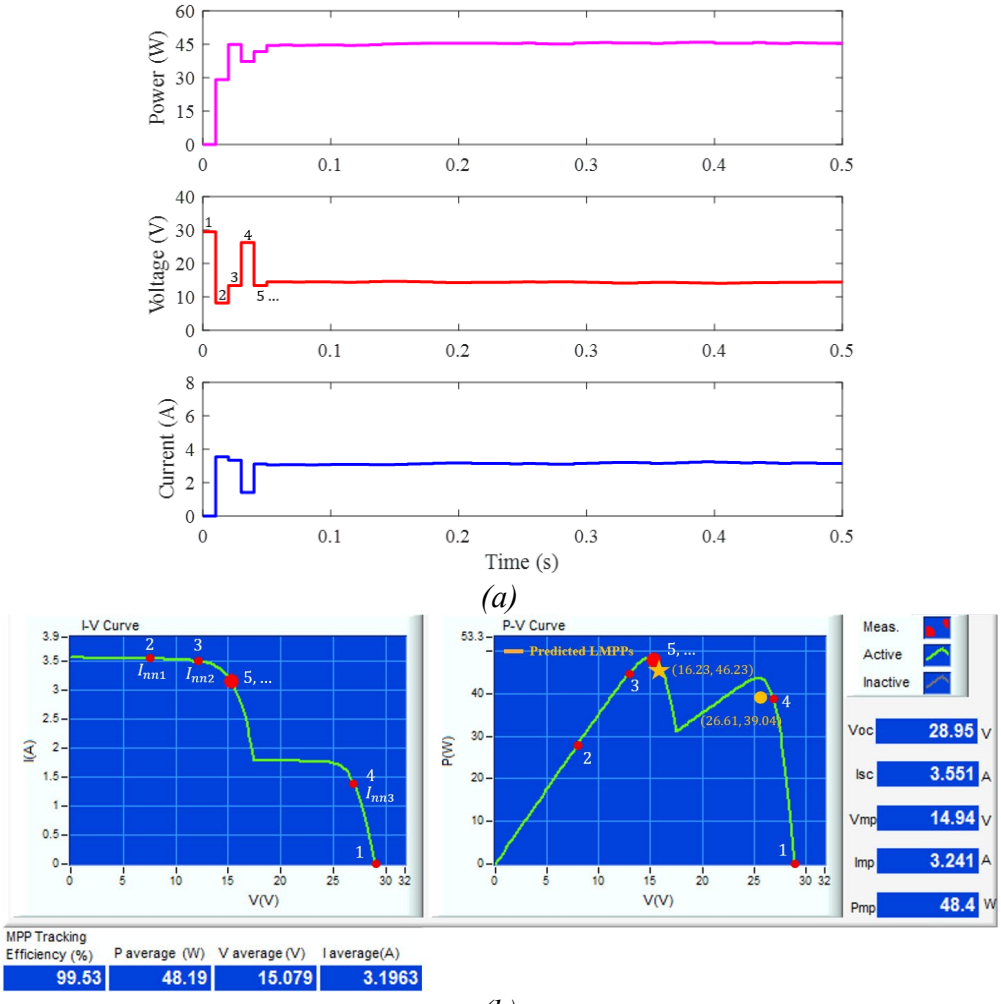


Figure 4.34. GMPPT performance of the proposed method under PSC6 with irradiance levels of 200, 400, and 400 W/m² at a temperature of 70°C. (a) The tracking waveform shows power (purple), voltage (red), and switching current (blue). (b) Screenshot of the soft panel GUI.

4.4.2 Hardware Prototype Analysis

The hardware results across six partial shading patterns (Patterns I–VI) show that the proposed algorithm with PFM control consistently achieves high TA of approximately 99%, closely matching simulation performance. The 1D-CNN model accurately predicts voltage and power, even under changing irradiance and temperature, with fast tracking times between 0.04 and 0.17 seconds. Notably, the algorithm maintains stable performance across different temperatures a feature of the proposed method which is not addressed in [10]. The slightly lower accuracy in Pattern I and V is due to hardware safety limits on switching frequency to avoid non-ZVS operation, not algorithm limitations. Overall, the method proves effective and reliable for real-world PV applications using LLC converters

Table 4.2. Summary of hardware prototype results.

Pattern	Actual MPP (W)	PI	Proposed PFM
I	48.7W	TC_{SP} (s)	0.03
		TT (s)	0.04
		NN_{PP} (%)	93.94
		NN_{VP} (%)	93.08
		TA (%)	98.87
II	60.7W	TC_{SP} (s)	0.03
		TT (s)	0.16
		NN_{PP} (%)	99.09
		NN_{VP} (%)	97.39
		TA (%)	99.15
III	47.5W	TC_{SP} (s)	0.03
		TT (s)	0.17
		NN_{PP} (%)	95.33
		NN_{VP} (%)	97.31
		TA (%)	99.21
IV	44.1W	TC_{SP} (s)	0.03
		TT (s)	0.17
		NN_{PP} (%)	92.97
		NN_{VP} (%)	95.84
		TA (%)	99.22
V	26.5W	TC_{SP} (s)	0.03
		TT (s)	0.04
		NN_{PP} (%)	96.04
		NN_{VP} (%)	95.64
		TA (%)	98.82
VI	48.4W	TC_{SP} (s)	0.03
		TT (s)	0.05
		NN_{PP} (%)	95.52
		NN_{VP} (%)	91.37
		TA (%)	99.53

4.5 Discussion on Proposed Two-Stage GMPPT Method

The proposed two-stage GMPPT algorithm demonstrates consistently high performance across all tested scenarios in both simulation and hardware implementation. In simulation, the method achieves tracking accuracy above 99% with fast convergence and reliable identification of the GMPP, confirming the effectiveness of the neural network model. Hardware results further support these findings, showing tracking accuracy ranging from 98.82% to 99.53% and tracking time (TA) between 0.04 and 0.17s which is comparable to the one presented in [10], [18], [75], [76]. The algorithm also maintains strong stability under varying irradiance and temperature conditions, achieving results closely aligned with simulation.

The 1D-CNN used in this study performs better than the traditional artificial neural network in understanding the P-V curve and accurately predicting the GMPP. Although it requires slightly more computation, the cost is acceptable since the prediction is performed only once per tracking cycle and is followed by a standard P&O algorithm. In the hardware test, slight oscillations were observed during steady-state operation when using the basic perturb and observe method. While these could be reduced by implementing an adaptive method such as the one proposed in [33].

Another important highlight is the effectiveness of the initialization method based on the first harmonic approximation. This approach, originally proposed in [18], has proven to be a practical way to start the P&O algorithm. It was further improved and integrated into the SS-PSM method used in simulation. This enhancement enabled faster tracking in most test cases compared to the PFM method. The consistent results in both simulation and hardware testing indirectly validate the correctness of the initialization method and the overall structure of the proposed algorithm.

In summary, the algorithm shows strong potential for practical deployment in PV systems using LLC resonant converters. It combines accurate prediction, fast response, and robust performance under diverse shading and temperature conditions, making it a reliable solution for real-world applications.

Chapter 5. Conclusion, Time Plan and Future Works

5.1 Conclusion

The growing demand for efficient and intelligent solar energy systems, particularly in standalone applications, motivated the development of this project. While LLC resonant converters are known for their high efficiency and compact design, their integration with advanced maximum power point tracking (MPPT) techniques remains limited, especially under challenging conditions like partial shading. Traditional MPPT methods often struggle to identify the true global peak, resulting in energy losses and reduced tracking efficiency. To improve the reliability and efficiency of PV systems, there is a clear need for smart, accurate, and adaptable MPPT algorithms that can operate effectively using the high efficiency converters in real-world environments.

Through this project, a smart MPPT system for an LLC resonant converter in PV applications has been successfully developed, with all project objectives fully achieved. An in-depth analysis of the modulation techniques of LLC resonant converters was conducted, providing a solid foundation for system design. A smart MPPT algorithm was developed, capable of maintaining high tracking efficiency under partial shading conditions (PSC), temperature variations, and within a 3S1P PV configuration. The algorithm was thoroughly validated through comprehensive simulations using software simulation tools, followed by successful implementation and testing on a hardware prototype, where dynamic PV conditions were tested using a programmable DC power supply.

The main novelty of the proposed two stage GMPPT algorithmis the adaptability to changing irradiance and temperature conditions, addressing limitations not fully explored in [10]. The neural network was evaluated using 1851 test samples, significantly more than the 252 cases in [10], highlighting its improved reliability and robustness. The 1D-CNN model developed in this project demonstrated better performance in predicting the GMPP, achieving 97.8% accuracy compared to 94.2% with the ANN. In addition, the overall algorithm attained tracking accuracy exceeding 99% in simulations and over 98% in hardware testing.

In conclusion, this study delivers a efficient and adaptable MPPT solution that effectively integrates deep learning with practical converter control to ensure optimal PV performance under varying temperature and irradiance condition and the proposed method eliminates the need for expensive irradiance and temperature sensors and can be implemented using a low-cost digital signal controller. The outcomes of this project represent a significant advancement in intelligent PV energy management and provide a strong foundation for future research and development in smart MPPT technologies.

5.2 Time Plan

Practical projects often come with unforeseen challenges, and this project was no exception. As detailed in Appendix A, the design and development of the smart GMPPT algorithm, along with the setup of the edge computing platform, required more time than originally anticipated. While these delays affected overall progress, they were gradually mitigated during the study week and Chinese New Year holidays. To account for these adjustments, a revised Gantt chart was developed, updating the initial project timeline. Further delays were encountered due to the process of borrowing a solar emulator from Asia Pacific University (APU), which was essential for conducting hardware testing under controlled PV conditions. Despite these challenges, the project successfully achieved its core objectives and delivered a functional, validated smart MPPT solution, demonstrating resilience, flexibility, and effective problem-solving throughout its execution.

Throughout the project, Lean Six Sigma principles were applied to drive continuous improvement. The DMAIC methodology (Define, Measure, Analyse, Improve, Control) was particularly useful in refining the system design. This approach was clearly demonstrated during the development of the hardware prototype, where collaboration with a PhD researcher helped address challenges related to the narrow gain of the converter (an issue beyond the defined scope of this project). The application of these problem-solving frameworks and creative approaches will be valuable for future engineering endeavors.

Despite the challenges, the primary objectives of the project were successfully achieved within the revised timeframe, resulting in the development of a Smart Maximum Power Point Tracking system for an LLC Converter. Several iterative experiments were conducted to validate system functionality and performance. However, delays were encountered during the deployment phase due to hardware failures caused by high power testing and excessive losses across switching devices, which led to component damage. To mitigate this, all experimental results were collected below 100W, acknowledging the hardware limitations, which were outside the project's scope.

5.3 Future Works

The performance of the proposed smart GMPPT system can be further improved through several key enhancements. One potential area is the implementation of the Secondary-Side Phase-Shift Modulation (SS-PSM) technique into hardware prototype, which is expected to increase efficiency by eliminating circulating currents and ensuring zero-voltage switching (ZVS) across the entire PV curve. Additionally, the accuracy of voltage and current sensing could be enhanced by integrating Direct Memory Access (DMA) in the STM32 microcontroller. This would allow higher sampling frequencies without burdening the CPU, enabling better detection of small transients and improving the effectiveness of hill-climbing algorithms, ultimately enhancing tracking efficiency. Another important improvement involves testing the converter at higher power levels. While this study was limited to a maximum of 80W due to hardware constraints, the converter is rated for 250W. Future work should focus on operating the system closer to its full capacity to better evaluate performance and reliability under real-world conditions. These advancements would contribute to increased efficiency, robustness, and practical applicability of the system in photovoltaic energy management.

Chapter 6. References

[1] M. Höök and K. Aleklett, "A review on coal-to-liquid fuels and its coal consumption," *Int J Energy Res*, vol. 34, no. 10, pp. 848–864, Aug. 2010, doi: https://doi.org/10.1002/er.1596.

[2] S. Busch, R. Kasdorp, D. Koolen, A. Mercier, and M. Spooner, "The Development of Renewable Energy in the Electricity Market," Directorate General Economic and Financial Affairs (DG ECFIN), European Commission, 2023. doi: DOI:

[3] "Transforming our world : the 2030 Agenda for Sustainable Development | Refworld." Accessed: Apr. 25, 2025. [Online]. Available: https://www.refworld.org/legal/resolution/unga/2015/en/111816

[4] "Renewables 2023 – Analysis - IEA." Accessed: Apr. 25, 2025. [Online]. Available: https://www.iea.org/reports/renewables-2023#overview

[5] N. Femia, G. Petrone, G. Spagnuolo, and M. Vitelli, "Optimization of perturb and observe maximum power point tracking method," *IEEE Trans Power Electron*, vol. 20, no. 4, pp. 963–973, 2005, doi: 10.1109/TPEL.2005.850975.

[6] Y.-C. Kuo, T.-J. Liang, and J.-F. Chen, "Novel maximum-power-point-tracking controller for photovoltaic energy conversion system," *IEEE Transactions on Industrial Electronics*, vol. 48, no. 3, pp. 594–601, 2001, doi: 10.1109/41.925586.

[7] J. Zeng, G. Zhang, S. S. Yu, B. Zhang, and Y. Zhang, "LLC resonant converter topologies and industrial applications — A review," *Chinese Journal of Electrical Engineering*, vol. 6, no. 3, pp. 73–84, 2020, doi: 10.23919/CJEE.2020.000021.

[8] H. Patel and V. Agarwal, "MATLAB-Based Modeling to Study the Effects of Partial Shading on PV Array Characteristics," *IEEE Transactions on Energy Conversion*, vol. 23, no. 1, pp. 302–310, 2008, doi: 10.1109/TEC.2007.914308.

[9] J. Bai, Y. Cao, Y. Hao, Z. Zhang, S. Liu, and F. Cao, "Characteristic output of PV systems under partial shading or mismatch conditions," *Solar Energy*, vol. 112, pp. 41–54, 2015, doi: https://doi.org/10.1016/j.solener.2014.09.048.

[10] S.-P. Ye, Y.-H. Liu, H.-Y. Pai, A. Sangwongwanich, and F. Blaabjerg, "A Novel ANN-Based GMPPT Method for PV Systems Under Complex Partial Shading Conditions," *IEEE Trans Sustain Energy*, vol. 15, no. 1, pp. 328–338, 2024, doi: 10.1109/TSTE.2023.3284866.

[11] S. Atin and R. Lubis, "Implementation of Critical Path Method in Project Planning and Scheduling," *IOP Conf Ser Mater Sci Eng*, vol. 662, no. 2, p. 022031, 2019, doi: 10.1088/1757-899X/662/2/022031.

[12] M. Massaoudi, H. Abu-Rub, S. S. Refaat, I. Chihi, and F. S. Oueslati, "Deep Learning in Smart Grid Technology: A Review of Recent Advancements and Future Prospects," *IEEE Access*, vol. 9, pp. 54558–54578, 2021, doi: 10.1109/ACCESS.2021.3071269.

[13] M. Rashid, *Power Electronics Handbook - Devices, Circuits, and Applications (4th Edition)*, 4th ed. Elsevier, 2014.

[14] Y. Jieb and E. Hossain, *Photovoltaic Systems: Fundamentals and Applications*. 2022. doi: 10.1007/978-3-030-89780-2.

[15] S. Ghosh, A. Alhatlani, and I. Batarseh, "MPPT of Grid-Tied Dual-PV LLC Converter using Adaptive Neuro Fuzzy Interface System," in *2022 IEEE Applied Power Electronics Conference and Exposition (APEC)*, 2022, pp. 1291–1297. doi: 10.1109/APEC43599.2022.9773584.

[16] Y. Zhuang, F. Liu, X. Zhang, X. Diao, J. Jiang, and J. Sun, "Direct Frequency Control Based MPPT Algorithm of LLC Resonant Converter for Photovoltaic System," in *2019 IEEE*

Energy Conversion Congress and Exposition (ECCE), 2019, pp. 3402–3406. doi: 10.1109/ECCE.2019.8912689.

[17] Q. Zhang *et al.*, "A Center Point Iteration MPPT Method With Application on the Frequency-Modulated LLC Microinverter," *IEEE Trans Power Electron*, vol. 29, no. 3, pp. 1262–1274, 2014, doi: 10.1109/TPEL.2013.2262806.

[18] C.-Y. Tang, H.-J. Wu, C.-Y. Liao, and H.-H. Wu, "An Optimal Frequency-Modulated Hybrid MPPT Algorithm for the LLC Resonant Converter in PV Power Applications," *IEEE Trans Power Electron*, vol. 37, no. 1, pp. 944–954, 2022, doi: 10.1109/TPEL.2021.3094676.

[19] B. Hu, J. Wang, Z. Ke, J. Yao, C. Zhang, and Z. J. Shen, "Simplified Hybrid Dual-Bridge LLC Resonant Converter With Reduced Passive Components and Loss for Emergency Power Supply," *IEEE J Emerg Sel Top Power Electron*, vol. 13, no. 1, pp. 254–265, 2025, doi: 10.1109/JESTPE.2024.3508082.

[20] S. Abdel-Rahman, "Resonant LLC Converter: Operation and Design," Sep. 2012.

[21] R. W. Erickson, *Fundamentals of Power Electronics*. Boston, MA: Boston, MA: Springer US, 1997. doi: 10.1007/978-1-4615-7646-4.

[22] S.-Y. Yu, "Power Tips: Designing an LLC Resonant Half-bridge Power Converter," Jul. 2015. [Online]. Available: www.ti.com

[23] J. Zhou and H. Ma, "Full-Bridge LLC Resonant Converter with Parallel-Series Transformer Connection and Voltage Doubler Rectifier," in *2019 10th International Conference on Power Electronics and ECCE Asia (ICPE 2019 - ECCE Asia)*, 2019, pp. 1–6. doi: 10.23919/ICPE2019-ECCEAsia42246.2019.8796966.

[24] F. Degioanni, "Dual-loop linear controller for LLC resonant converters," 2018. [Online]. Available: https://open.library.ubc.ca/collections/24/items/1.0368722

[25] Y. Wei, Q. Luo, and A. Mantooth, "Overview of Modulation Strategies for LLC Resonant Converter," *IEEE Trans Power Electron*, vol. 35, no. 10, pp. 10423–10443, 2020, doi: 10.1109/TPEL.2020.2975392.

[26] H. Wang and Z. Li, "A PWM LLC Type Resonant Converter Adapted to Wide Output Range in PEV Charging Applications," *IEEE Trans Power Electron*, vol. 33, no. 5, pp. 3791–3801, 2018, doi: 10.1109/TPEL.2017.2713815.

[27] H. Wu, T. Mu, X. Gao, and Y. Xing, "A Secondary-Side Phase-Shift-Controlled LLC Resonant Converter With Reduced Conduction Loss at Normal Operation for Hold-Up Time Compensation Application," *IEEE Trans Power Electron*, vol. 30, no. 10, pp. 5352–5357, 2015, doi: 10.1109/TPEL.2015.2418786.

[28] M. Killi and S. Samanta, "Modified Perturb and Observe MPPT Algorithm for Drift Avoidance in Photovoltaic Systems," *IEEE Transactions on Industrial Electronics*, vol. 62, no. 9, pp. 5549–5559, 2015, doi: 10.1109/TIE.2015.2407854.

[29] A. K. Abdelsalam, A. M. Massoud, S. Ahmed, and P. N. Enjeti, "High-Performance Adaptive Perturb and Observe MPPT Technique for Photovoltaic-Based Microgrids," *IEEE Trans Power Electron*, vol. 26, no. 4, pp. 1010–1021, 2011, doi: 10.1109/TPEL.2011.2106221.

[30] Srishti, D. P. Samajdar, and P. K. Padhy, "Incremental Conductance Based Model Predictive Control Algorithm for Solar PV Module for Tracking of MPP," in *2023 IEEE Region 10 Symposium (TENSYMP)*, 2023, pp. 1–6. doi: 10.1109/TENSYMP55890.2023.10223623.

[31] S. K. Kollimalla and M. K. Mishra, "Variable Perturbation Size Adaptive P&O MPPT Algorithm for Sudden Changes in Irradiance," *IEEE Trans Sustain Energy*, vol. 5, no. 3, pp. 718–728, 2014, doi: 10.1109/TSTE.2014.2300162.

[32] H. G. Ali, R. Vilanova, and J. Pelez-Restrepo, "Perturb & Observe based Adaptive Sliding Mode MPPT Control of Solar Photovoltaic System," in *2020 IEEE International Conference on Environment and Electrical Engineering and 2020 IEEE Industrial and Commercial Power Systems Europe (EEEIC / I&CPS Europe)*, 2020, pp. 1–6. doi: 10.1109/EEEIC/ICPSEurope49358.2020.9160539.

[33] H. Rezk, M. Aly, M. Al-Dhaifallah, and M. Shoyama, "Design and Hardware Implementation of New Adaptive Fuzzy Logic-Based MPPT Control Method for Photovoltaic Applications," *IEEE Access*, vol. 7, pp. 106427–106438, 2019, doi: 10.1109/ACCESS.2019.2932694.

[34] S. Ahmed, S. Mekhilef, M. Mubin, K. S. Tey, and M. Kermadi, "An Adaptive Perturb and Observe Algorithm With Enhanced Skipping Feature for Fast Global Maximum Power Point Tracking Under Partial Shading Conditions," *IEEE Trans Power Electron*, vol. 38, no. 9, pp. 11601–11613, 2023, doi: 10.1109/TPEL.2023.3285243.

[35] Q. Shen, J.-H. Jiang, J. Tao, G. Shen, and R.-Q. Yu, "Modified Ant Colony Optimization Algorithm for Variable Selection in QSAR Modeling: QSAR Studies of Cyclooxygenase Inhibitors," *J Chem Inf Model*, vol. 45, no. 4, pp. 1024–1029, Jul. 2005, doi: 10.1021/ci049610z.

[36] M. Seyedmahmoudian *et al.*, "State of the art artificial intelligence-based MPPT techniques for mitigating partial shading effects on PV systems – A review," *Renewable and Sustainable Energy Reviews*, vol. 64, pp. 435–455, 2016, doi: https://doi.org/10.1016/j.rser.2016.06.053.

[37] M. D. Toksari, "Ant colony optimization for finding the global minimum," *Appl Math Comput*, vol. 176, no. 1, pp. 308–316, 2006, doi: https://doi.org/10.1016/j.amc.2005.09.043.

[38] L. L. Jiang, D. L. Maskell, and J. C. Patra, "A novel ant colony optimization-based maximum power point tracking for photovoltaic systems under partially shaded conditions," *Energy Build*, vol. 58, pp. 227–236, 2013, doi: https://doi.org/10.1016/j.enbuild.2012.12.001.

[39] R. Eberhart and J. Kennedy, "A new optimizer using particle swarm theory," in *MHS'95. Proceedings of the Sixth International Symposium on Micro Machine and Human Science*, 1995, pp. 39–43. doi: 10.1109/MHS.1995.494215.

[40] Y.-H. Liu, S.-C. Huang, J.-W. Huang, and W.-C. Liang, "A Particle Swarm Optimization-Based Maximum Power Point Tracking Algorithm for PV Systems Operating Under Partially Shaded Conditions," *IEEE Transactions on Energy Conversion*, vol. 27, no. 4, pp. 1027–1035, 2012, doi: 10.1109/TEC.2012.2219533.

[41] L. P. N. Jyothy and M. R. Sindhu, "An Artificial Neural Network based MPPT Algorithm For Solar PV System," in *2018 4th International Conference on Electrical Energy Systems (ICEES)*, 2018, pp. 375–380. doi: 10.1109/ICEES.2018.8443277.

[42] M. Mao, X. Feng, J. Xin, and T. W. S. Chow, "A Convolutional Neural Network-Based Maximum Power Point Voltage Forecasting Method for Pavement PV Array," *IEEE Trans Instrum Meas*, vol. 72, pp. 1–9, 2023, doi: 10.1109/TIM.2022.3227552.

[43] Y. Lecun and Y. Bengio, *Convolutional Networks for Images, Speech, and Time-Series*. 1995.

[44] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet Classification with Deep Convolutional Neural Networks," in *Advances in Neural Information Processing Systems*, F. Pereira, C. J. Burges, L. Bottou, and K. Q. Weinberger, Eds., Curran Associates, Inc., 2012. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2012/file/c399862d3b9d6b76c8436e924a68c45b-Paper.pdf
[45] B. Mehlig, *Machine Learning with Neural Networks: An Introduction for Scientists and Engineers*. Cambridge University Press, 2021. doi: 10.1017/9781108860604.
[46] A. Karpathy, "CONNECTING IMAGES AND NATURAL LANGUAGE," Aug. 2016.
[47] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*. MIT Press, 2016.
[48] N. Srivastava, G. Hinton, A. Krizhevsky, I. Sutskever, and R. Salakhutdinov, "Dropout: a simple way to prevent neural networks from overfitting," *J. Mach. Learn. Res.*, vol. 15, no. 1, pp. 1929–1958, Jan. 2014.
[49] X. Glorot, A. Bordes, and Y. Bengio, "Deep Sparse Rectifier Neural Networks," in *Proceedings of the Fourteenth International Conference on Artificial Intelligence and Statistics (AISTATS)*, G. Gordon, D. Dunson, and M. Dudik, Eds., Fort Lauderdale, FL, USA: JMLR Workshop and Conference Proceedings, 2011, pp. 315–323. [Online]. Available: http://proceedings.mlr.press/v15/glorot11a/glorot11a.pdf
[50] C. Szegedy *et al.*, "Going Deeper with Convolutions," 2014. [Online]. Available: https://arxiv.org/abs/1409.4842
[51] K. He, X. Zhang, S. Ren, and J. Sun, "Deep Residual Learning for Image Recognition," 2015. [Online]. Available: https://arxiv.org/abs/1512.03385
[52] K. Simonyan and A. Zisserman, "Very Deep Convolutional Networks for Large-Scale Image Recognition," 2015. [Online]. Available: https://arxiv.org/abs/1409.1556
[53] S. Hershey *et al.*, "CNN Architectures for Large-Scale Audio Classification," 2017. [Online]. Available: https://arxiv.org/abs/1609.09430
[54] S. Abdoli, P. Cardinal, and A. L. Koerich, "End-to-End Environmental Sound Classification using a 1D Convolutional Neural Network," 2019. [Online]. Available: https://arxiv.org/abs/1904.08990
[55] Z. Zihao, J. Geng, and W. Jiang, "A Time Series Classification Method Based on 1DCNN-FNN," in *2021 33rd Chinese Control and Decision Conference (CCDC)*, 2021, pp. 1566–1571. doi: 10.1109/CCDC52312.2021.9602164.
[56] A. O. Ige and M. Sibiya, "State-of-the-Art in 1D Convolutional Neural Networks: A Survey," *IEEE Access*, vol. 12, pp. 144082–144105, 2024, doi: 10.1109/ACCESS.2024.3433513.
[57] S. Ruder, "An overview of gradient descent optimization algorithms," 2017. [Online]. Available: https://arxiv.org/abs/1609.04747
[58] D. P. Kingma and J. Ba, "Adam: A Method for Stochastic Optimization," 2017. [Online]. Available: https://arxiv.org/abs/1412.6980
[59] D. E. Rumelhart, G. E. Hinton, and R. J. Williams, "Learning representations by back-propagating errors," *Nature*, vol. 323, no. 6088, pp. 533–536, 1986, doi: 10.1038/323533a0.
[60] Y. LeCun, Y. Bengio, and G. Hinton, "Deep learning," *Nature*, vol. 521, no. 7553, pp. 436–444, 2015, doi: 10.1038/nature14539.
[61] H. Allamy, "METHODS TO AVOID OVER-FITTING AND UNDER-FITTING IN SUPERVISED MACHINE LEARNING (COMPARATIVE STUDY)," Dec. 2014.

[62] X. Glorot and Y. Bengio, “Understanding the difficulty of training deep feedforward neural networks,” in *Proceedings of the Thirteenth International Conference on Artificial Intelligence and Statistics*, Y. W. Teh and M. Titterington, Eds., in Proceedings of Machine Learning Research, vol. 9. Chia Laguna Resort, Sardinia, Italy: PMLR, Apr. 2010, pp. 249–256. [Online]. Available: https://proceedings.mlr.press/v9/glorot10a.html
[63] Zhejiang BLD Solar Technology Co. Ltd., “BLD250-290 60M Monocrystalline Solar Panel Datasheet,” 2020.
[64] L. ZHANG, S. S. YU, T. FERNANDO, H. H.-C. IU, and K. P. WONG, “An online maximum power point capturing technique for high-efficiency power generation of solar photovoltaic systems,” *Journal of Modern Power Systems and Clean Energy*, vol. 7, no. 2, pp. 357–368, 2019, doi: 10.1007/s40565-018-0440-2.
[65] A. Geron, *Hands-on Machine Learning With Scikit-Learn, Keras and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems*, Third edition. O’Reilly, 2022.
[66] D. Vinnikov, A. Chub, O. Korkh, E. Liivik, F. Blaabjerg, and S. Kouro, “MPPT performance enhancement of low-cost PV microconverters,” *Solar Energy*, vol. 187, pp. 156–166, 2019, doi: https://doi.org/10.1016/j.solener.2019.05.024.
[67] Q. Zhao, J. Zhang, C. Fu, Y. Chen, and Q. Yang, “A Structure-Reconfigurable LLC Resonant Converter With Wide Gain Range,” *IEEE J Emerg Sel Top Power Electron*, vol. 11, no. 4, pp. 4057–4067, 2023, doi: 10.1109/JESTPE.2023.3281929.
[68] S. Zhang, Y. Wang, B. Chen, F. Han, and Q. Wang, “Studies on a Hybrid Full-Bridge/Half-Bridge Bidirectional CLTC Multi-Resonant DC-DC Converter with a Digital Synchronous Rectification Strategy,” *Energies (Basel)*, vol. 11, no. 1, 2018, doi: 10.3390/en11010227.
[69] GaN Systems Inc., “GS61008T Top-side cooled 100 V E-mode GaN transistor Datasheet,” Apr. 2020.
[70] STMicroelectronics, “MB1551-U585I-C02 Schematic.” Accessed: May 01, 2025. [Online]. Available: https://www.st.com/resource/en/schematic_pack/mb1551-u585i-c02_schematic.pdf
[71] Y. Liu, “High Efficiency Optimization of LLC Resonant Converter for Wide Load Range,” Faculty of the Virginia Polytechnic Institute and State University, Blacksburg, Virginia, 2007.
[72] Infineon, “GaN-Based 3KW Full Bridge LLC Resonant Converter Reference Design.” Accessed: May 01, 2025. [Online]. Available: https://www.infineon.com/cms/en/product/evaluation-boards/gs-evb-llc-3kw-gs/
[73] STMicroelectronics, “STM32U585AI Datasheet,” Mar. 2024.
[74] Frederik Dostal, “Switching Regulator Noise Reduction with an LC Filter.” Accessed: May 01, 2025. [Online]. Available: https://www.analog.com/en/resources/technical-articles/switching-regulator-noise-reduction-with-an-lc-filter.html
[75] S. J. Zand, S. Mobayen, H. Z. Gul, H. Molashahi, M. Nasiri, and A. Fekih, “Optimized Fuzzy Controller Based on Cuckoo Optimization Algorithm for Maximum Power-Point Tracking of Photovoltaic Systems,” *IEEE Access*, vol. 10, pp. 71699–71716, 2022, doi: 10.1109/ACCESS.2022.3184815.
[76] B.-R. Peng, K.-C. Ho, and Y.-H. Liu, “A Novel and Fast MPPT Method Suitable for Both Fast Changing and Partially Shaded Conditions,” *IEEE Transactions on Industrial Electronics*, vol. 65, no. 4, pp. 3240–3251, 2018, doi: 10.1109/TIE.2017.2736484.

Chapter 7. Appendices

7.1 Appendix A

A Smart Maximum Power Point Tracking for LLC Converter

Supervisor: Ir. Dr. Freddy Tan Kheng Suan
Moderator: Prof. T. Nandha Kumar

Student Name: Donald Ho Guan Hui
Student ID: 20278538

Legends: Tasks Planned Week: Tasks Delay Week: Project Milestone: Submission Dateline:

Tasks	Start Week	End Week	Total Week
Project Initiation and Planning			
Conduct Preliminary Research	Week 4	Week 6	3
Prepare Project Planning Report	Week 6	Week 7	2
Review and Finalize Project Planning Report	Week 7	Week 9	3
Submit Project Planning Documentation	Week 9	Week 9	1
Design and Development of Two Stage GMPPT Algorithm			
Define PV Configuration Parameters	Week 8	Week 9	2
Analyze LLC Resonant Converter Topology	Week 9	Week 11	3
Study Pulse Frequency Modulation (PFM) Techniques	Week 9	Week 10	2
Study Secondary-Side Phase Shift Modulation (SS-PSM) Techniques	Week 28	Week 29	2
Design Current Sampling Methods	Week 13	Week 23	4
Generate Synthetic Data (PSC Variations)	Week 14	Week 23	3
Apply Data Generalization Techniques (Temperature Variation)	Week 27	Week 28	2
Train Artificial Neural Network (ANN)	Week 11	Week 12	2
Train 1D-Convolutional Neural Network (1D-CNN)	Week 13	Week 23	4
Perform Quantitative Analysis of Neural Networks	Week 14	Week 23	3
Implement Second-Stage Initialization and Hill Climbing Method	Week 23	Week 24	2
Software Simulation and Validation			
Configure LLC Resonant Converter in Simulation Environment	Week 9	Week 11	3
Generate Switching Signals for Switching Devices	Week 11	Week 11	1
Set Up Simulated PV System Environment	Week 11	Week 12	2
Integrate Deep Neural Network into Simulation Framework	Week 13	Week 14	2
Implement P&O Algorithm for Baseline Comparison	Week 14	Week 15	2
Deploy and Validate Smart MPPT Algorithm using PFM Modulation	Week 15	Week 27	6
Deploy and Validate Smart MPPT Algorithm using SS-PSM Modulation	Week 28	Week 30	3
Edge Computing Deployment			
Establish UART Communication Protocol on STM32-B-U585I-IOT02A	Week 12	Week 23	5
Integrate Feedback Sensors with Microcontroller	Week 14	Week 24	4
Configure Complementary PWM Signals for PFM Operation	Week 14	Week 24	4
Code Hill Climbing Algorithm on Edge Device	Week 23	Week 24	2
Code Two-Stage GMPPT Algorithm (PFM Mode) on Microcontroller	Week 24	Week 25	2
Hardware Implementation and Testing (PFM Only)			
Verify Hardware Prototype and Circuit Integrity	Week 23	Week 24	2
Conduct Open-Loop Testing using PFM Modulation	Week 23	Week 26	2
Test Hill Climbing Algorithm in Hardware Environment	Week 27	Week 32	1
Test Smart MPPT Algorithm (PFM Mode) in Real-Time Operation	Week 29	Week 33	1
Collect and Analyze System Performance Data under PSC and Temperature Variations	Week 29	Week 34	1
Thesis, Documents and Final Presentation			
Draft Thesis Writing	Week 30	Week 32	2
Submit Draft Thesis for Supervisor Review	Week 32	Week 32	1
Final Thesis Writing	Week 32	Week 35	4
Submit Final Thesis, Logbooks, and Project Files	Week 35	Week 35	1
Prepare and Deliver Final Presentation	Week 35	Week 35	1
Presentation	Week 36	Week 36	1
Return Project Equipment and Complete Project Closure	Week 36	Week 37	2

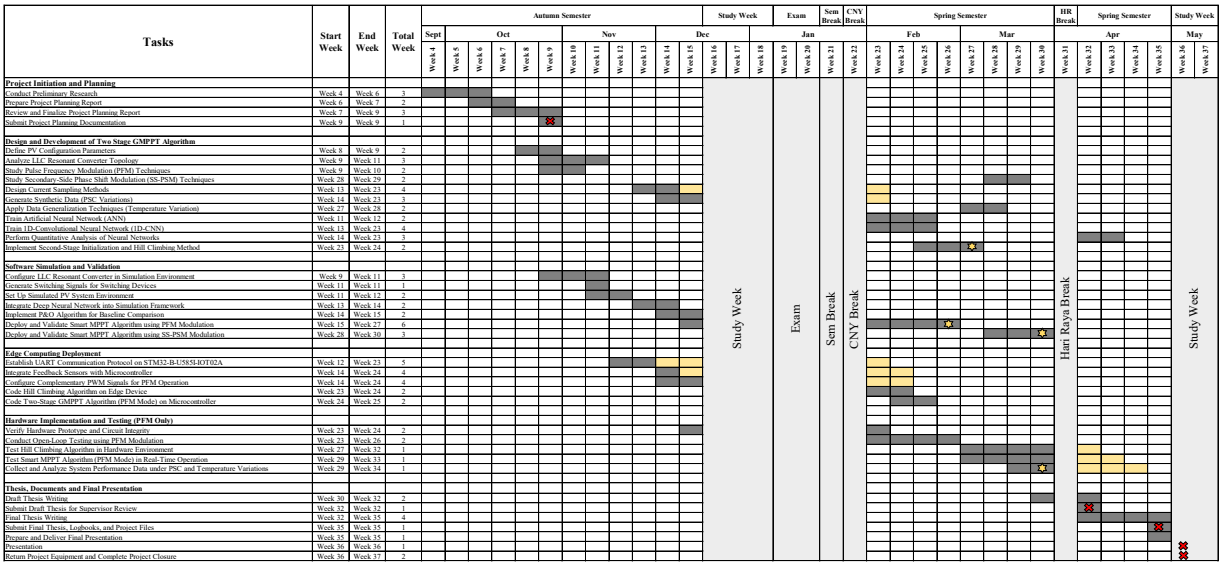


Figure 7.1. Actual Gantt chart for the project.

7.2 Project Proforma

7.2.1 Project Proforma 1

Project Review Proforma – Review number: 1 Date: 27/11/2024

Summarised Planned State of Project:	**Actual Progress Since Last Review**
Literature Review: 1. Research covers operation of LLC resonance converter, global maximum power point tracking (GMPPT) algorithm, and edge computer. 2. Research covers software tools, models, and architecture for development and deployment for GMPPT algorithm. Setting up LLC resonance converter with photovoltaic panel (PV) in Simulink: 1. Analyse the circuit waveform of the converter under different input voltage and load condition. 2. Interfacing PV panel with LLC resonance converter using P&O algorithm for local maximum power tracking. Setting up B-U585I-IOT02A microcontroller: 1. Programming on edge computer. 2. Generate complementary PWM signal using ADC values from microcontroller.	Literature Review: 1. Research was carried out on development of GMPPT algorithm onto edge computer and GMPPT algorithm using Boost and LLC converter with specific AI models. Setting up LLC resonance converter with photovoltaic panel (PV) in Simulink: 1. Basic waveforms of LLC converter and hill climbing method (P&O) are analysed and demonstrated to supervisor.
Next Steps 1. According to my Gantt chart planning, I am slightly behind schedule. The delayed tasks include open loop testing of the LLC converter, voltage and current sensor reading using ADC, generation of complementary PWM signals, and creation of synthetic data for AI model prediction. Therefore, I need to expedite these tasks before the next project review (Proforma 2). 2. Focusing on a journal that prediction power and voltage of 3 substring PV panel using ANN training the neural network in Google Collab, deploy the developed TensorFlow Lite model inside Simulink and B-U585I-IOT02A microcontroller for testing. **Supervisor Feedback** Overall, the project is progressing well, and the student has been able to keep up with the project.	

Figure 7.2. Project proforma 1.

7.2.2 Project Proforma 2

Project Review Proforma – Review number: 2 Date: 13/2/2025

<table>
<tr>
<td>Summarised Planned State of Project:
Hardware Testing using B-U585I-IOT02A Edge Computer:
1. Open loop testing of the LLC converter.
2. Setup of 1kHz Timer for sampling and processing of GMPPT algorithm.
3. Setup of voltage and current sensor reading using ADC.
4. Generation of complementary PWM signals.

Development of GMPPT ANN model:
1. Generation of synthetic data for ANN prediction.
2. Implementation of ANN model inside Matlab environment.
3. Deployment of ANN model into B-U585I-IOT02A Edge Computer.</td>
<td>Actual Progress Since Last Review
Hardware Testing using B-U585I-IOT02A Edge Computer.
1. Open loop testing of the LLC converter under different operating condition.
2. Configuration of the ADC for reading voltage values from the voltage and current sensor.
3. Generation of complementary PWM signals with deadtime.

Development of stage first stage MPPT algorithm:
1. Explored adaptive P&O MPPT algorithm using Fuzzy Logic Controller and tested in Simulink environment.

Development of stage second stage MPPT algorithm:
1. Generated 12341 synthetic data points using a three-substring PV panel, with irradiance values ranging from 1 to 1000 in increments of 25, utilizing combinations with replacement.
2. Implementation of ANN model to predict sine wave inside Matlab and Simulink environment.</td>
</tr>
<tr>
<td colspan="2">Next Steps
1. According to my Gantt chart planning, I am now on schedule.
2. Calibrate the values obtained from voltage and current sensors using linear regression method.
3. Development of a GMPPT ANN model using three current values 0.4Voc, 0.4 + Voc, and 0.2 + 2Voc of the three-substring PV panel as inputs, to output the voltages and power of the three local maxima.
4. Integration with GMPPT ANN model, P&O algorithm, and LLC converter in Simulink.

Supervisor Feedback
Student is progressing well but is encouraged to place greater emphasis on the novelty of the project, particularly the use of neural network approaches for MPPT. Less focus should be placed on classical methods such as PSO and Fuzzy Logic, as they are not the core contribution of the work. Additionally, the supervisor recommends that the second-stage MPPT algorithm be lightweight and computationally efficient to ensure practical implementation.</td>
</tr>
</table>

Figure 7.3. Project proforma 2.

7.2.3 Project Proforma 3

Project Review Proforma – Review number: 3 Date: 6/3/2025

<table>
<tr>
<td>Summarised Planned State of Project:
Development of GMPPT ANN Algorithm:
1. Coding of GMPPT neural network model in Google Collab.
2. Convert trained model into TensorFlow file.

Integrated System Simulink:
1. Deployment of GMPPT neural network model inside Simulink.
2. Convert the predicted voltage and power at global maximum into initial switching frequency.

Deployment of into Edge Computer:
1. Calibration of sensors values using linear regression model.
2. Deployment of GMPPT neural network model into B-U585I-IOT02A Edge Computer.</td>
<td>Actual Progress Since Last Review
Development of GMPPT ANN Algorithm:
1. Normalized the input values using Isc, Voc, and Pmax of the PV panel.
2. Trained a neural network with three inputs and six outputs, featuring two hidden layers with 10 neurons each.

Integrated System Simulink:
1. Validate ANN predictions within Simulink.
2. Combine FHA method in initializing switching frequency with P&O (Hill Climbing) method in continuously tracking the global maximum.

Deployment into Edge Computer:
1. This task is still ongoing.</td>
</tr>
<tr>
<td colspan="2">Next Steps
1. According to my Gantt chart planning, I am now on schedule.
2. The next step would be to test deploy the algorithm onto the hardware prototype.

Supervisor Feedback
The supervisor suggested exploring the performance of the algorithm under varying temperature conditions, as it has already shown promising results under different irradiance levels with a fixed temperature. This would help further validate the robustness and adaptability of the proposed method.</td>
</tr>
</table>

Figure 7.4. Project proforma 3.

7.2.4 Project Proforma 4

Project Review Proforma – Review number: 4 Date: 27/3/2025

Summarised Planned State of Project:	**Actual Progress Since Last Review**
Development of MPPT using both PFM and SS-PSM modulation technique: 1. Input parameter generalization method for neural network (to account for varying temperature of the PV array). 2. Optimize current sampling algorithm, NN prediction, P&O algorithm for both modulation technique. Hardware Prototype Testing: 1. Testing functionality and reliability of two stage GMPPT method in STM32 ensuring that the result obtain is identical to SIMULINK. 2. Test the functionality of the algorithm using programmable DC power supply. Thesis and Documents: 1. Writing Draft thesis	Development of Two-Stage GMPPT Algorithm in SIMULINK: 1. Able to get normalized parameter Isc, denormalized parameter Pmpp and Voc for the neural network under varying temperature condition. 2. Derived the initialization Equation for SS-PSM method. 3. Managed to sample input current values for the neural network, initialize P&O algorithm using the predicted GMPP value obtained from neural network using both modulation technique. Extending Gain Range of the Converter: 1. Adding VDQRR circuit at secondary side to increase the gain of the converter. 2. Using Xilinx to code and reconfigure the existing full bridge into half bridge. Testing MPPT using hardware prototype: 1. This task is still ongoing.
Next Steps 1. Testing the algorithm using programmable DC power supply. 2. Collect necessary data for draft thesis. **Supervisor Feedback** The student is progressing well and is on track to meet the project objectives. In response to the limited gain range of the existing converter, the student proposed fabricating a new SS-PSM board and the fabrication process is currently in progress.	

Figure 7.5. Project proforma 4.

7.2.5 Project Proforma 5

Project Review Proforma – Review number: 5 Date: 24/4/2025

Summarised Planned State of Project:	**Actual Progress Since Last Review**
Thesis, Documents and Presentation: 1. Submission of draft thesis. 2. Writing final thesis. Hardware Implementation and Testing (Using PFM method): 1. Testing P&O algorithm using programmable DC power supply. 2. Fully functional GMPPT algorithm using PFM method.	Hardware Implementation and Testing (using PFM method): 1. Managed to perform P&O algorithm using chroma programmable power supply. 2. Algorithm managed to track GMPP under varying temperature and irradiance condition. 3. Managed to collect data under varying temperature and irradiance condition. PFM and SS-PSM method in Simulink: 1. Managed to collect data under varying temperature and irradiance condition for PFM method. 2. Managed to collect data under varying temperature and irradiance condition for SS-PSM method. Thesis, Documents and Presentation: 1. The draft thesis is completed and submitted. 2. Working on final thesis
Next Steps 1. Overall, the project is completed on time with all the objectives met although the project timeline is delayed compared to the initially planned timeline. 2. Finalizing thesis and presentation slide for project presentation. **Supervisor Feedback** The student has met all project objectives, with the prototype performing reliably and delivering excellent results when tested against a programmable power supply.	

Figure 7.6. Project proforma 5.